



6.2 Winding Better Rolls

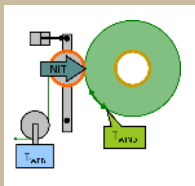
Winding Process, Roll Structure, Winder Design

Timothy J. Walker, TJWalker + Associates Inc.

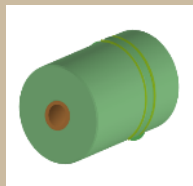
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tjwalker@tjwa.com www.webhandling.com +1 651.686.5400

WH Fundamentals

6: Winding: Process, Roll Quality, Equipment Design



**Winding
Process**



**Roll
Quality**



**Winder
Design**

+ Winding, Rolls, Winders

- ✓ What determines the pressure inside a wound roll?
- ✓ What causes roll and web defects?

6.0 Winding

6.1. Winding Success

6.2. Material Properties (Web and Core)

6.3. Winding a Roll

- ▶ Starting a Roll: Roll Transfers
- ▶ Managing Initial Contact
- ▶ Tension and Torque
- ▶ Winder Tensioning
- ▶ Co-Axial Winding

6.4. Roll Structure and Web Quality

- ▶ Wound-In Stresses (Pressures)
- ▶ Lateral Errors
- ▶ Cross Machine Variations (Gauge Bands and Bagginess)
- ▶ In-Roll Buckling

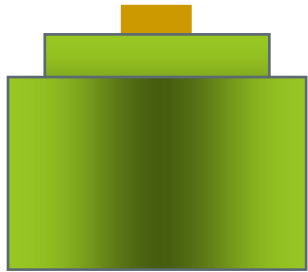
6.6. Wind-Ability

6.4 Roll Structure and Web Quality

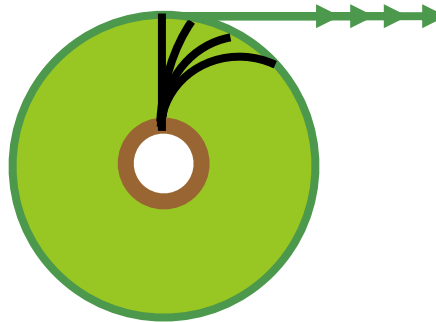
- Wound-In Stresses 
- Lateral Errors
- In-Roll Buckling
- Transverse Variations
(Gauge Bands and Bagginess)

Why is Roll Pressure Important?

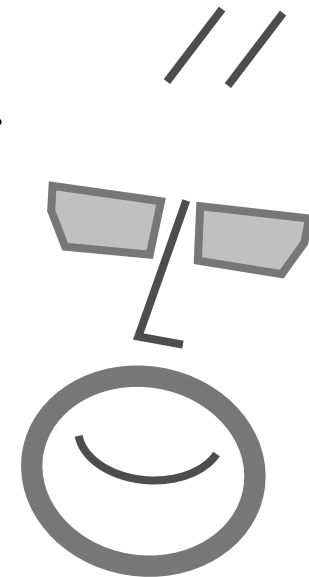
Roll pressure is required to hold a roll's layers together
(unless your product is self-wound adhesive coated).



Avoid slip in
handling



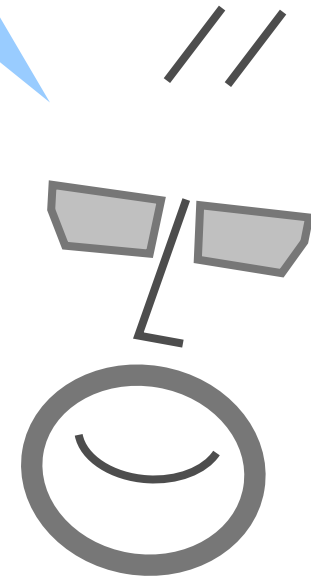
Avoid slip at
unwinding



Winding Process Questions

What is the pressure in a roll?

How does pressure build up as roll size increases?



Pressure Buildup in Stacking Bricks

Pressure under
1 brick

$$P_1 = W/A$$

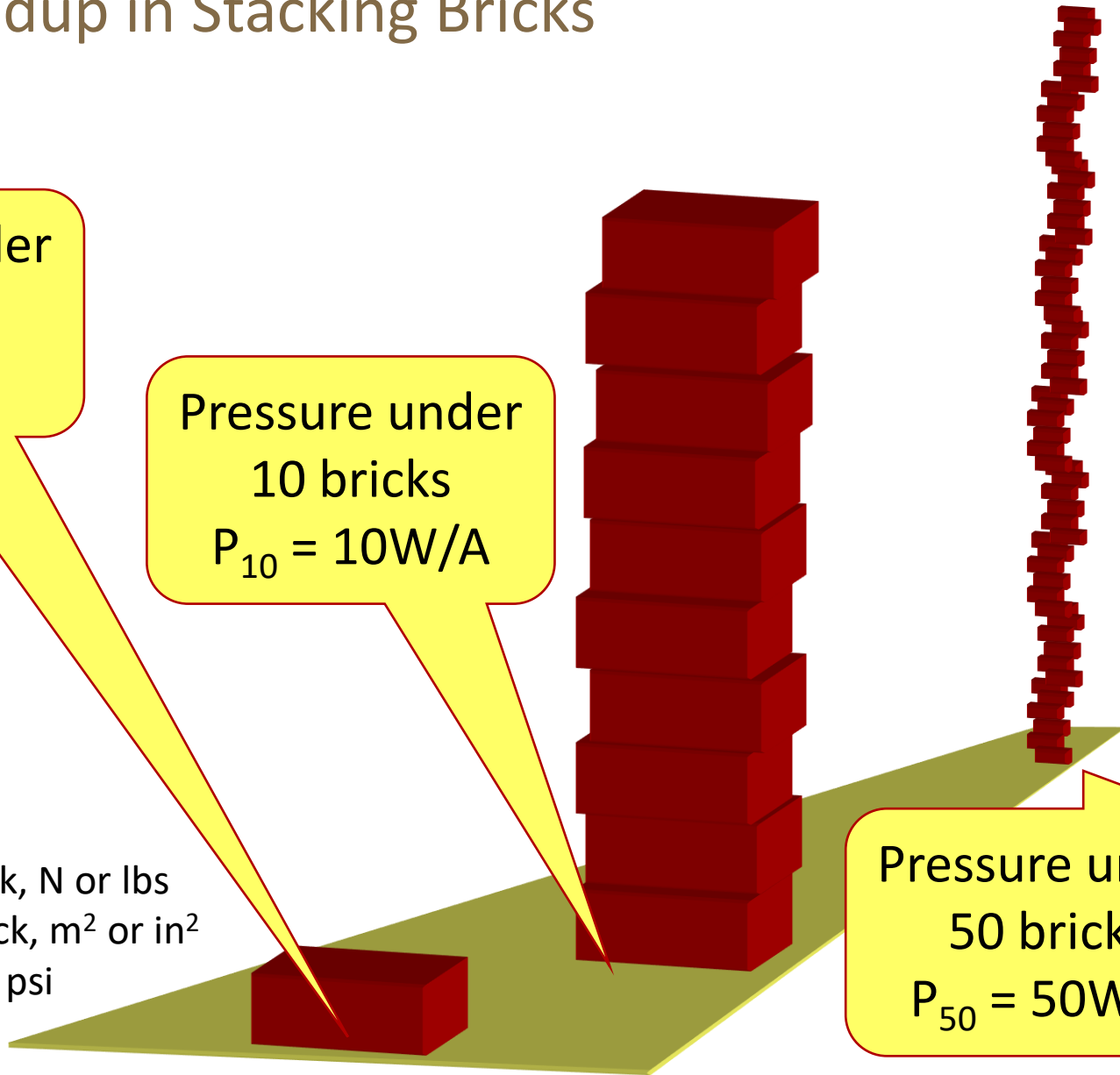
Pressure under
10 bricks

$$P_{10} = 10W/A$$

W = Weight of brick, N or lbs
A = Area under brick, m² or in²
P = Pressure, Pa or psi

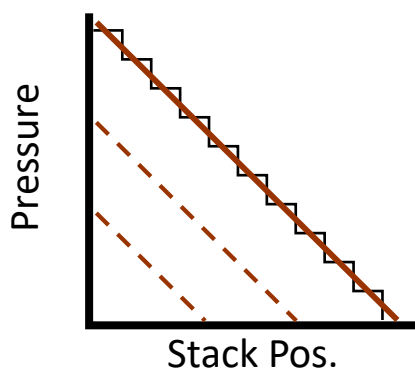
Pressure under
50 bricks

$$P_{50} = 50W/A$$

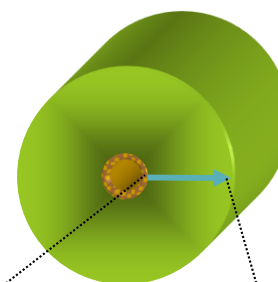
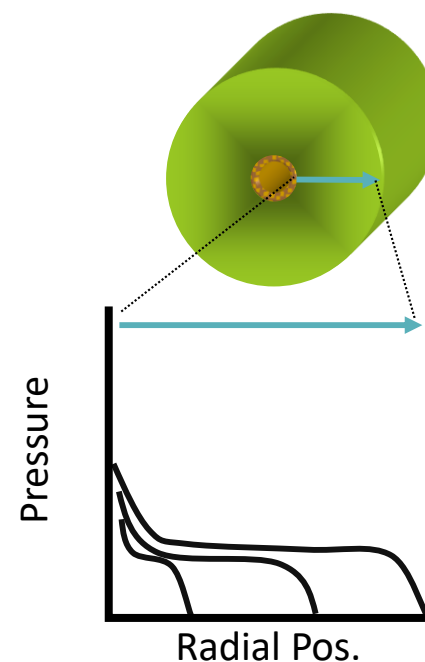
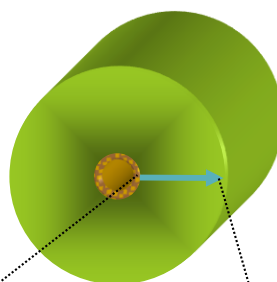
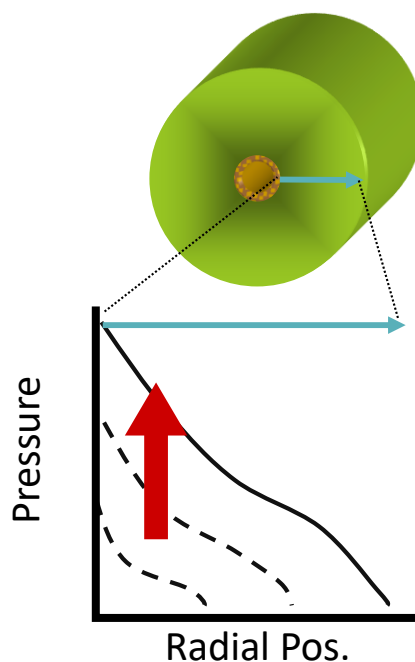


Pressure vs. Stack

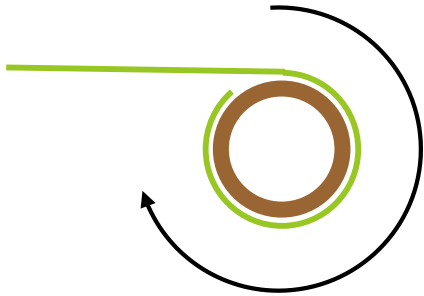
In brick stacks, the pressure is created by gravity. Stack compression or shifting does not change the force of gravity or stack pressure.



In wound rolls, the pressure is created by tension, which is directly proportional to strain. Compression or radial shifting of layers will reduce strain and reduce the pressure with the roll stack.

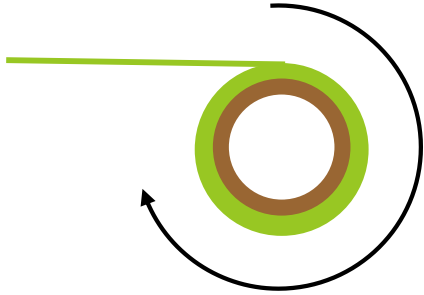


Pressure from Tension



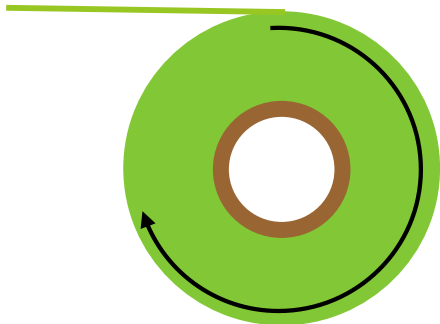
$$P = \frac{T}{r}$$

A tensioned web wrapping a core creates pressure.



$$P_{CORE} = n \frac{T}{r}$$

With added layers pressure may increase like stacking bricks.



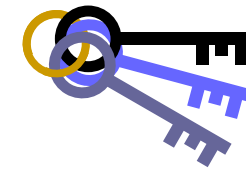
$$P_{CORE,n} < \sum_{1}^n \frac{T_x}{r_x}$$

Eventually, adding layers does not increase core pressure proportional to T/r.

T = Tension, N/m or lb/in

n = Number of layers

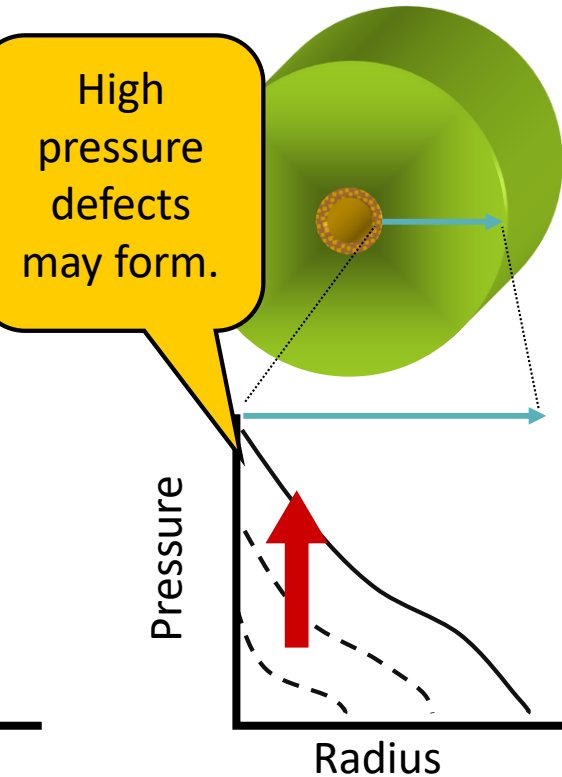
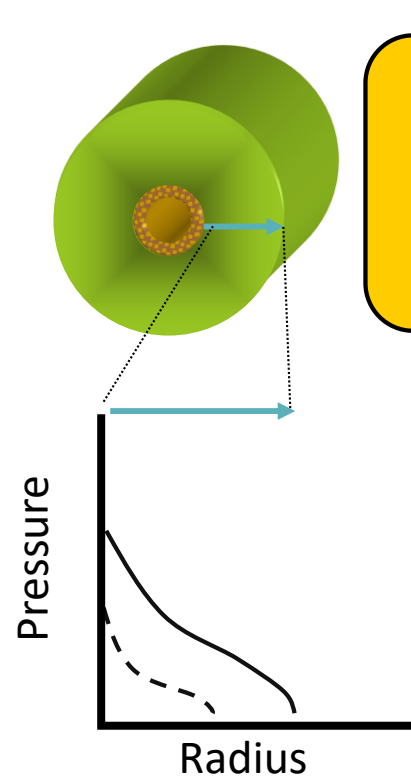
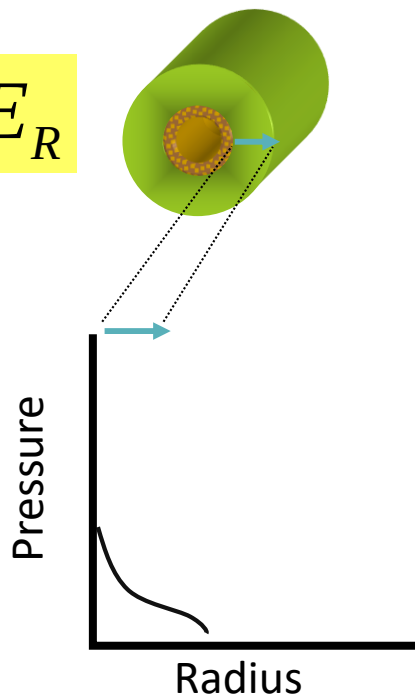
r = Radius, m or in



Tourniquet Effect

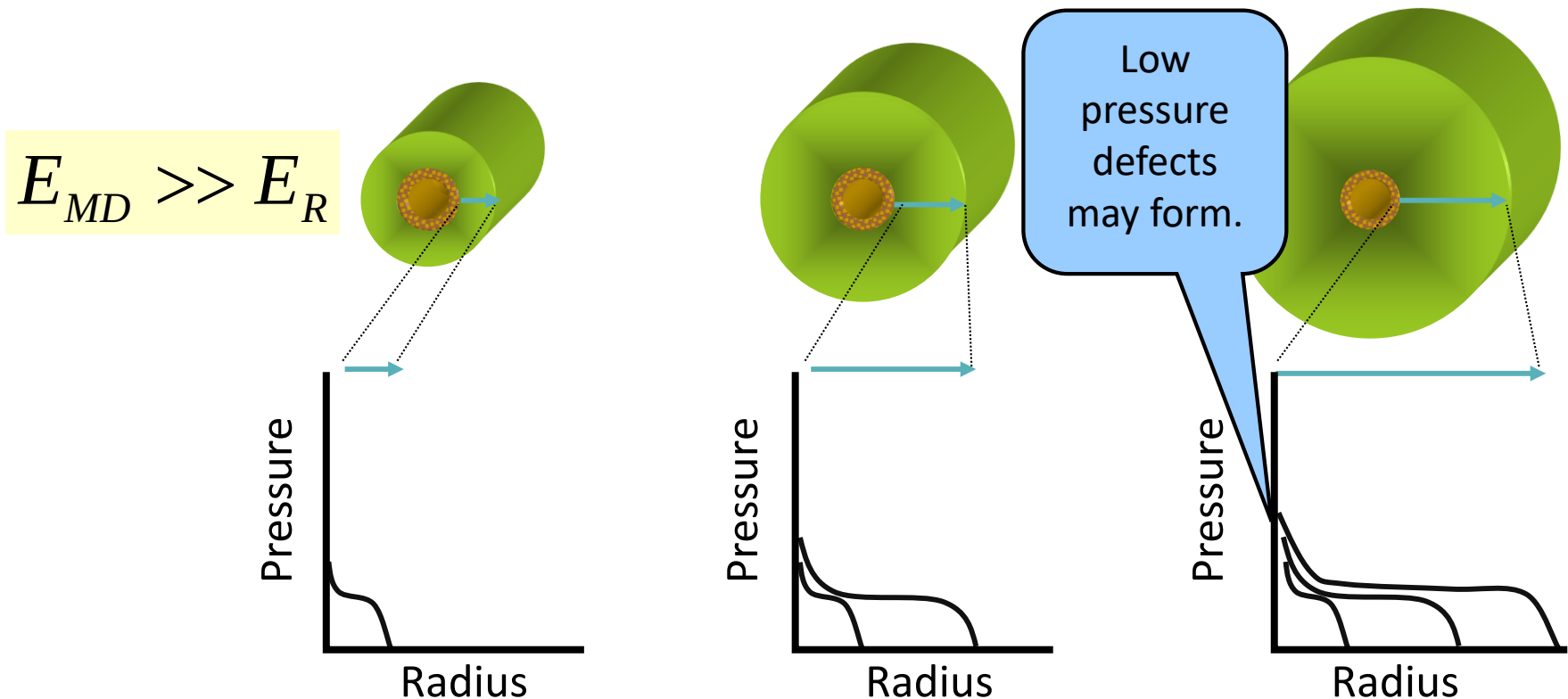
Isotropic rolls, where MD and radial moduli are similar, will have a significant tourniquet effect, increasing core pressure with every additional wrap.

$$E_{MD} \cong E_R$$



Minimal Tourniquet Effect

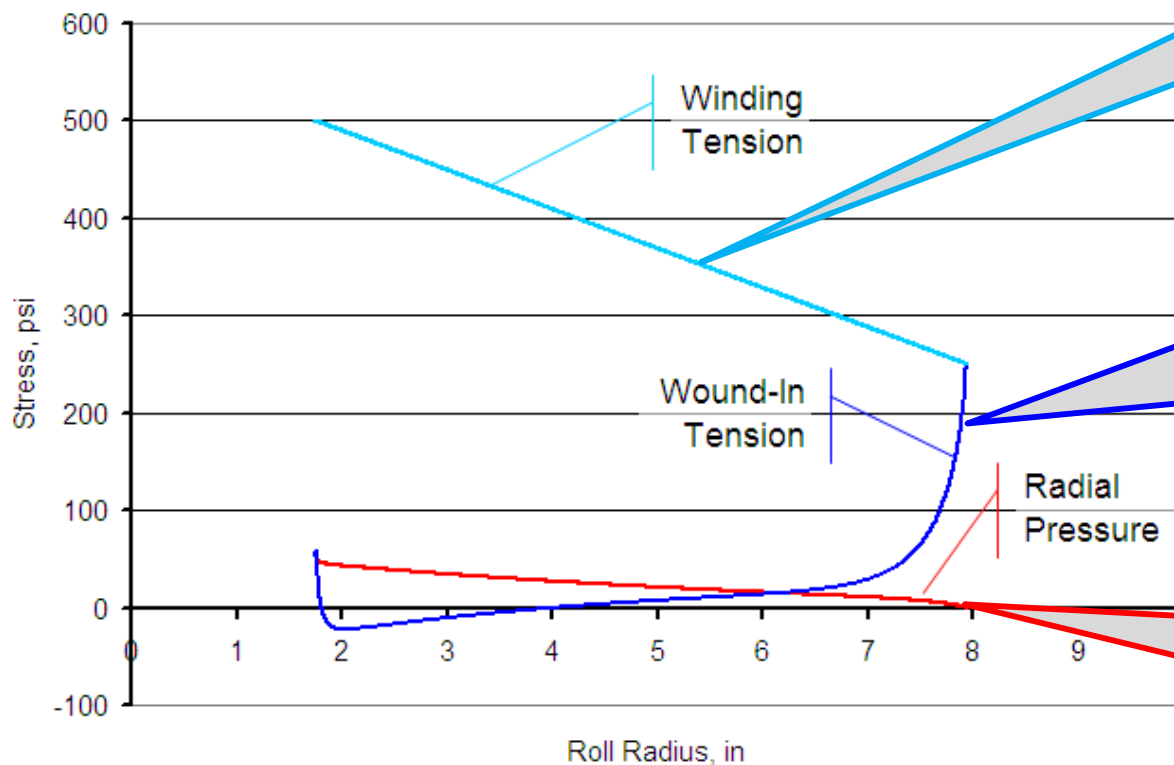
Anisotropic rolls, where MD modulus is much high than radial modulus, will have a minimal tourniquet effect - additional wraps will add little to core pressure.



Winding vs. Roll Stresses

This graph shows the winding process conditions and the final stresses of the finished roll, both tensions (hoop stresses) and pressures (radial stresses) vs roll radius.

Roll and Winding Stresses



Initial tension from torque and nipping (here showing taper tension vs radius).

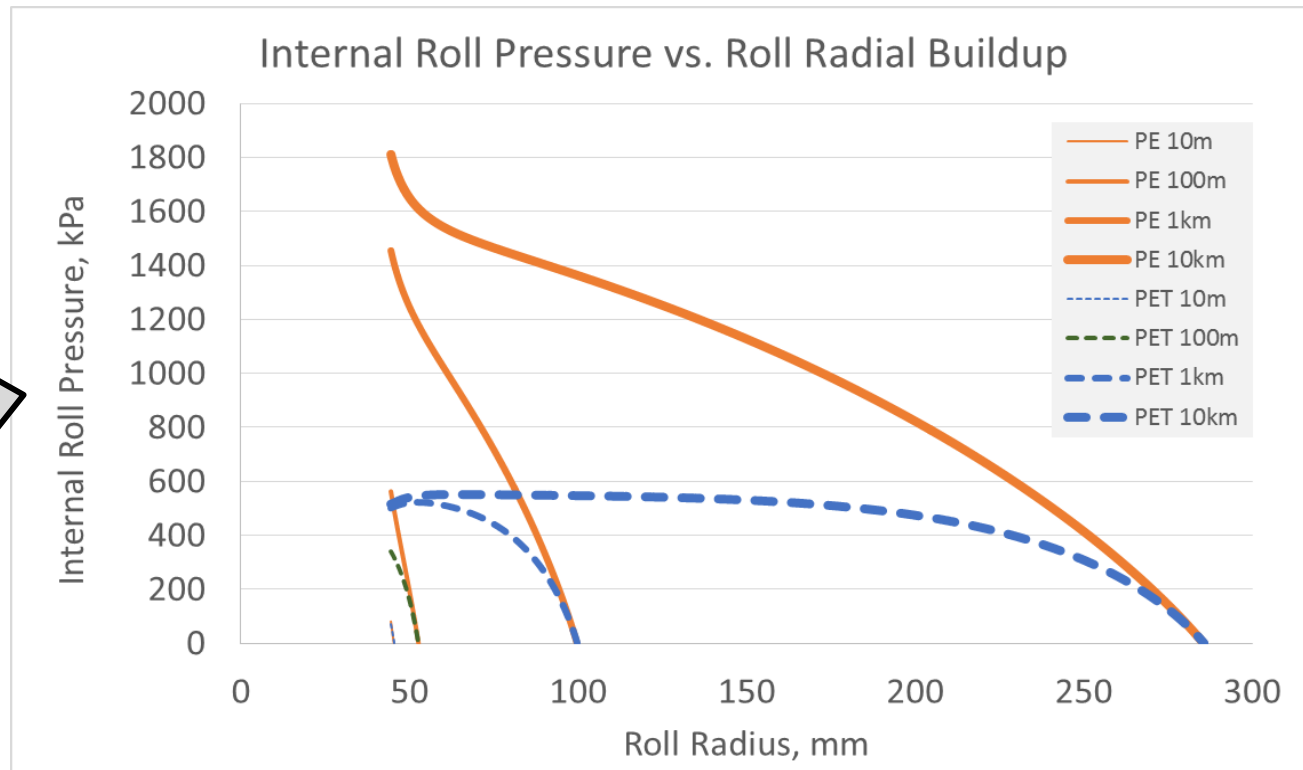
Tensions of finished roll are usually much lower than winding tensions due to strain loss of compression.

Pressures in the finished roll are highest at or near the core.

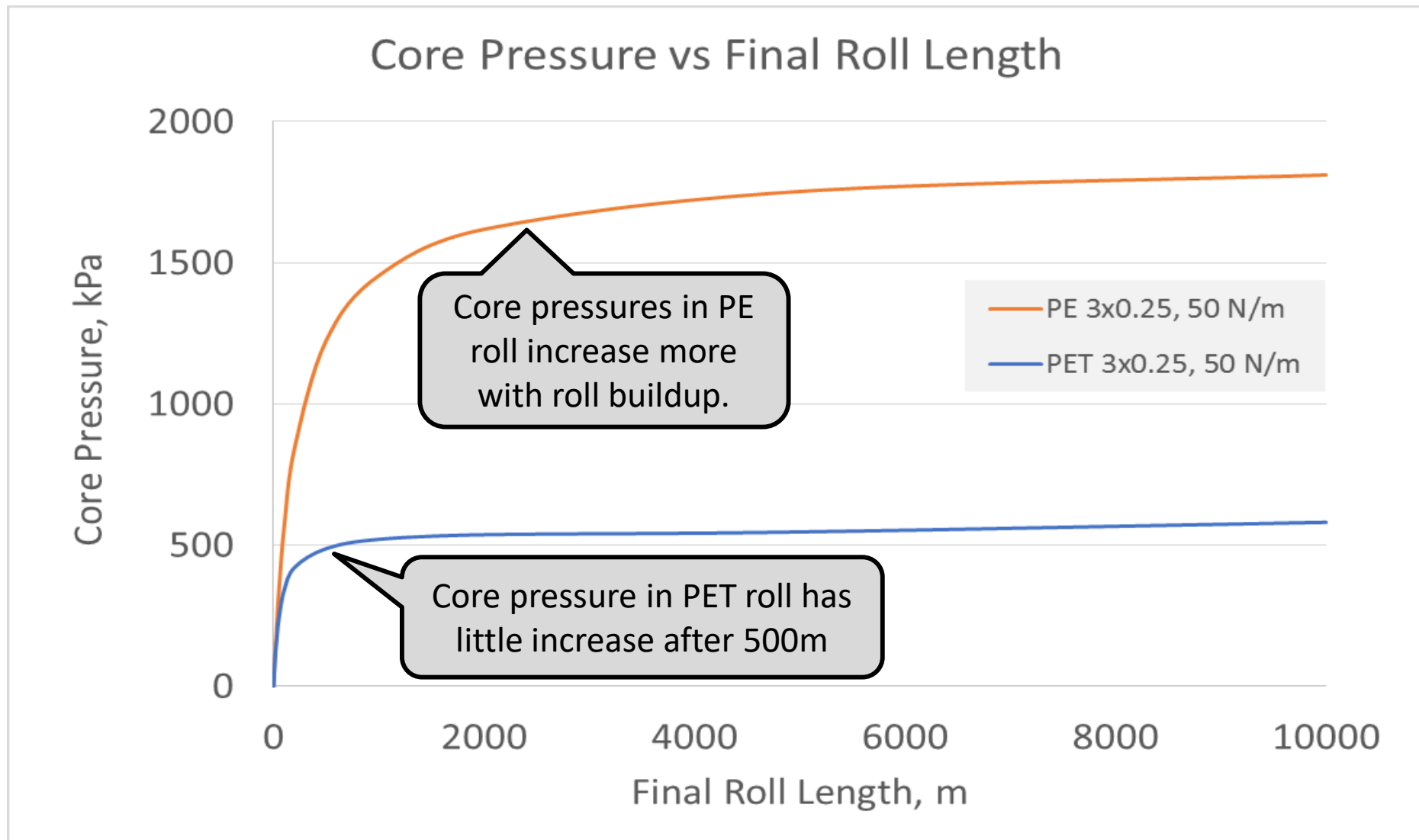
Internal Pressures of PE vs PET Films

Like stacking bricks, more layers in a roll will create more pressure. Unlike a stack of bricks, the pressure increase will eventually be non-linear vs. buildup. High modulus materials are more likely to have lower pressure when small compression relieve low strains.

Polyethylene films pressure rises more than polyester films wound to the same length.



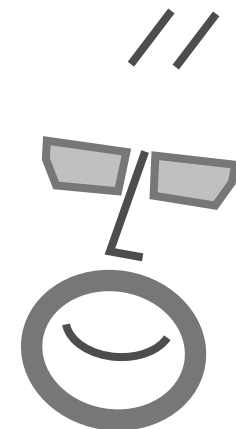
Core Pressure Increase vs Roll Length (PE vs PET)



Winding Solutions Qs

How do you measure pressure within a roll?

- Frictional inserts and force gauges
- Electronic pressure sensing laminate

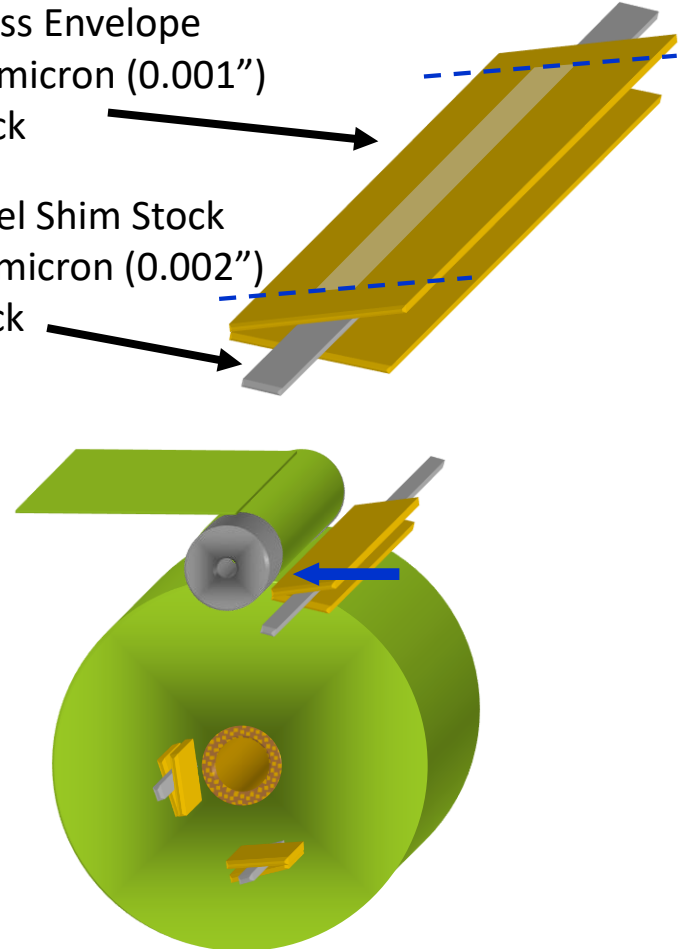


Roll Pressure from Friction Force Measurements

1. Make a friction 'sandwich' with thin steel strip in brass foil envelope.
2. Insert brass-steel foil sandwich into winding roll (preferably without stopping winding process (be safe)).
3. Remove roll from winder.
4. Connect force gauge to steel and measure force to slip steel relative to brass envelope.
5. Calculate pressure from frictional force.

Brass Envelope
25 micron (0.001")
thick

Steel Shim Stock
50 micron (0.002")
thick



D.M. Kedl, 3M Co., "Using a Two Dimensional Winding Model to Predict Wound Roll Stresses that Occur Due to Circumferential Steps in Core Diameter or to Cross-Web Caliper Variation", Proceedings of the First International Conference on Web Handling, 1993.

IWEB 93

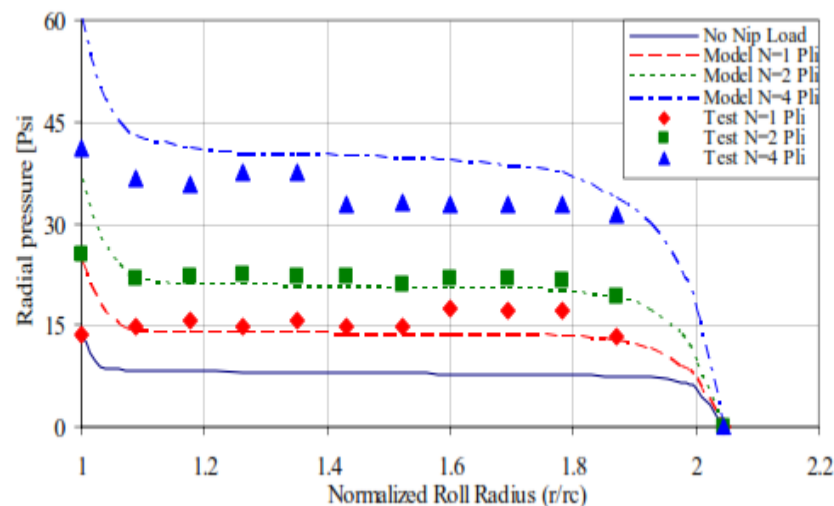
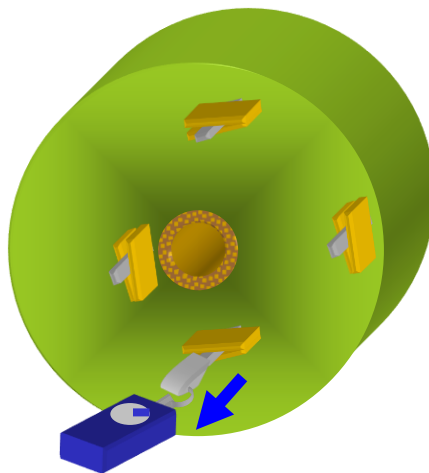
Roll Pressure from Friction Force Measurements

$$F = \mu N$$

$$F = \mu P A$$

$$F = \mu P (2w_S w_R)$$

$$P = \frac{F}{2\mu w_S w_R}$$



F = Frictional force, N or lbs

μ = Brass-steel coefficient of friction (~ 0.25)

P = Roll pressure, Pa or psi

A = Area of frictional contact (two sides of steel)

w_S = Width of steel strip

w_R = Width of roll (or length of steel strip for wide rolls).

Example: Roll pressure vs. radius
vs. winding nip load estimated
from friction sandwich forces.

**The Development of Wound-On
Tension in Webs Wound Into Rolls,**
B. K. Kandadai, Oklahoma St. Univ.
PhD Thesis, Dec. 2006

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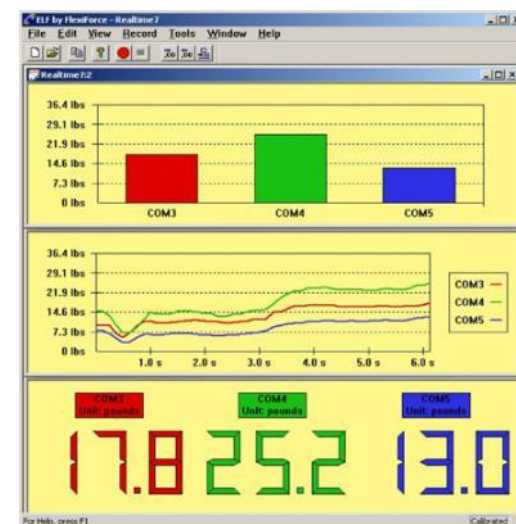


**Force
Range**

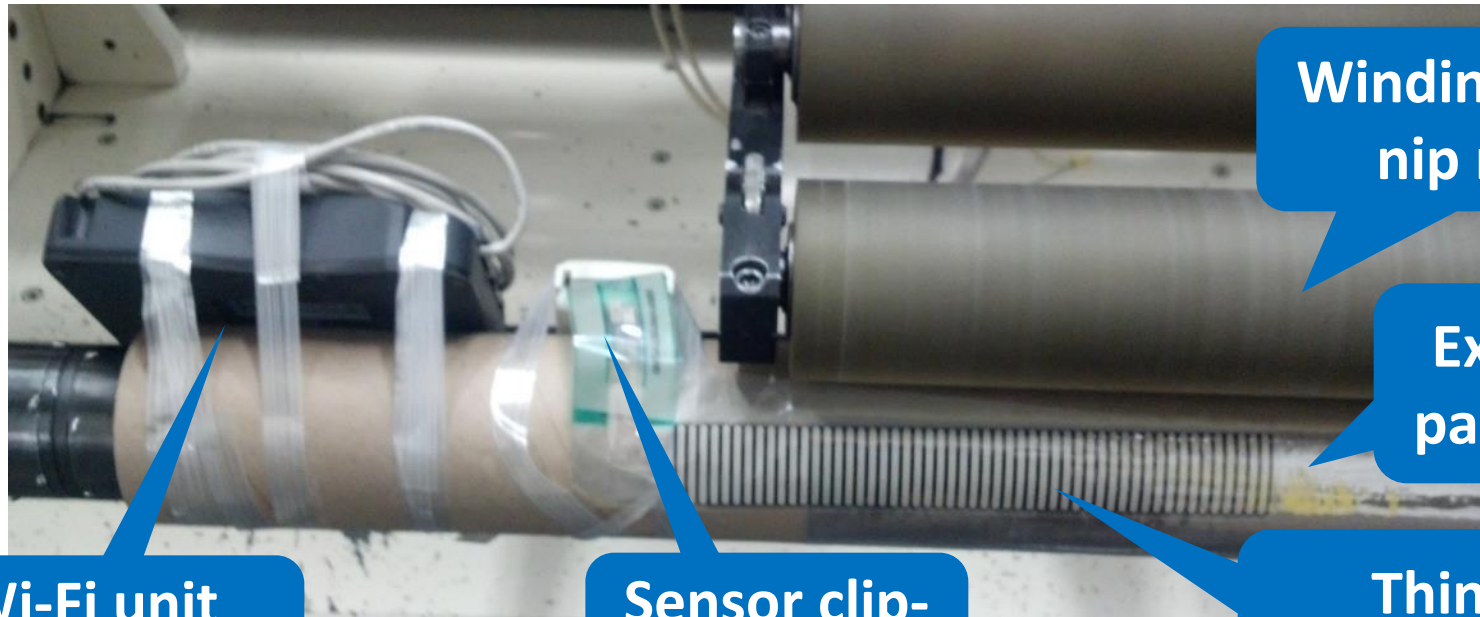
**1
25
100**

**Pressure
Range**

**5
225
906**



Wi-Fi Winding Measurement Exp'l Setup #1



Winding lay-on
nip roller

Extended
paper core

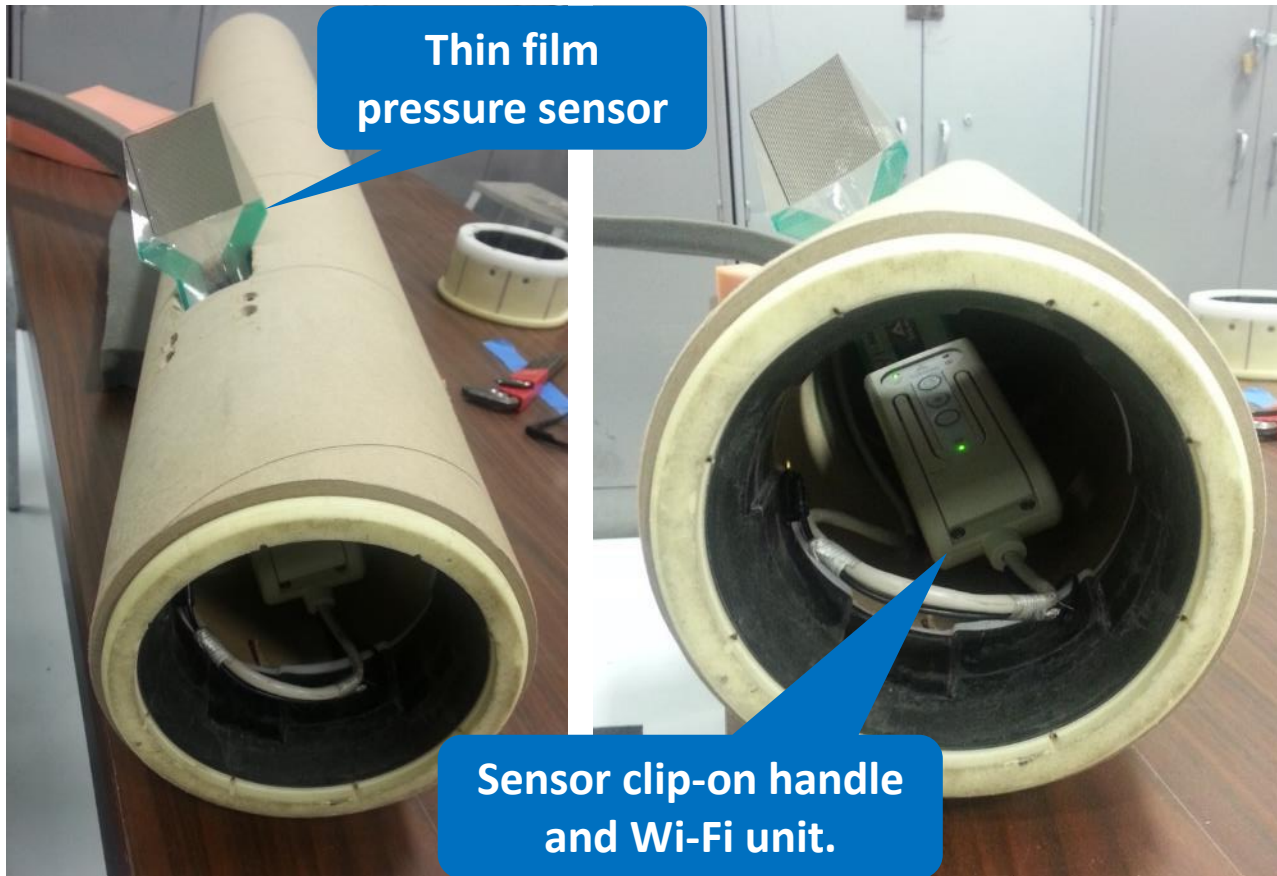
Wi-Fi unit
with battery

Sensor clip-
on 'handle'

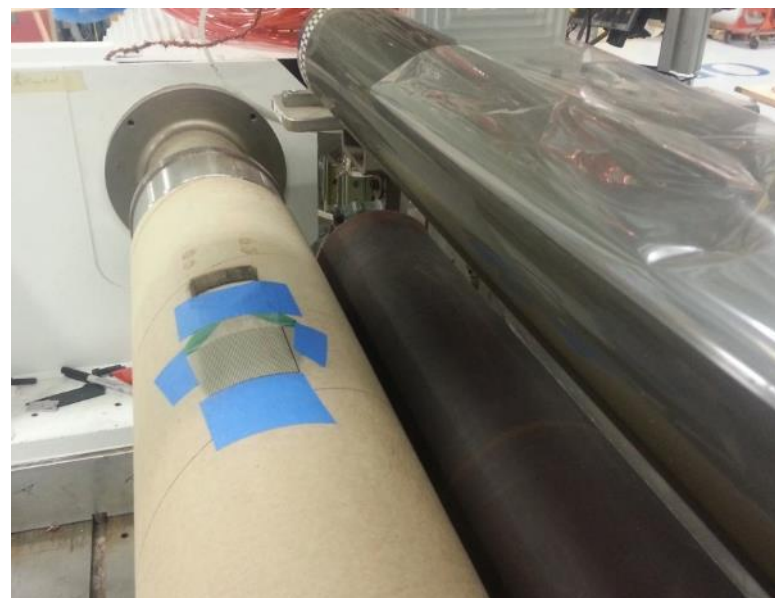
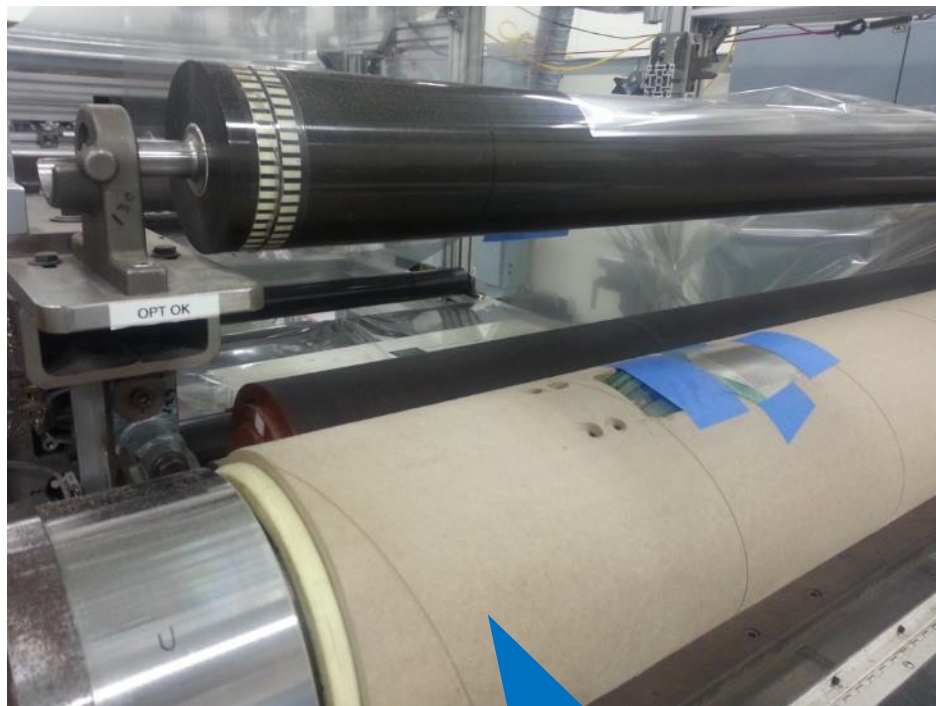
Thin film
pressure sensor



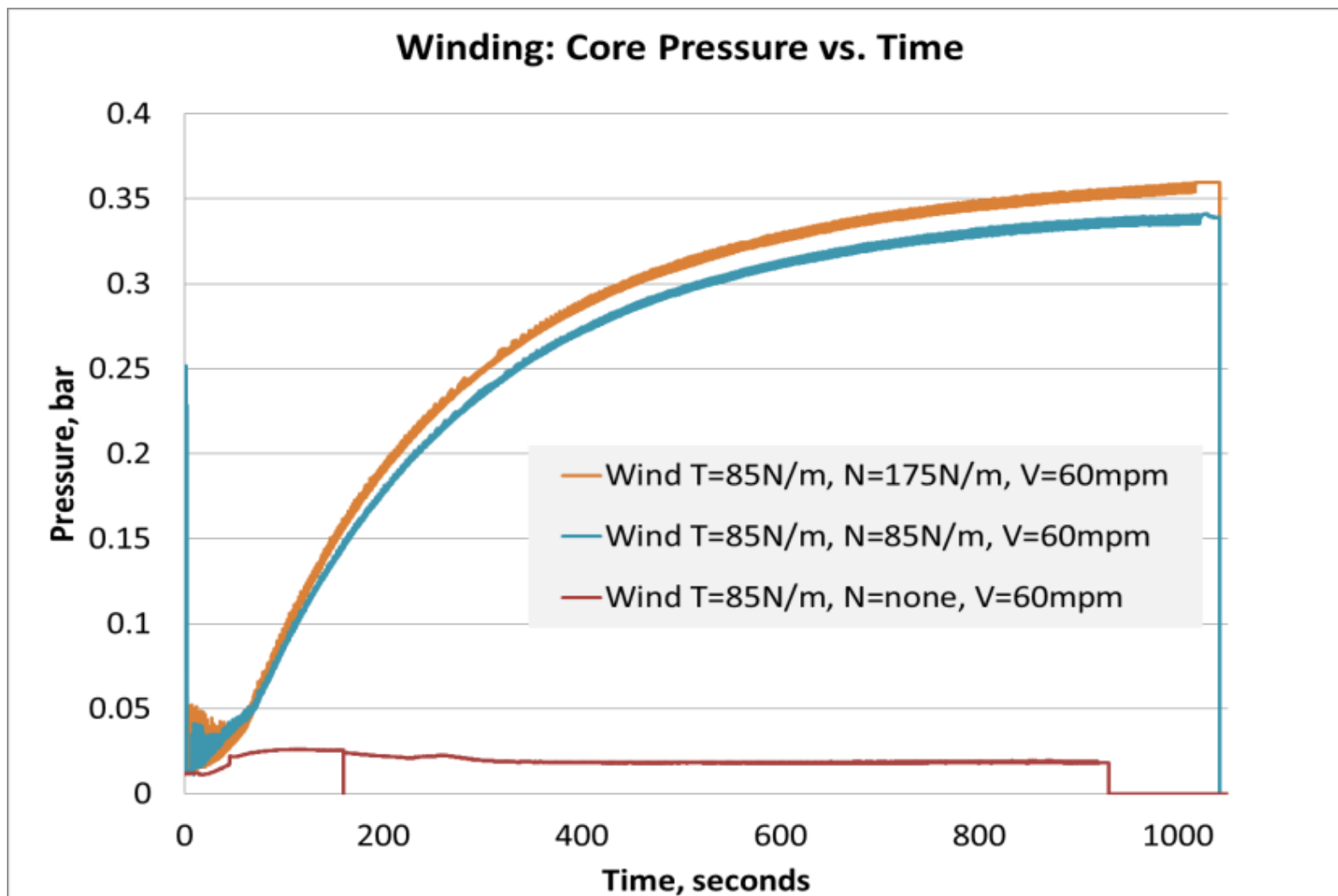
Experimental Setup #2

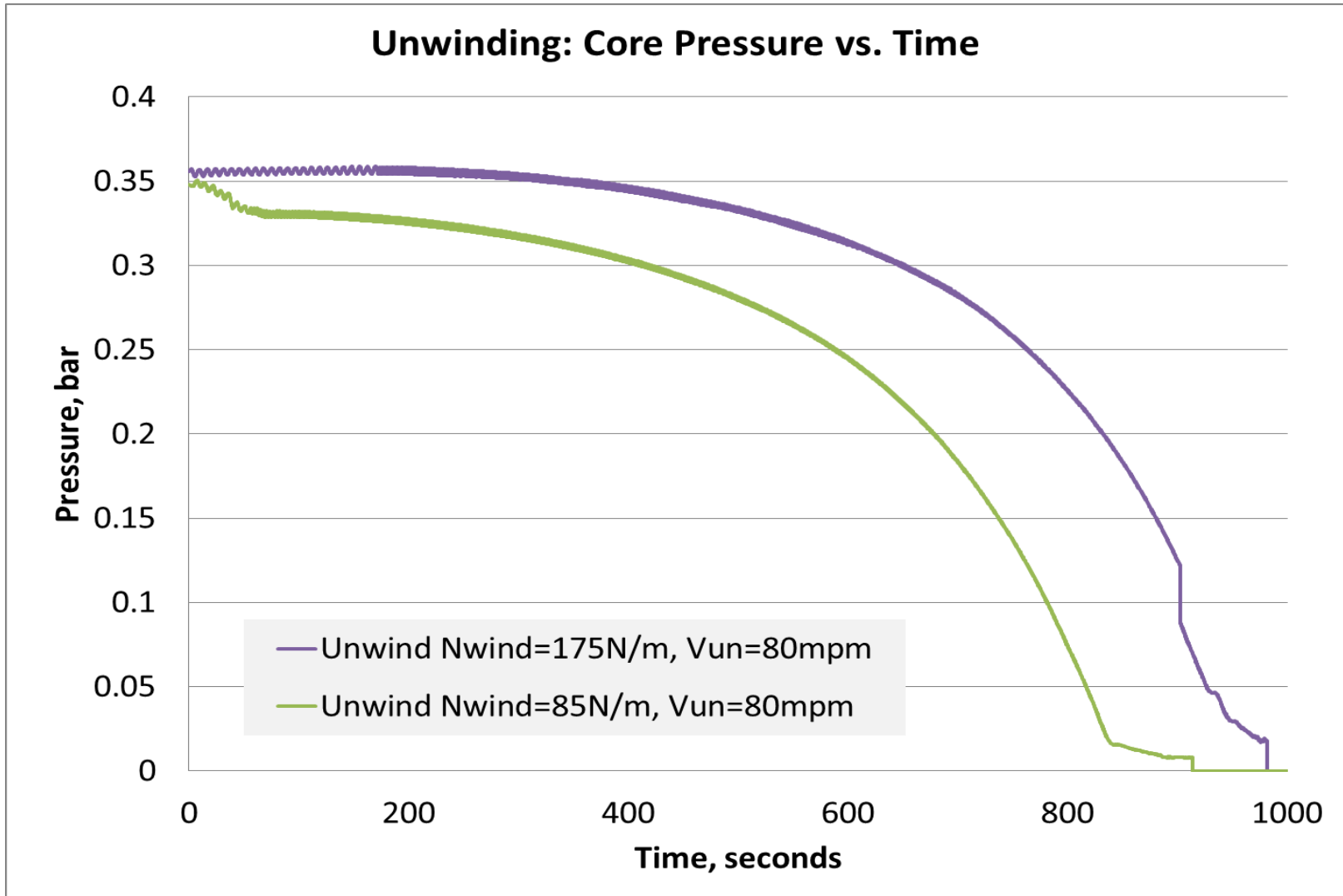


Experimental Setup #2 on Winder



**Wi-Fi system inside core
mounted on winder with end
chucks (no through shaft).**





Web Handling (including Winding) Models

TopWeb 2.0 from Rheologic
£299/annual license

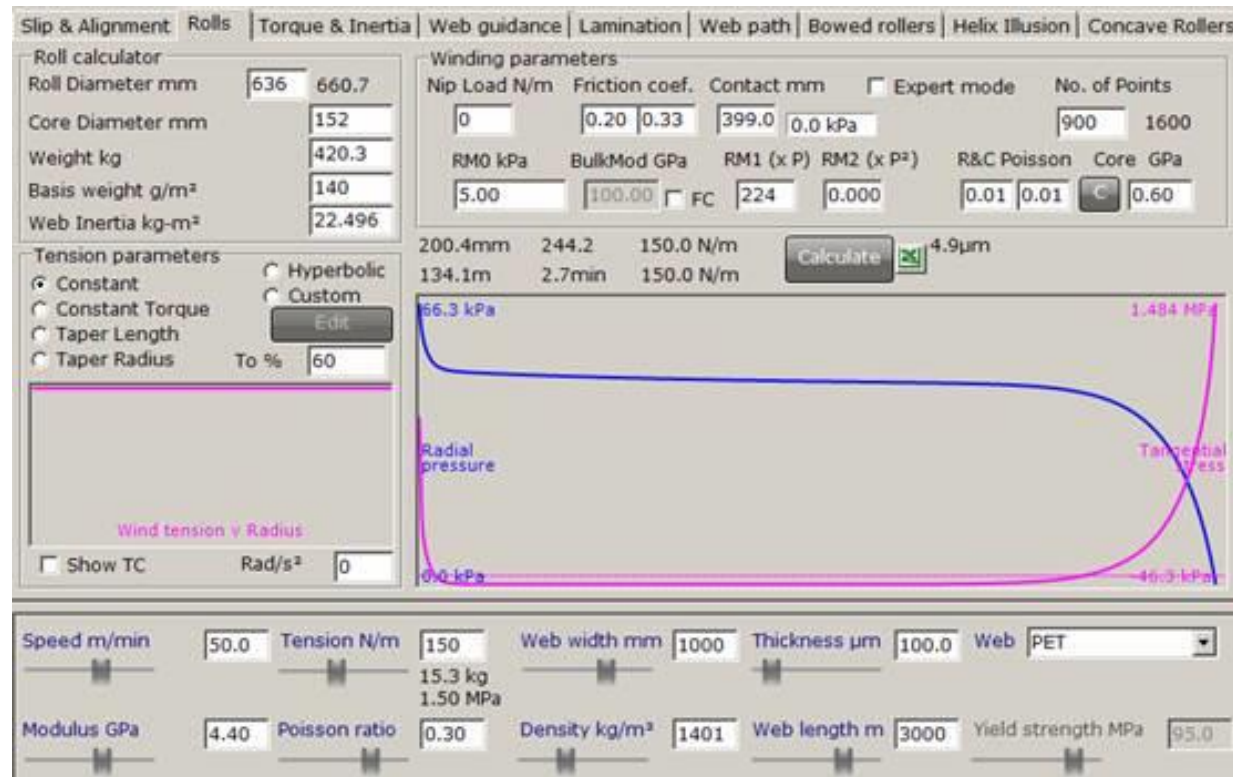


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Or via email:
support@RheoLogic.co.uk
Or through their web site:
www.RheoLogic.co.uk

TopWeb 2.0 software
includes models for:

- Roll stresses
- Wrinkling
- Bearing drag
- Roller inertia
- Laminate curl
- Bowed roller spreading
- Concave roller design



Wound Roll Stress Models

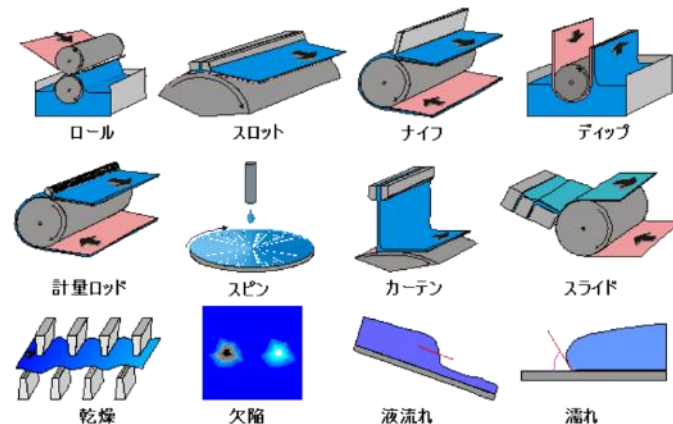
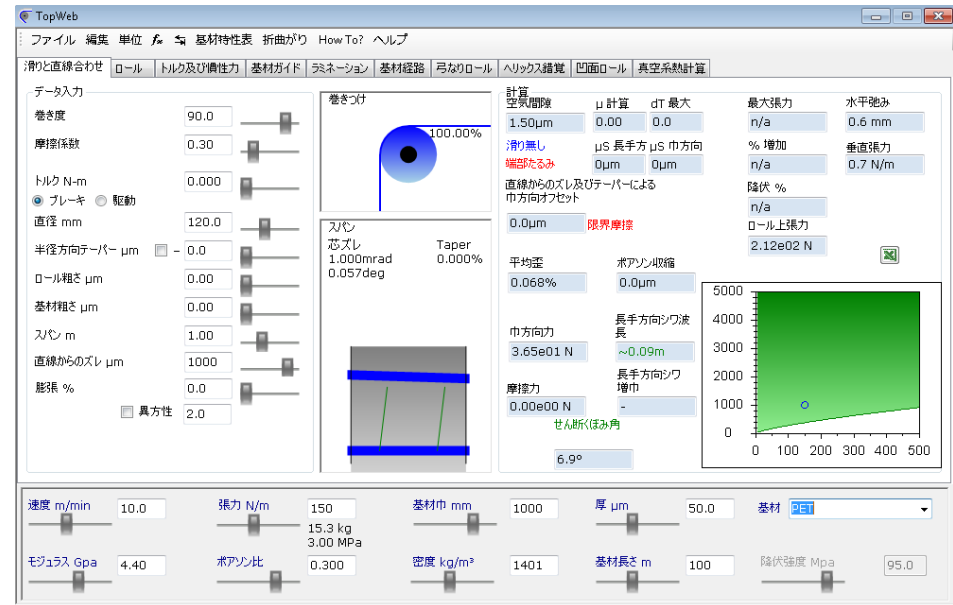
TopWeb J from Rheologic
Everything in TopWeb 2.0
with full Japanese
translation.

£469/annual license

Japanese contact:

SFuchigami@rheologic.jp

Rheologic also has coating
model software with
Japanese translation:
TopCoat J



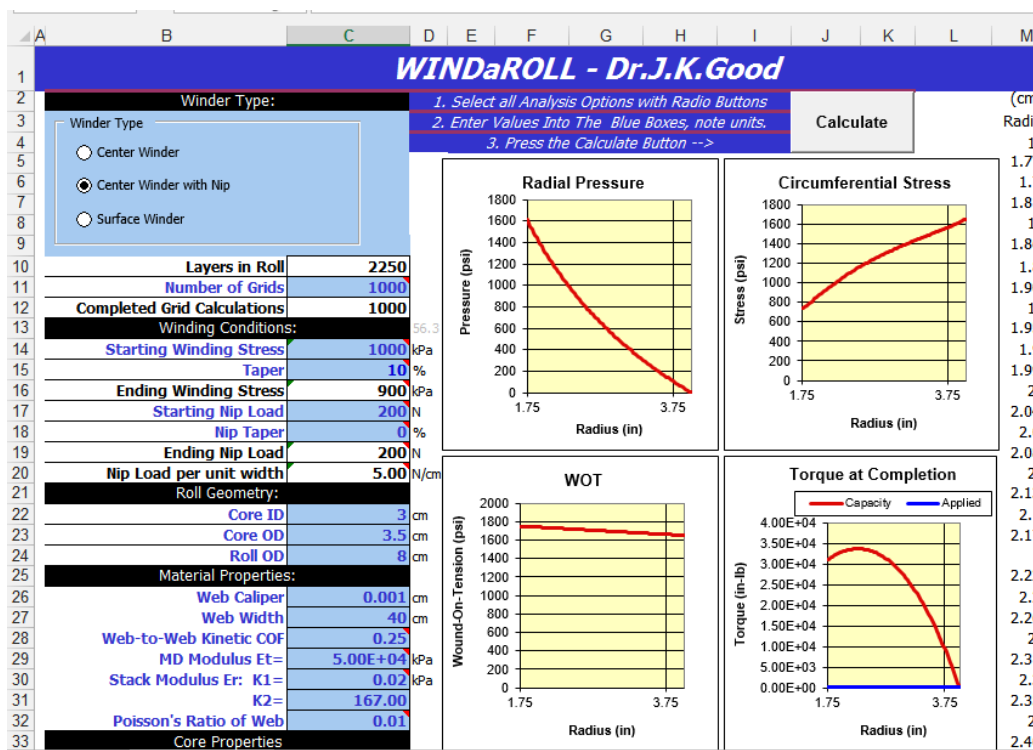
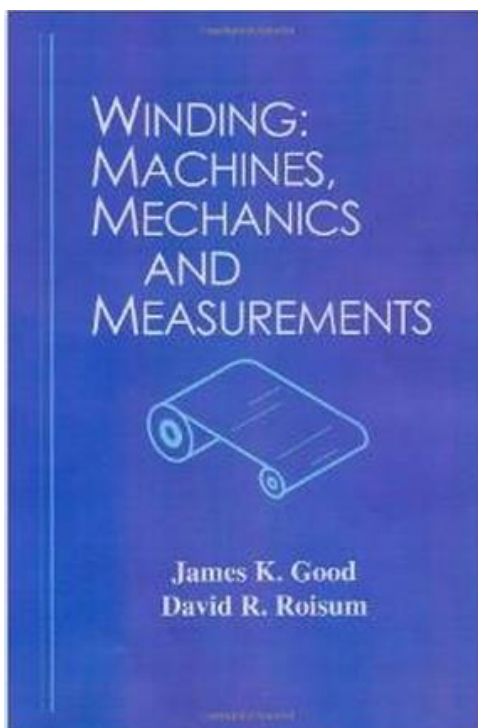
Wound Roll Stress Models

Winding: Machines, Mechanics, and Measurements

James K. Good and David R. Roisum

ISBN-13: 978-1932078695 (Listed at Amazon.com for \$175)

Book in includes CD with WINDaROLL model (an Excel file running Visual Basic macro).



Useful Application of Roll Models

Understanding winding pressure and stress changes as a function of:

- ▶ Material properties (thickness, stack modulus, friction, roughness)
- ▶ Winding conditions (tension, taper, nip load)
- ▶ Roll geometry (width, length or outer diameter)
- ▶ Core properties (core size, core material)

Roll Model Limitations

Models will not predict defects, but help to understand changes to likely reduce defects, including:

- ▶ High pressure defects (blocking, core crushing)
- ▶ Low pressure defects (MD slip, torque transmission capacity)
- ▶ Compression defects (CD buckles, MD buckles)
- ▶ Variable strain defects (bagginess)

Models do not predict:

- ▶ Slip knot, convolution buckles
- ▶ Wrinkling
- ▶ Acceleration and inertia effects

PE Winding Stresses

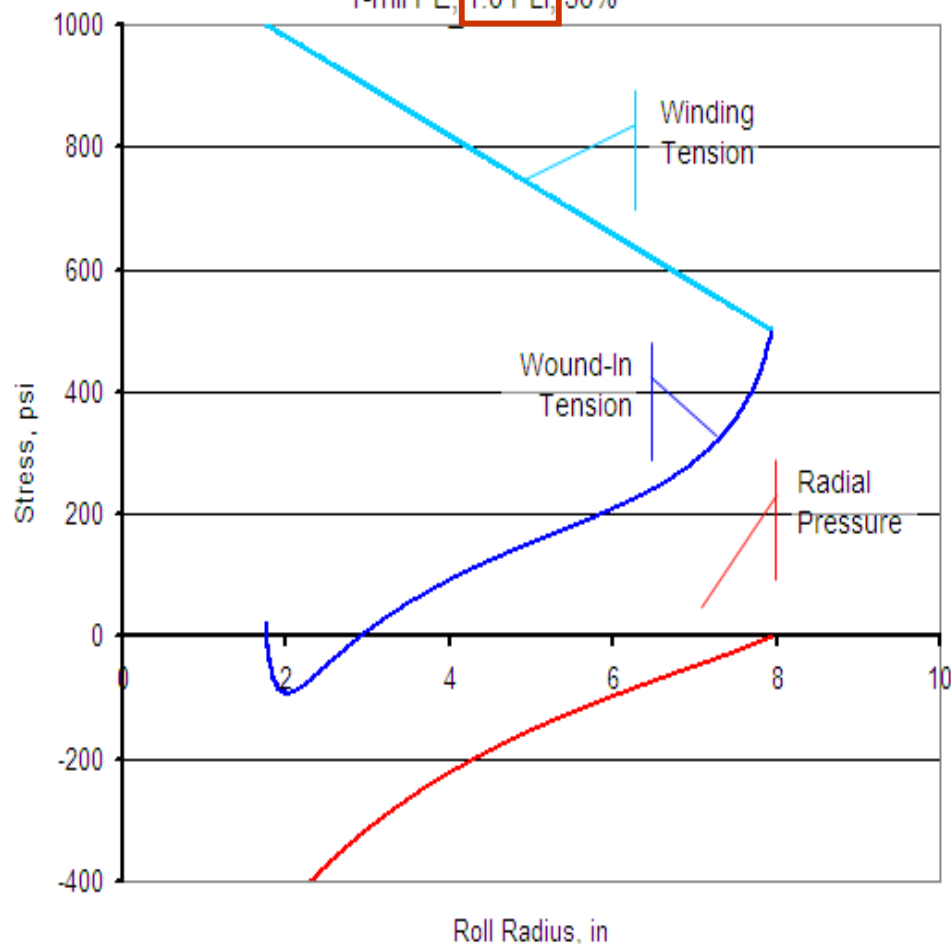
Roll and Winding Stresses

1-mil PE, 0.5 PLI, 50%



Roll and Winding Stresses

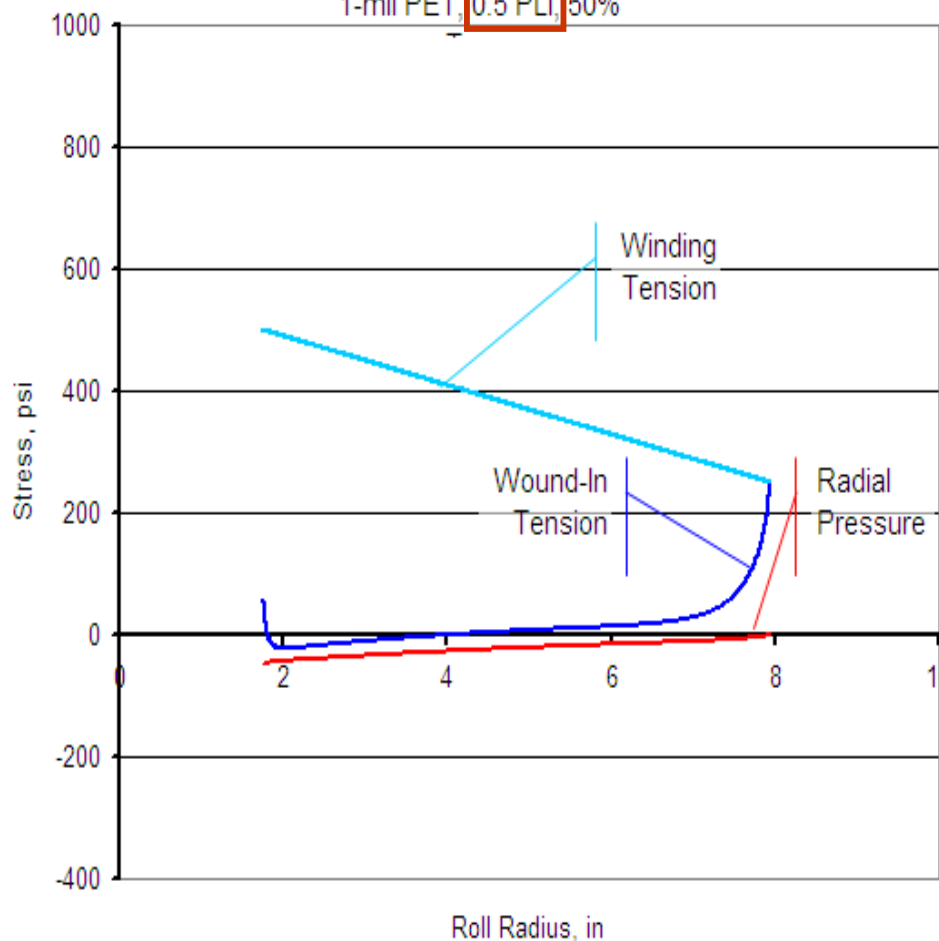
1-mil PE, 1.0 PLI, 50%



PET Winding Stresses

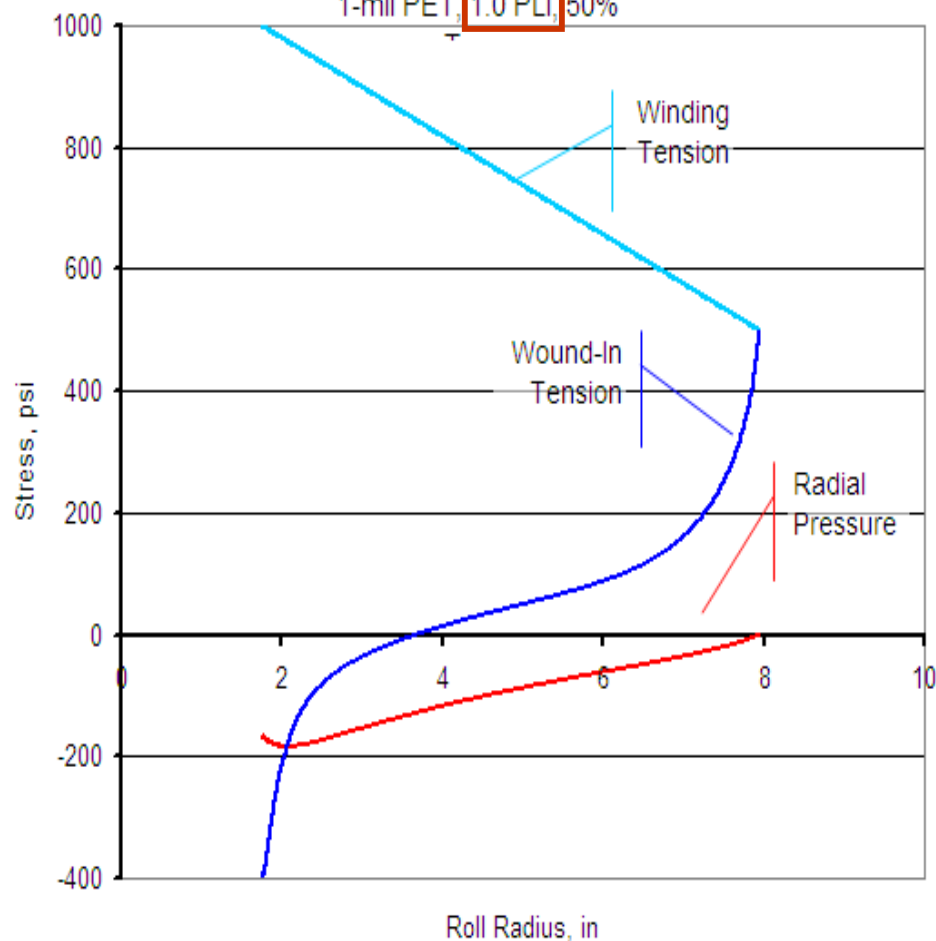
Roll and Winding Stresses

1-mil PET, 0.5 PLI, 50%

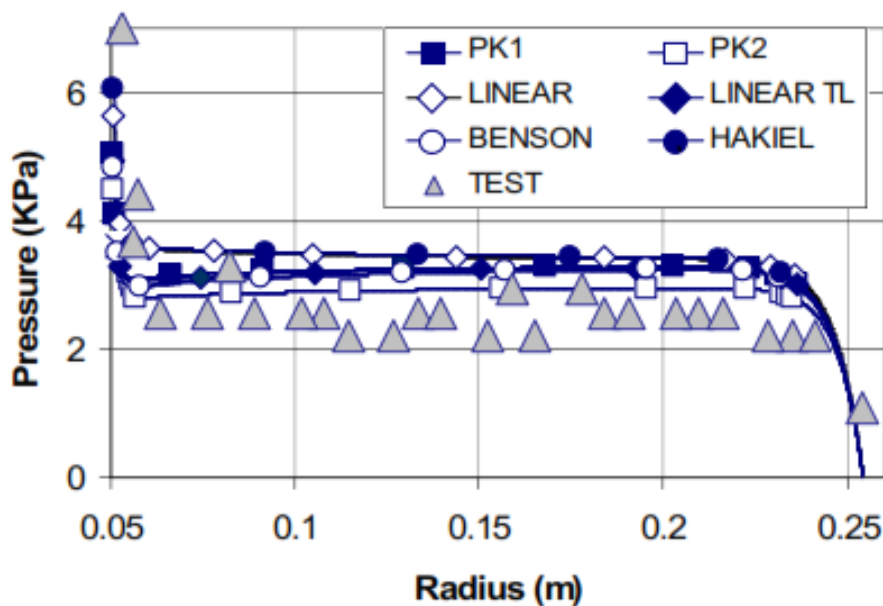


Roll and Winding Stresses

1-mil PET, 1.0 PLI, 50%



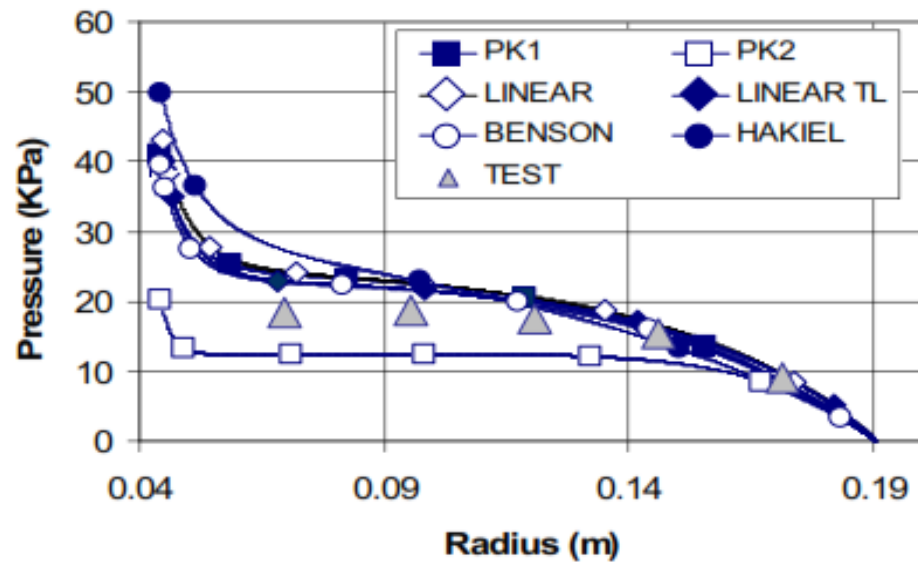
In-Wound Roll Pressure – Spun-Bound Non-Woven



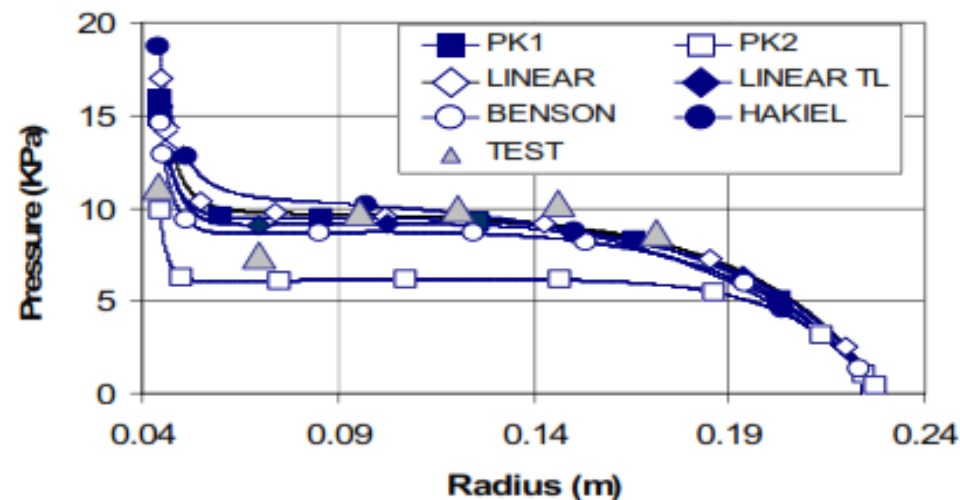
Core inner radius	1.27 cm	J_{CMD}	16550 KPa
Core outer radius	5.08 cm	J_{core}	$6.9 \cdot 10^8$ KPa
Roll final radius	24.1 cm	v_{core}	0.3
K_1	1.32 KPa	α	2.25 KPa
K_2	13.39	β	9.6
J_0	55160 KPa	T_w	115 KPa

Figure 3.4 Radial Pressures for Spun-bond Non-Woven – $T_w = 115$ KPa

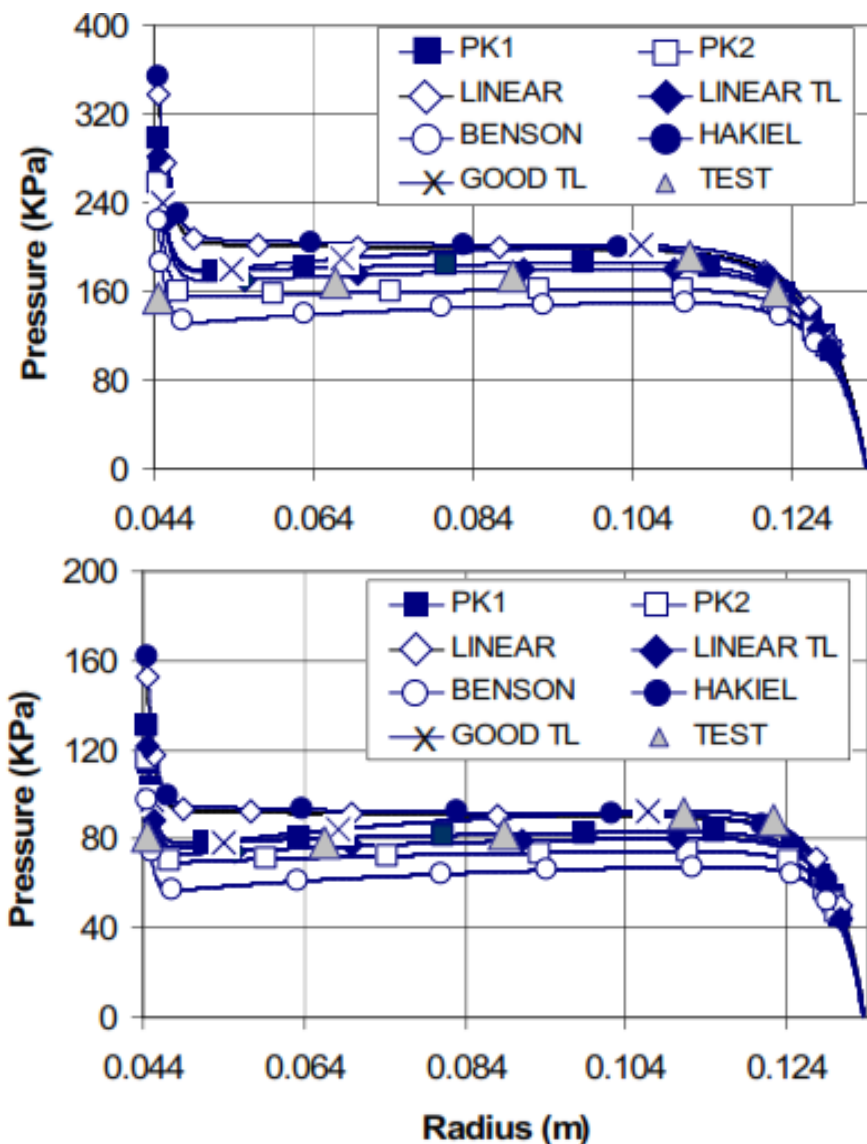
In-Wound Roll Pressure – Bath Tissue



Core inner radius	1.27 cm	J_{CMD}	3337 KPa
Core outer radius	4.45 cm	J_{core}	$6.9 \cdot 10^8$ KPa
Roll final radius	19 and 22.9 cm	v_{core}	0.3
K_1	0.258 KPa	α	0.44
K_2	13.474	β	9.67
J_0	3337 KPa	T_w	59.2 and 92.4 KPa



In-Wound Roll Pressure – Newsprint



A 2D AXIS-SYMMETRIC WOUND ROLL MODEL

INCLUDING NIP EFFECTS

CAGRI MOLLAMAHMUTOGLU

Bachelor of Science in Civil Engineering
Istanbul Technical University
Istanbul, Turkey
2002

Master of Science in Civil Engineering
Istanbul Technical University
Istanbul, Turkey
2005

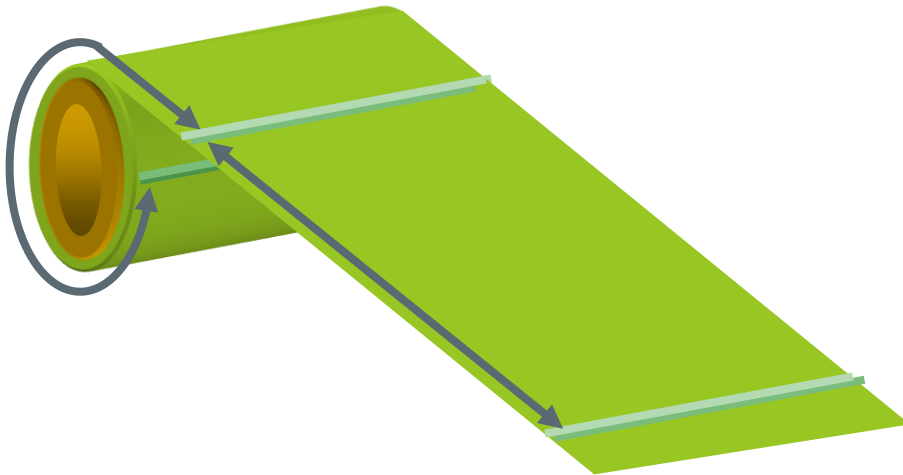
Core inner radius	1 cm	J_{CMD}	3.37 GPa
Core outer radius	4.45 cm	J_{core}	$6.9 \cdot 10^8$ KPa
Roll final radius	13.35 cm	v_{core}	0.3
K_1	1.175 KPa	α	1.54 KPa
K_2	45.14	β	39.72
J_0	3.37 GPa	T_w	3.45, 5.17 MPa

Core Impressions

Web distortions associated with core surface, core start, tape, or web tail.

Mechanism:

- High pressures near the core emboss (imprinting distortion or coining) any stress concentration features through many layers.
- Defect MD period will align to one revolution.



Remedies:

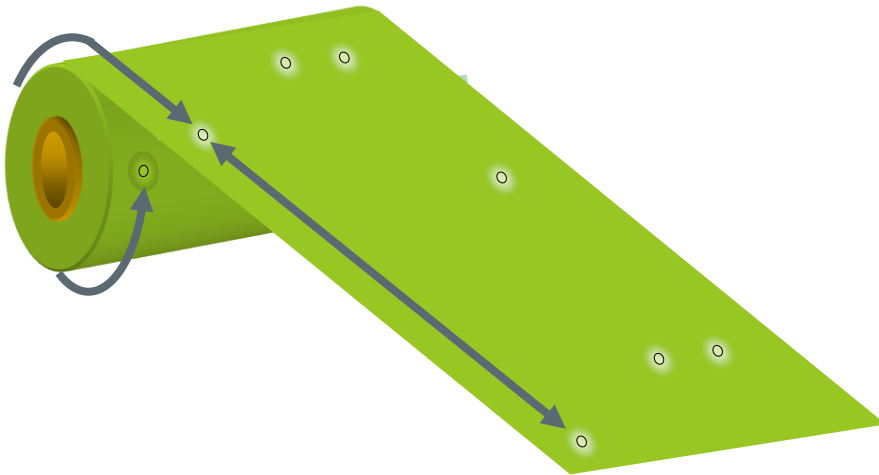
- Reduce tension, increase taper tension
- Reduce core modulus
- Add soft layer to core
- Add sacrificial material near core.
- Change material properties to reduce tourniquet effect
- Wind on larger diameter cores.

Contamination Dimples

Web distortions associated with surface debris, gouge, or asperity.

Mechanism:

- High pressure in the roll embosses any stress concentration feature through many layers.
- Defect MD period will align to one revolution.
- Dimples will be concave below and convex above a particle, usually imprinting more above than below.

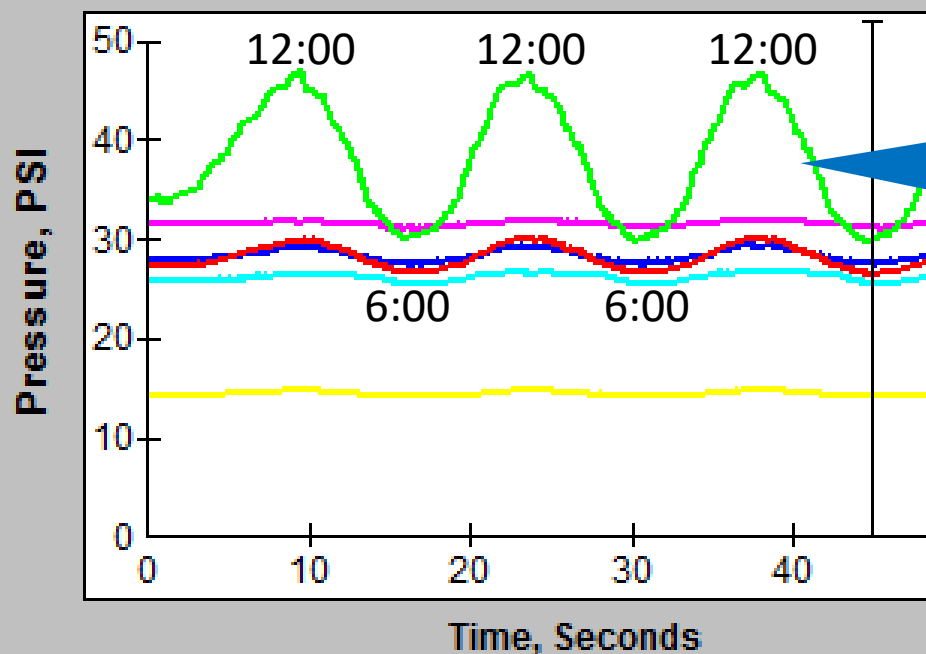


Remedies:

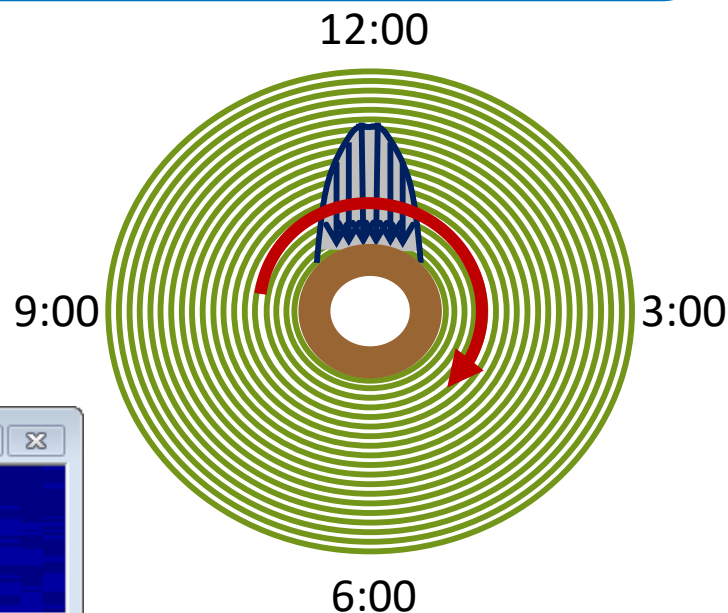
- Prevent debris, gouging, etc. by cleaning the equipment regularly, especially rollers.
- Remove roll wraps from dusty roll storage away from the machine.
- Consider web cleaning.

Core Pressure vs. Roll Rotation (Center Support)

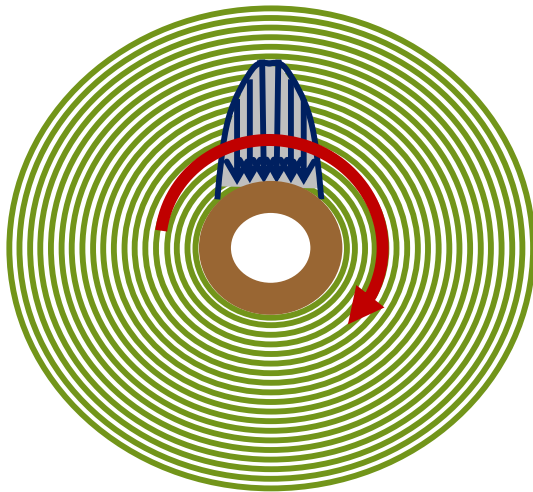
Pressure vs. Time



Core pressure will vary by rotational position, increase at 12:00 and decreasing at 6:00 by the saddle pressure.



Saddle Pressure



$$P_{\text{saddle}} = \text{Roll Web Weight} / \text{Saddle Area}$$

$$A_{\text{saddle}} = \text{Core Diameter } (D_c) \times \text{Web Width } (W_{\text{web}})$$

$$W_{\text{web}} = \text{Density } (\rho) \times \text{Volume}$$

$$V_{\text{web}} = \text{Width} \times \text{Pi} \times (R_{\text{final}}^2 - R_{\text{core}}^2)$$

$$P_{\text{saddle}} = \frac{\rho W_{\text{web}} \pi (R_f^2 - R_c^2)}{D_c W_{\text{web}}}$$

$$P_{\text{saddle}} = \frac{\rho \pi (R_f^2 - R_c^2)}{D_c}$$

This is an average pressure, not considering the parabolic shape of Hertzian stresses

Column Pressure



$$P_{\text{column}} = \text{Roll Web Weight} / \text{Contact Area}$$

$$A_{\text{contact}} = \text{Pi} \times (R_{\text{final}}^2 - R_{\text{core}}^2)$$

$$W_{\text{web}} = \text{Density } (\rho) \times \text{Volume}$$

$$V_{\text{web}} = \text{Width} \times \text{Pi} \times (R_{\text{final}}^2 - R_{\text{core}}^2)$$

$$P_{\text{column}} = \frac{\rho W_{\text{web}} A_{\text{contact}}}{A_{\text{contact}}}$$

$$P_{\text{column}} = \rho W_{\text{web}}$$

6.4 Roll Structure and Web Quality

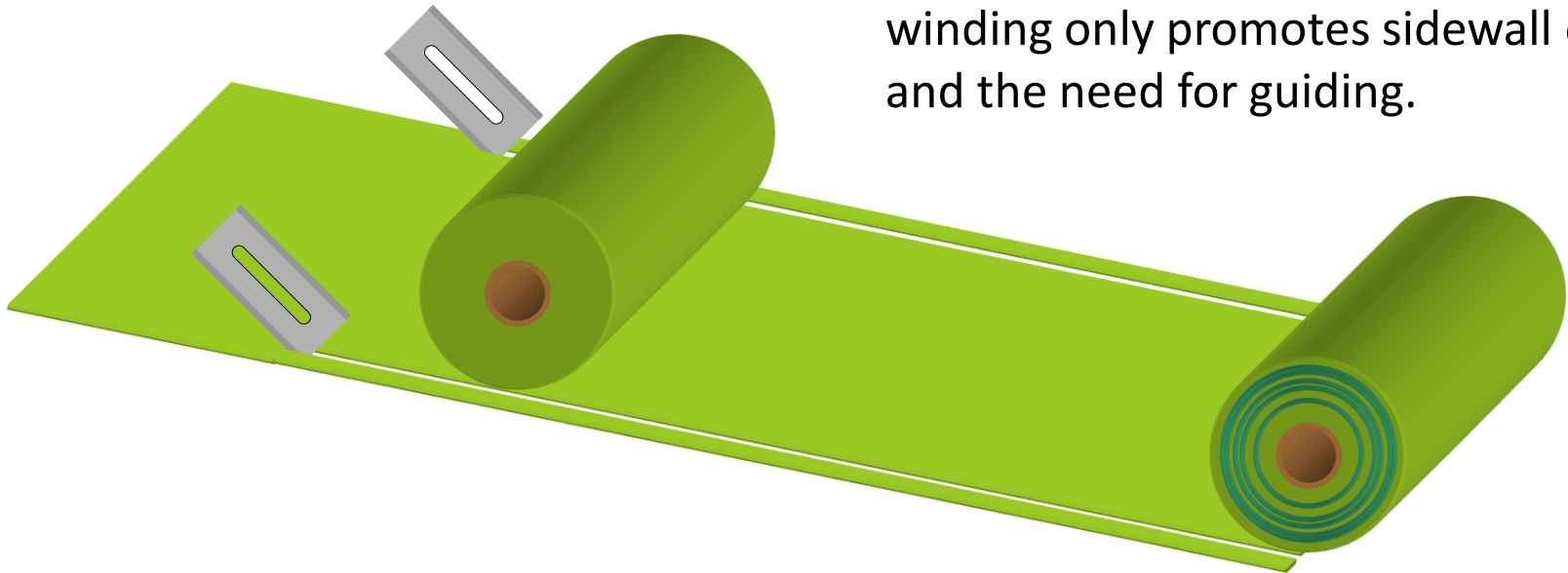
- Wound-In Stresses
- Lateral Errors 
- In-Roll Buckling
- Transverse Variations
(Gauge Bands and Bagginess)

Best Guiding: Slitting

The most accurate web positioning method is slitting.

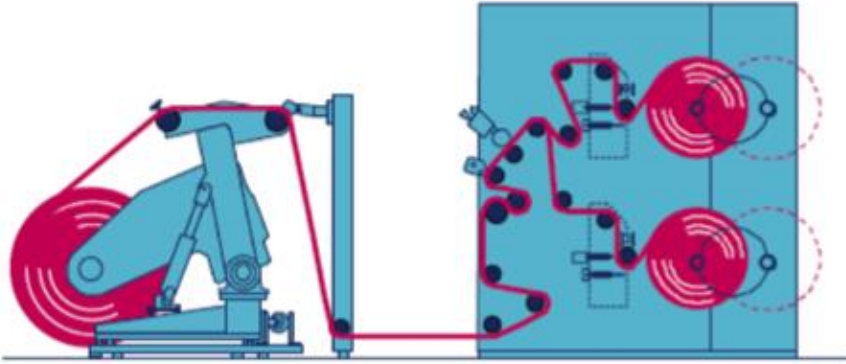
Immediately after slitting, the exact web position is known better than any automatic guide can deliver.

Any web length between slitting and winding only promotes sidewall error and the need for guiding.

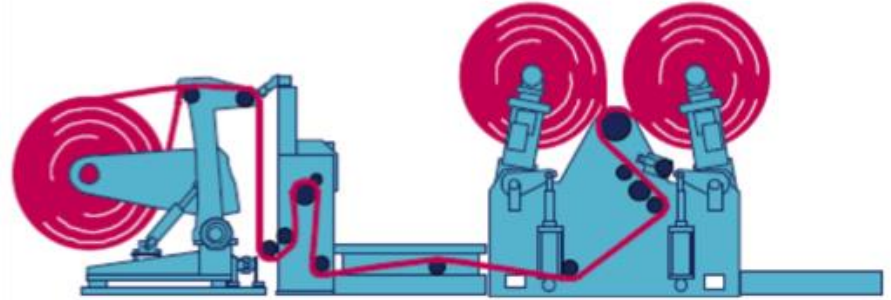


Slit-to-Wind Web Path

Long slit-to-wind web path



Short slit-to-wind web path

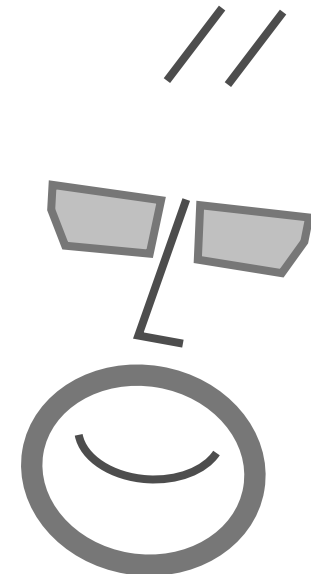


Images courtesy of ilamco - www.ilamco.com

Winding Solutions Qs

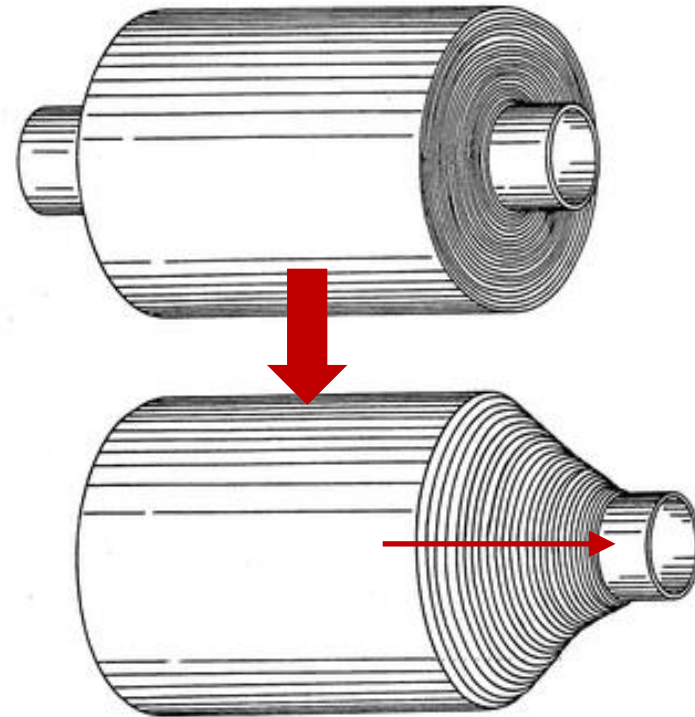
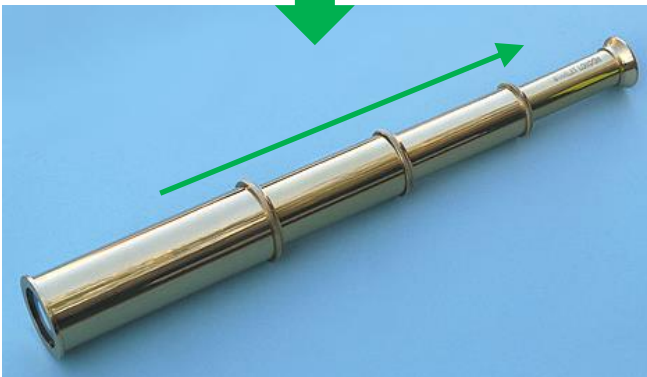
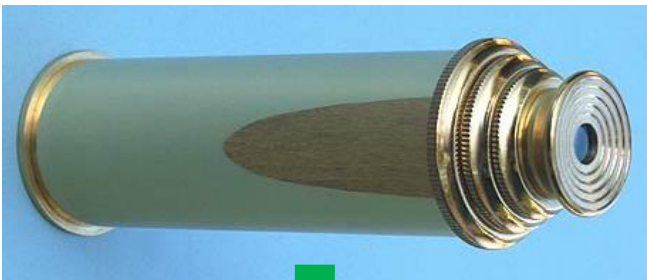
What causes shifted layers at winding?

- **Telescoping**
- **Web Tracking Problems**
- **Core Position or Width Problems**
- **Web Width Problems**



What is Telescoping?

Telescoping is the lateral sliding of layers within a wound roll. The layers begin in the correct lateral position, but then shift within the roll during winding, handling, storage, or unwinding.



Web Shifting Lateral Roll Errors

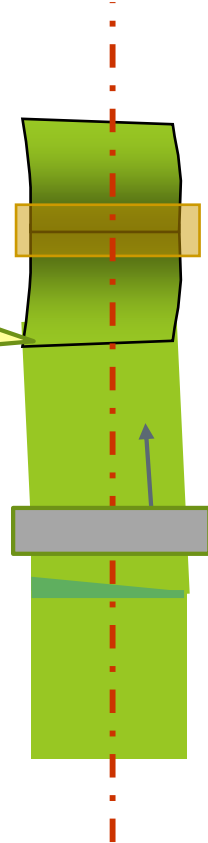
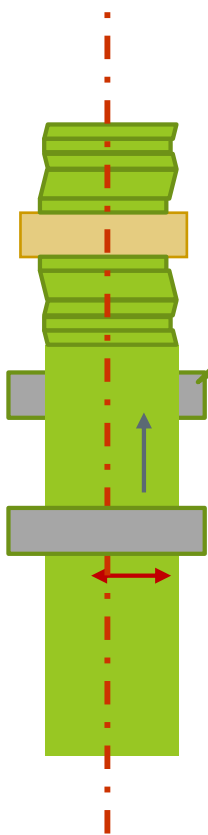
Pre-Wind
Web
Shift

Layers inside the roll do not telescope.

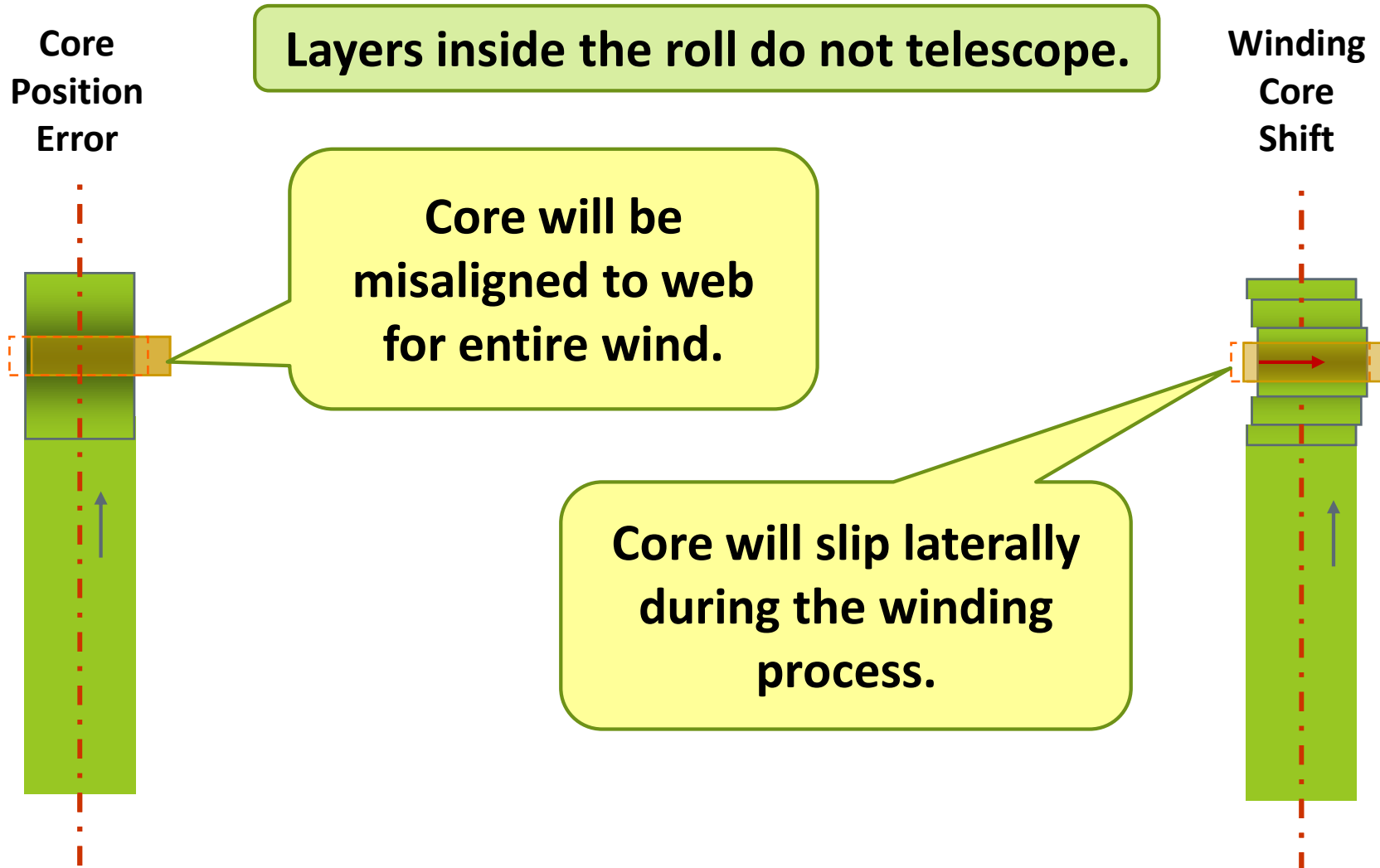
**Web will shift on
upstream rollers.**

**Web will shift between
last roller and contact
winding roll.**

Winding
Entry
Shift

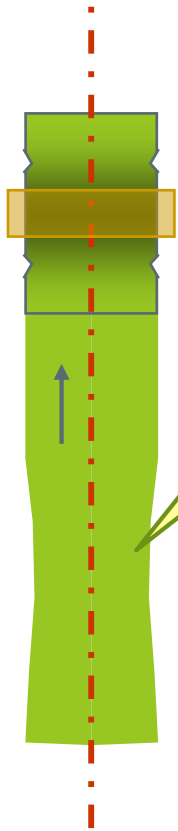


Lateral Errors



Lateral Errors

Pre-Wind
Width
Change

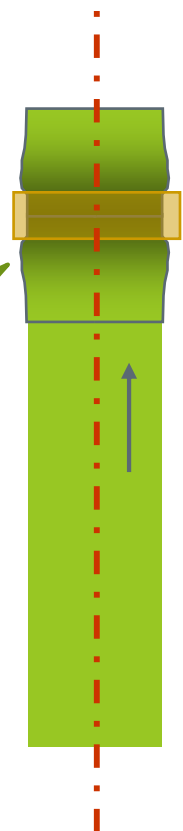


Layers inside the roll do not telescope.

**Web width will vary
as the web contacts
the winding roll.**

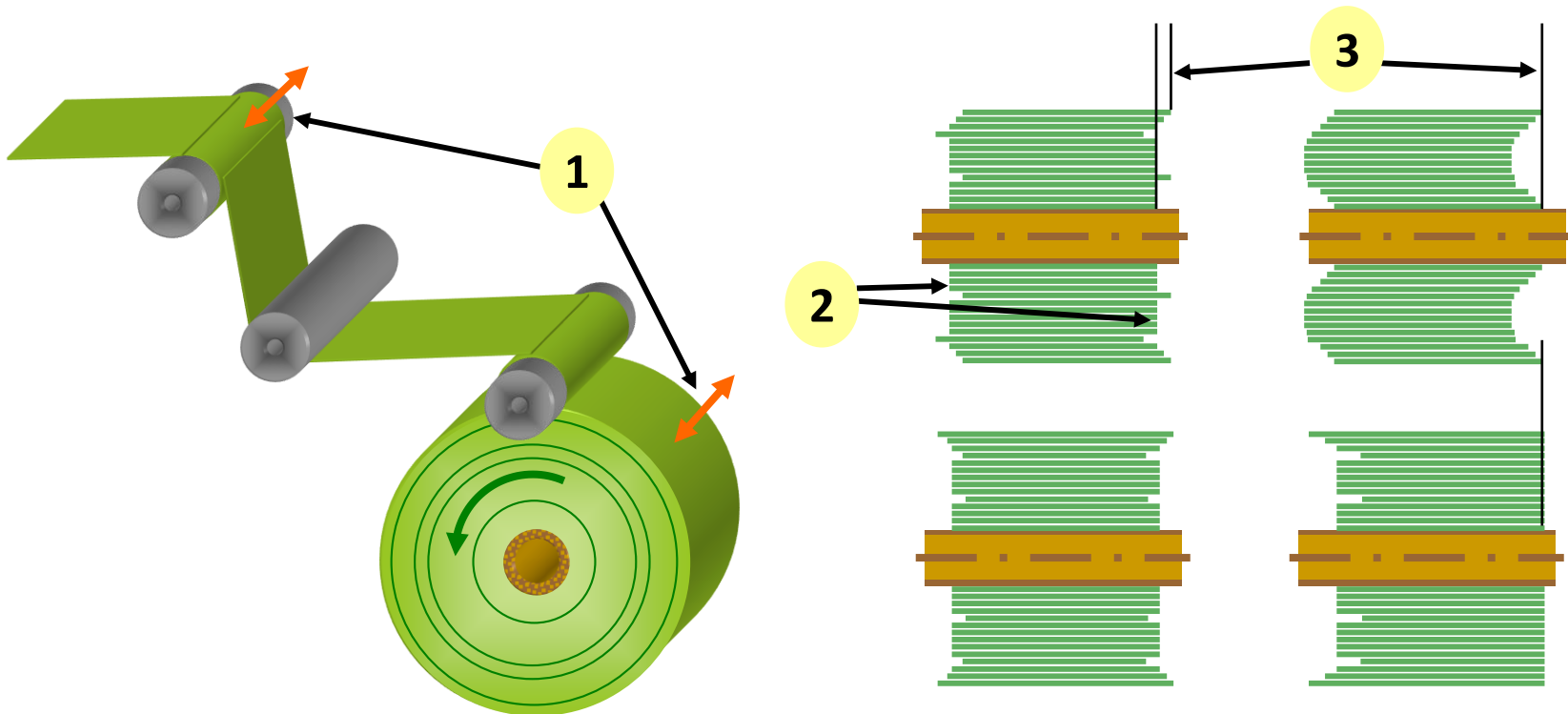
**Web width is constant
upstream and in the
roll's outer layers, but
expands due to high
pressure near the core.**

In-Wind
Width
Change



Diagnosing Shifted Layers

1. Compare lateral shifting of incoming web to wound roll shifting.
2. Compare shifting on both sides of the roll to check for width variations.
3. Note entering web position relative to core position.
4. Mark the core position and ensure it doesn't shift laterally.

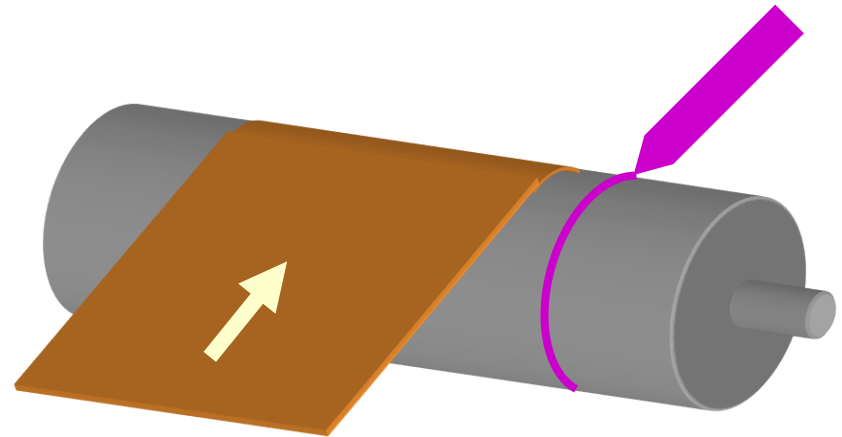


Lateral Shifting Before Winding?

How can you detect if shifting occurs before winding or within the winding roll?

Draw a line at the web's edge position on one of the last rollers prior to entering the winding roll.

If the web motion relative to the web mark is small compared to the winding roll's sideways error, then the shifting is occurring after the web enters the winding roll.



Entry Errors

Pre-Wind Lateral Shifting

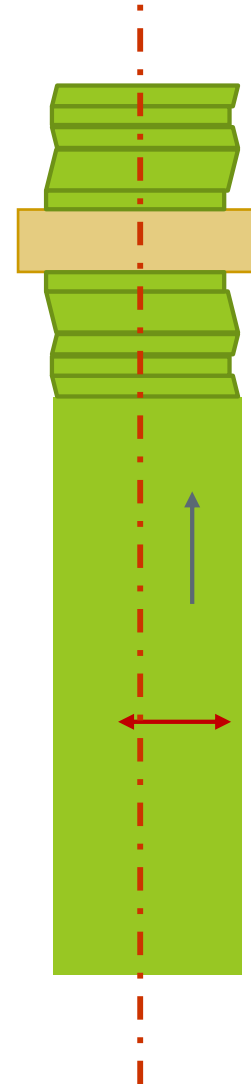
Web enters roll off core centerline

Errors For All Pre-Wind Tracking

- Tracking from diameter variation of roller or winding roll/core (includes “dishing”)
- Tracking from misalignment roller or winding roll/core
- Wander from web skew
- Web-roller attraction / uneven “peeling”

Errors Related to Post-Slit Tracking

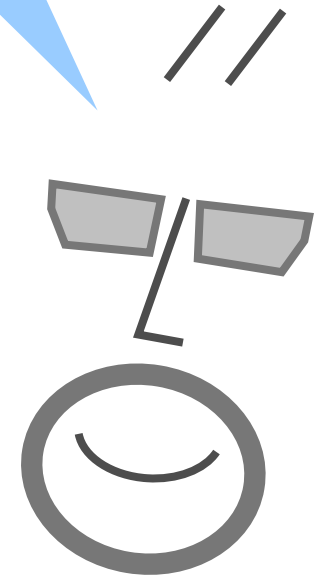
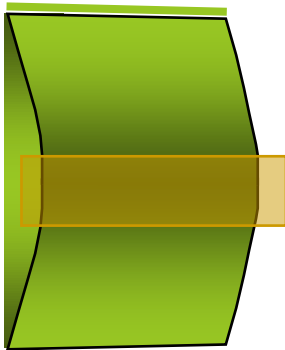
- Contact with neighboring strands
- Slackness from web bag
- Slackness from winding accumulation variation



Winding Solutions Qs

What is dishing? *What causes dishing?*

In dished rolls, the entering web has a gradually increasing CD misalignment to the core either to web entry error or core shifting.

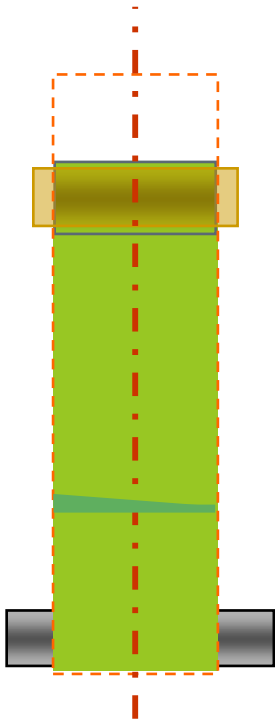


Dishing in Free-Span Center Winding

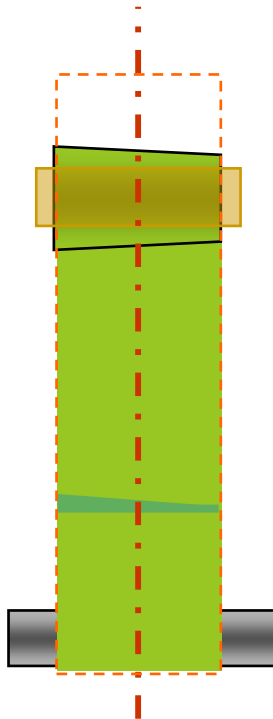
Dishing

Roll layers enter off core centerline.

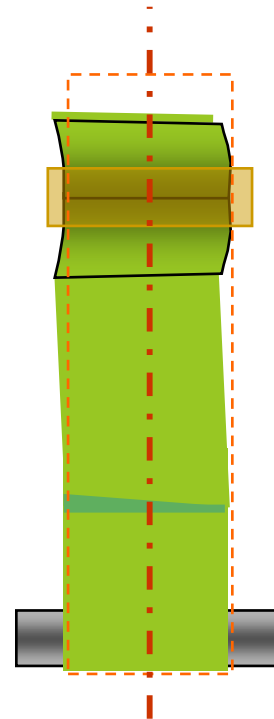
No error at
core start.



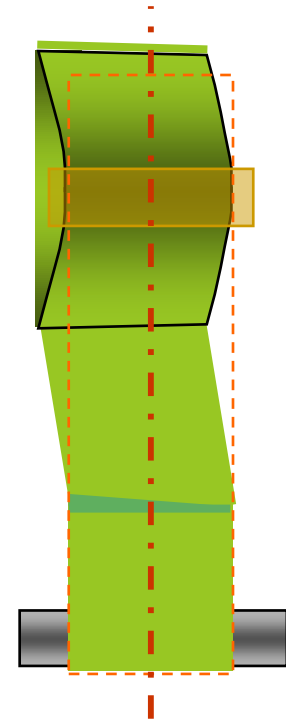
Roll begins to
cone from
thickness var.



Web tracks on
roll diam.
variation.



Offset worsens as
roll and variations
grow

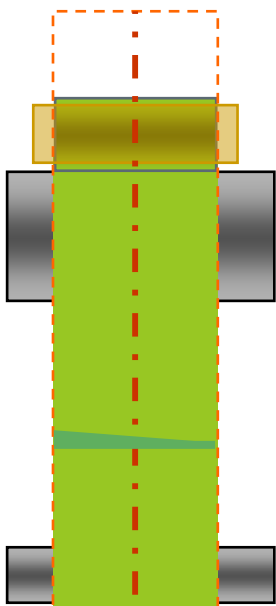


Dishing from Core Shifting

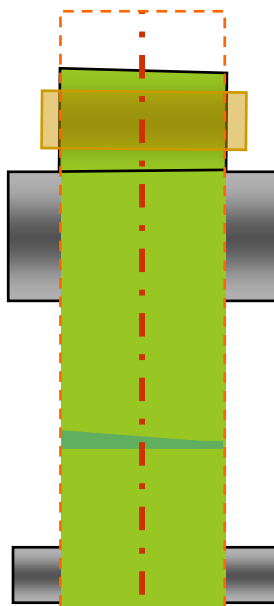
Dishing

Surface winding core tracks off angle.

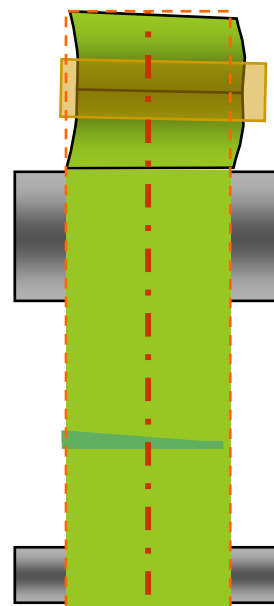
No error at
core start.



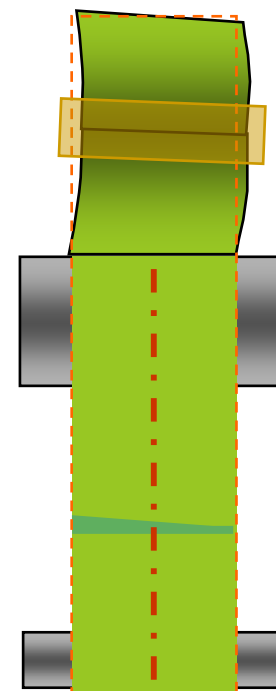
Roll begins to
cone from
thickness
variation



Core tracks on
roll diam.
variation.

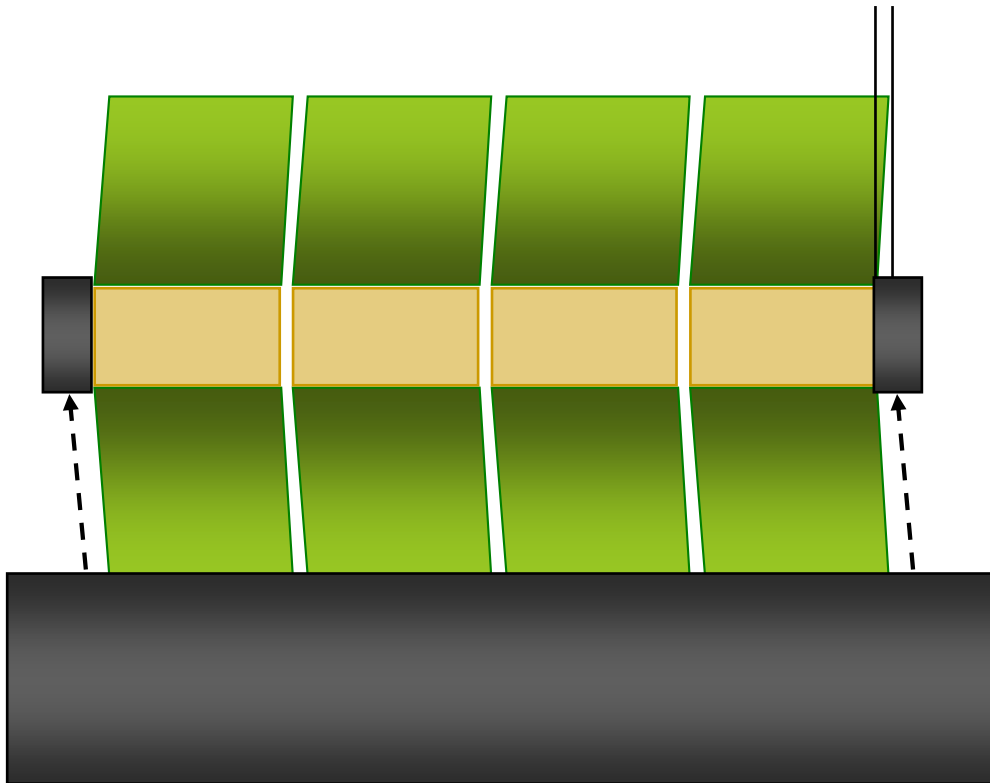


Offset worsens as
roll and variations
grow



Edge Dishing - Coordinated

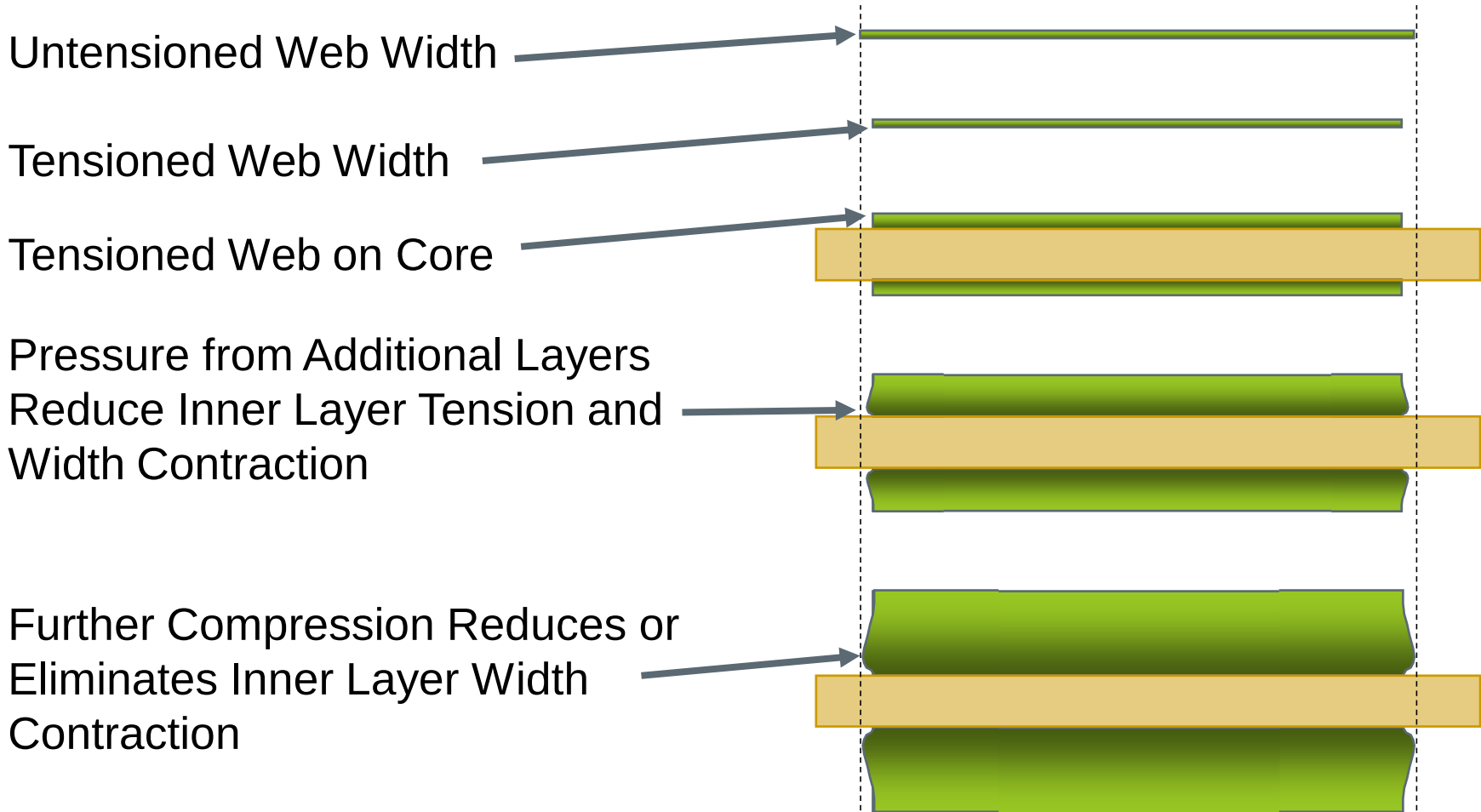
In a set of rolls, the outer layers of each roll have a unidirectional tapered offset relative to the core position.



Mostly likely caused by central drum winder with misaligned arms shifting laterally as the roll grows.

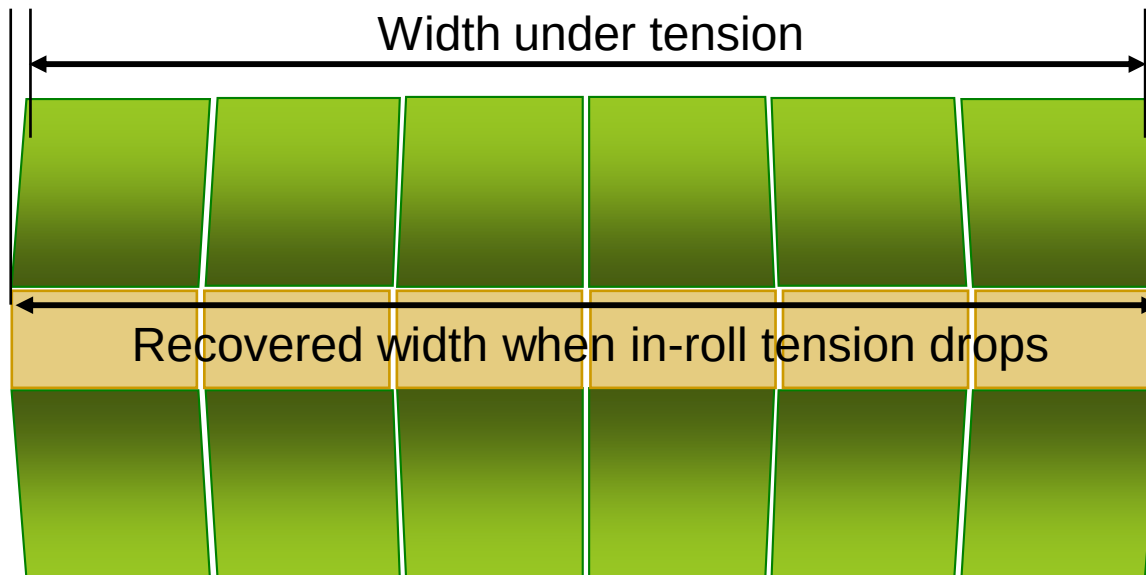
Width Contraction and Recovery

Width Expansion / Contraction Recovery Lateral Error



Edge Dishing - Progressive

Progressive dishing is unique to two-drum surface winding where cores are free to move laterally.

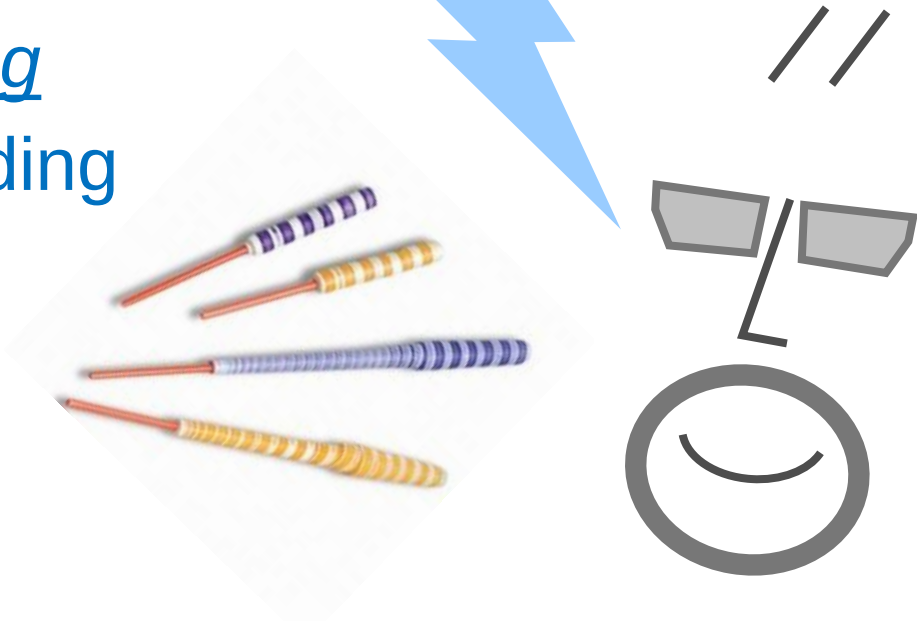


As width returns near core, the extra width pushes neighboring roll outward laterally.

Winding Solutions Qs

What are the four causes / types of telescoping?

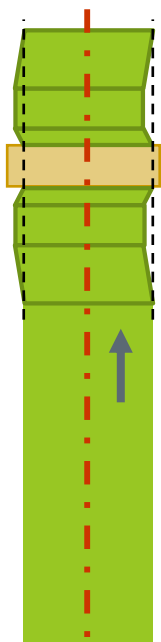
- 1A. Cinching at Winding
- 1B. Cinching at Unwinding
- 2. PSA Creep
- 3. Roll Handling
- 4. Air Lubrication



*I'm using the number codes of David Roisum and TAPPI Roll/Web Defect book.

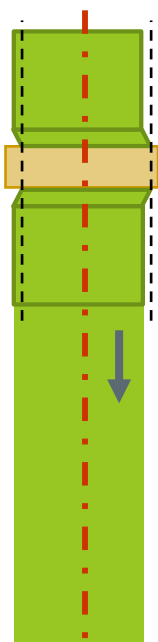
Telescoping Lateral Errors (1-4)

**Winding
Cinch
Telescoping**



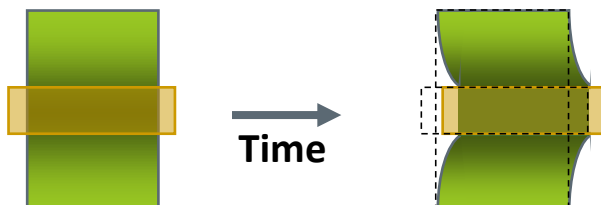
1A

**Unwinding
Cinch
Telescoping**

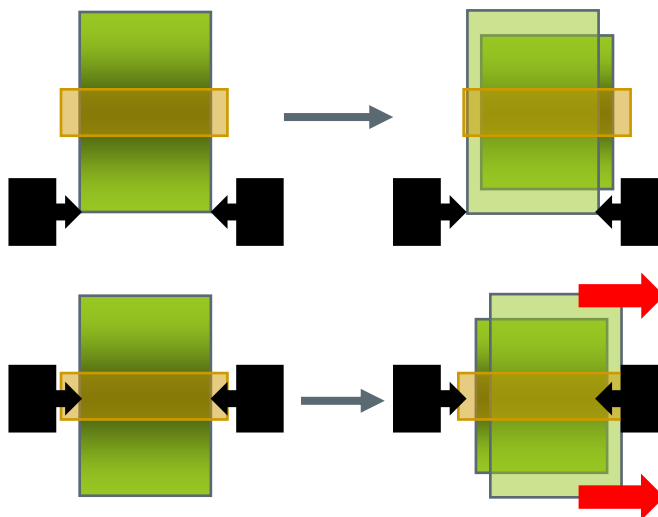


1B

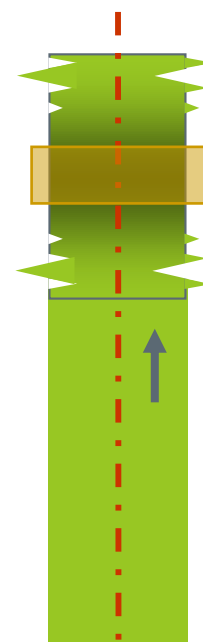
Roll Creep Telescoping, 2



**Roll Handling Telescoping, 3
(Core Shift vs. Core Fixed)**



**Winding Air
Lubrication
Telescoping**

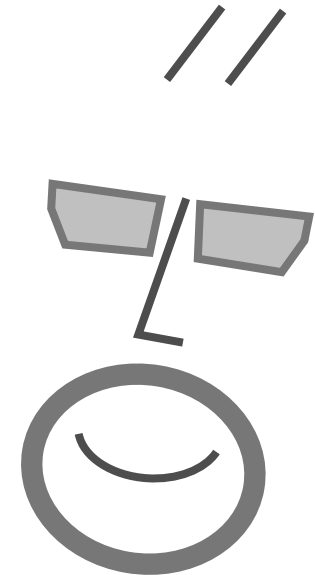
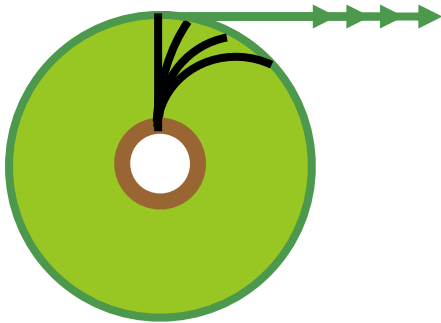


4

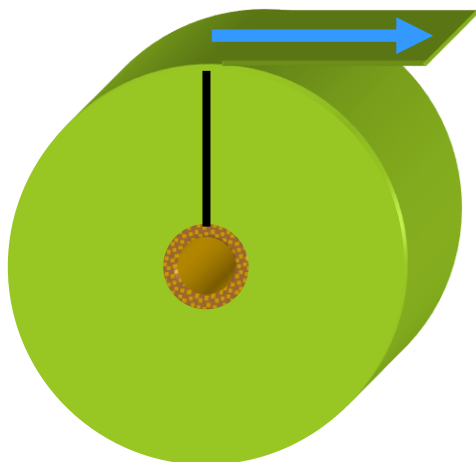
Winding Solutions Qs

What is cinching?

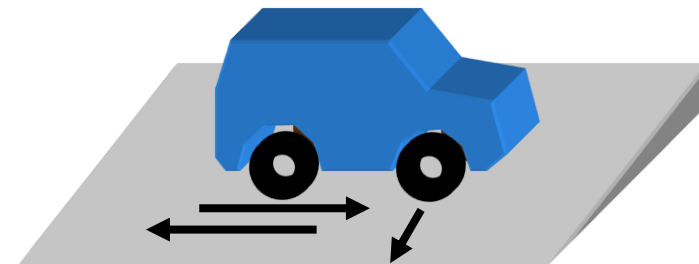
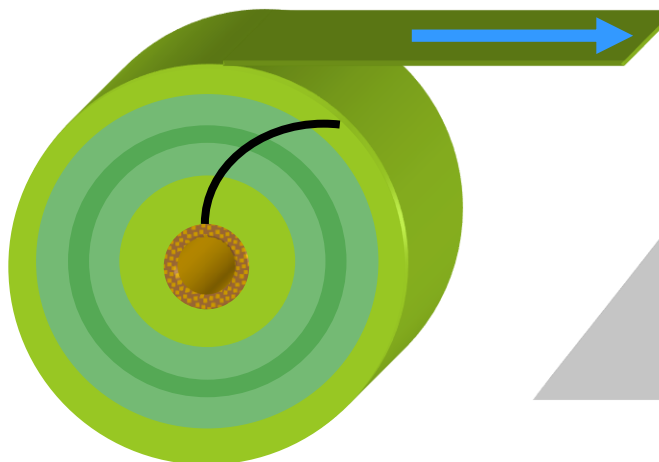
What is cinching-induced telescoping



Radial+Lateral Error: Clockspringing



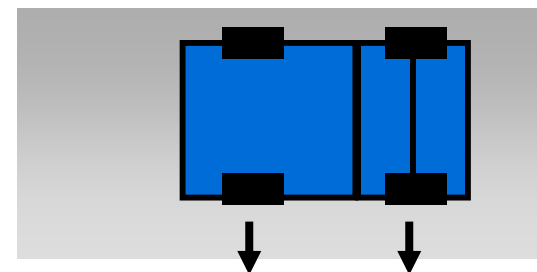
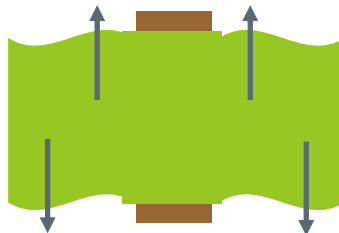
Layers in the wound roll slip radially when cinching (MD slip) occurs.



Similar to how a car with spinning tires...



When all available friction is used radially there is no friction to maintain lateral stability.

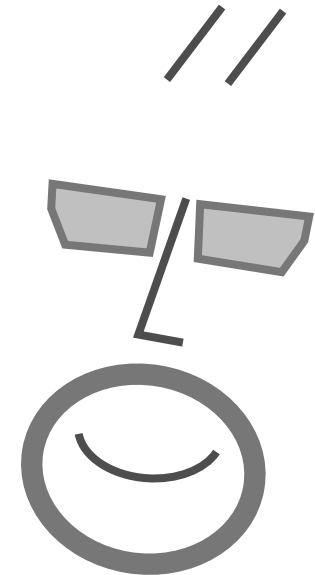


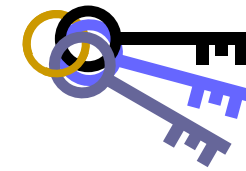
...won't stay up on a side-sloped road.

Winding Solutions Qs

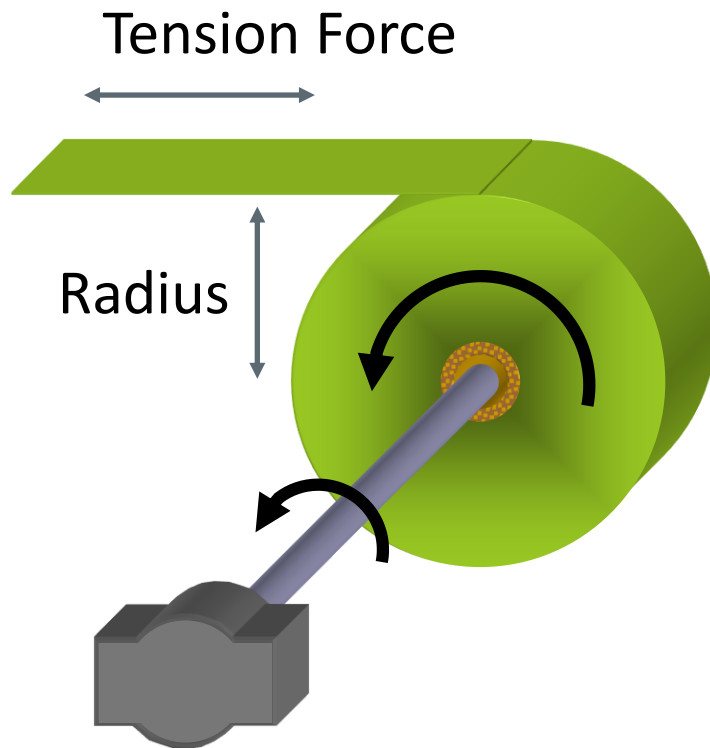
What is a roll's torque transmission capacity?

How is torque capacity improved?





Torque Capacity

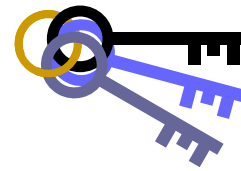


$$\text{Torque} = \text{Force} \times \text{Radius}$$

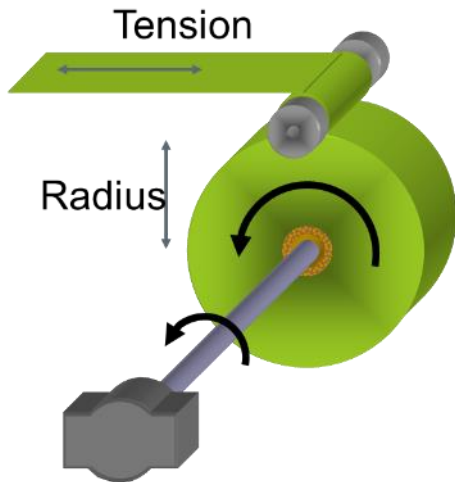
In center winding and unwinding, torque (M) is applied at the roll's center and transmitted through all the layers of the roll to create the tension at the outer radius.

Each layer must have the friction capacity to transmit the torque.

Applied torque often increases with radius, requiring more torque transmission capacity for larger diameter rolls.



Torque Capacity



A layer's torque transmission capacity is friction times radius.

Friction comes from internal roll pressures x a layer's area of contact.

$$M_{CAP,X} = F_X r_X$$

$$= \mu P_X A_X r_X = 2\pi \mu P_X w r_X^2$$

$M_{CAP,X}$ = Torque capacity at radius X, in-lbs or N-m

μ = friction coefficient side A-B

P_X = internal pressure at radius x, psi or Pa

A_X = area at radius x, in² or m²

F_X = friction available at radius x, lbs or N

r_X = any given radius x, in. or m

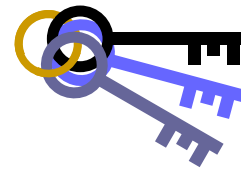
r_{OD} = outer radius of the winding roll, in. or m

w = roll width, in. or m

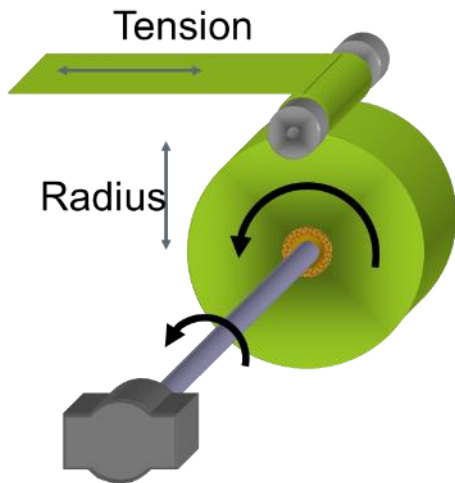
T = tension, lbf/in or N/m

M_{CAP} is proportional to radius squared.

Small radius, near-core layers will have the lowest M_{CAP} .



Torque Capacity



To NOT cinch or MD slip, all layer in a roll must have sufficient M_{CAP} .

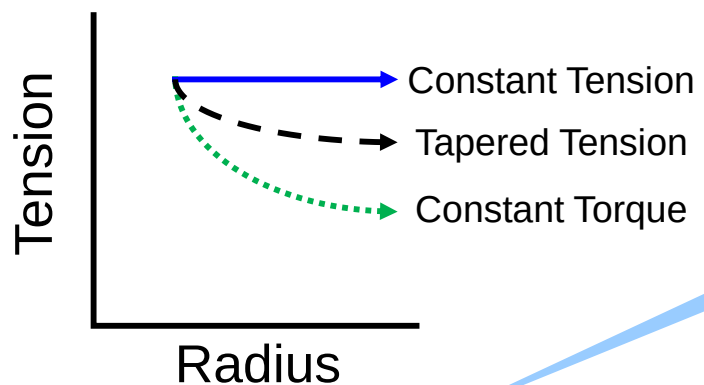
Frictional Torque > Applied Torque

$$\mu P_X A_X r_x > T w r_{OD}$$

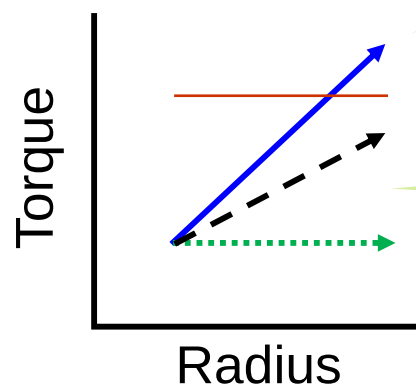
$$P_X > \frac{T r_{OD}}{2 \pi \mu r_x^2} \leftarrow$$

Each layer has a minimum pressure required to prevent cinching and telescoping. The layers of smaller radius (near core) need the highest pressure to avoid slippage.

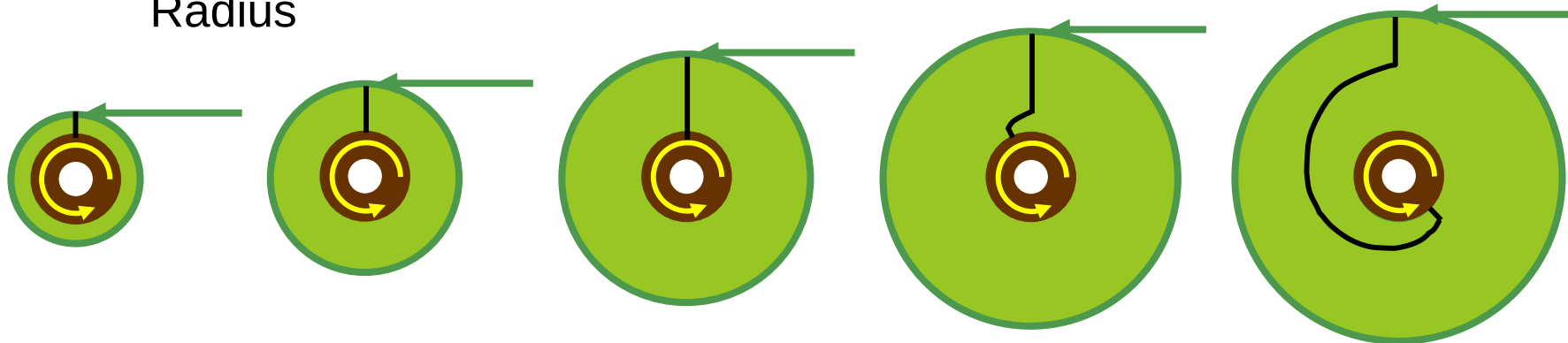
Cinching vs. Roll Radius



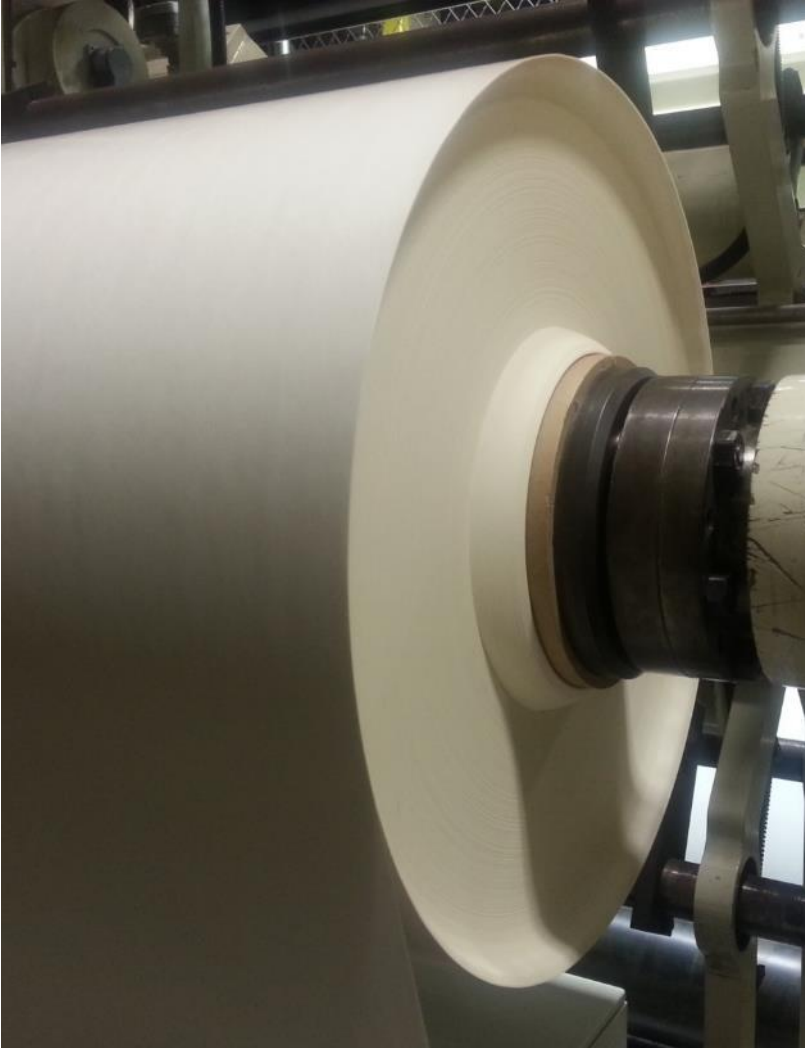
Constant tension requires increasing torque, increasing the likelihood of cinching.



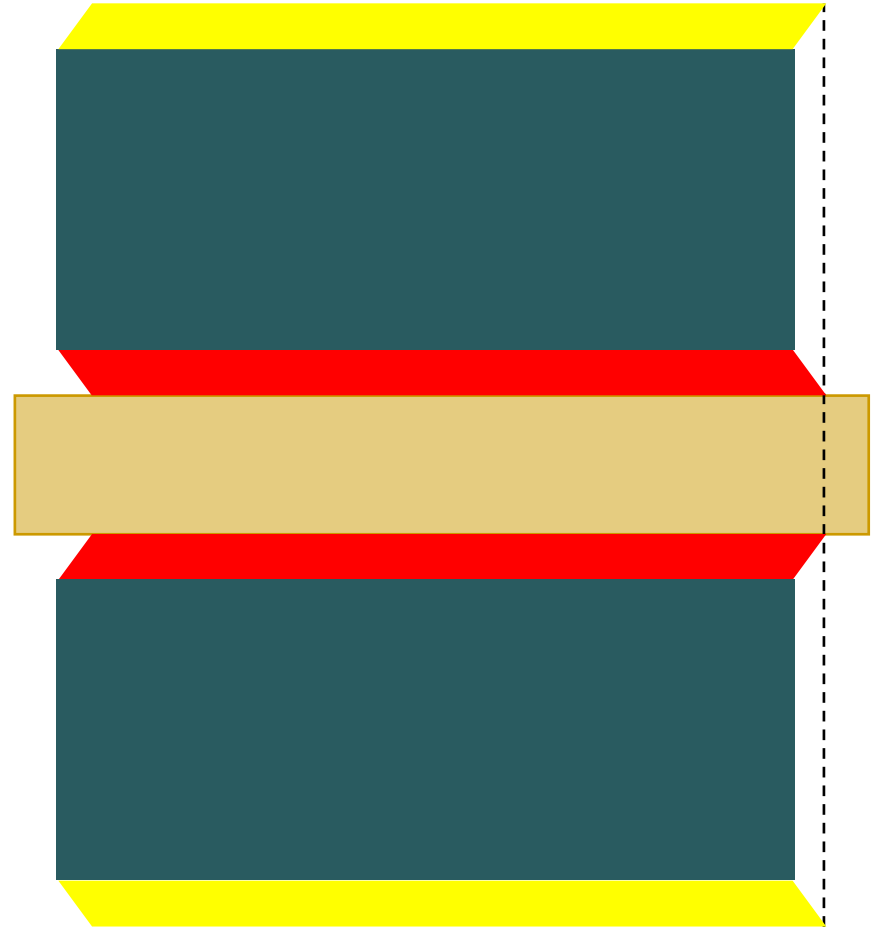
Tapered tension requires less torque increase, which may avoid exceeding a roll's torque capacity and cinching.



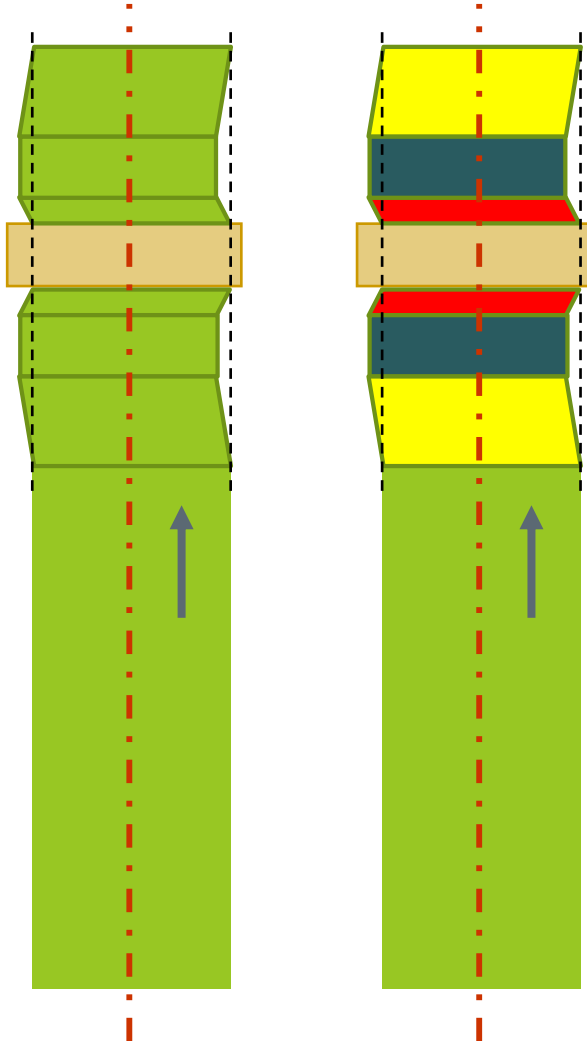
Center Torque Cinching-Induced Telescoping



Winding



Center Torque Cinching-Induced Telescoping

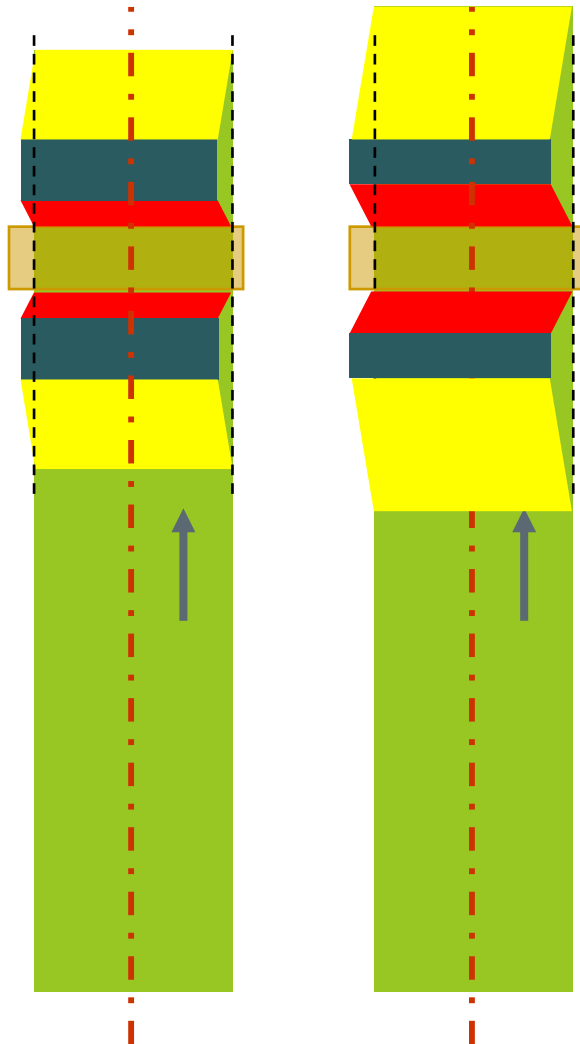


All layer-to-layer friction is used by MD slip, leaving no friction to oppose to CD shifting.

The outer layers wound before the near core slippage and outside the cinch zone, shift laterally as a 'block.'

New layers continue to join the winding roll, forming a tapered profile as the cinch zone and 'block' layers continue their lateral shift. These layers are wound with a tapered edge profile and do not slip laterally.

Center Torque Cinching-Induced Telescoping



As the roll continues to build, the slip zone may grow over time as near core tightening layers move to smaller radial position, dropping pressure in layers immediately outside the slip zone.

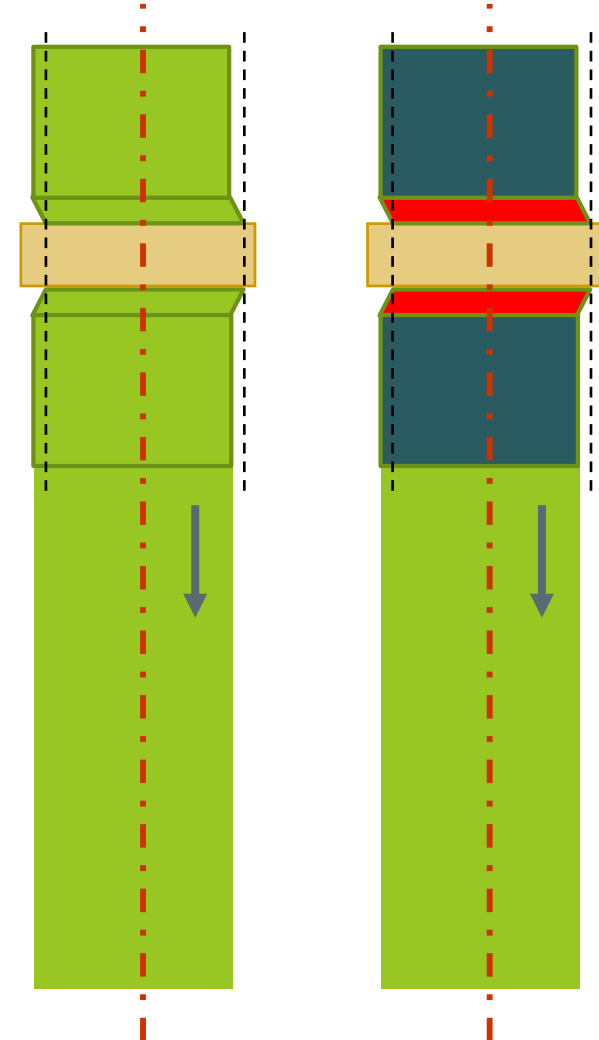
Cinching-Induced Telescoping

Internal layer-to-layer friction is exhausted by MD slip, leaving no opposition to CD shifting.

Entire 'block' of layers outside the slip zone will shift laterally relative to the core.

The cinching slip zone may grow over time as the inner layers tighten and shift to a smaller radius, the low pressure zone of poor torque capacity will expand.

Unwinding



Torque Capacity Testing

Test a roll's torque capacity:

Step 1: Draw spoke lines on the roll from core to outer diameter.

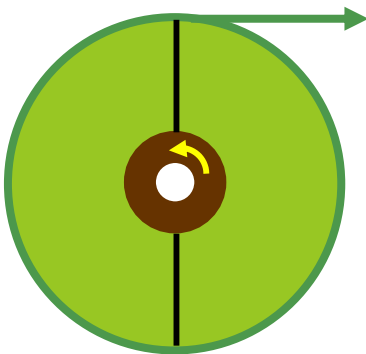
Step 2: Apply torque (via tension x radius) to unwinding roll.

Step 3: Increase tension (and torque) until layers slip.

- or -

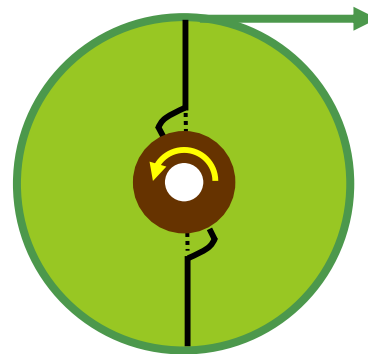
Step 2: Apply torque to the core with a torque wrench and core chuck.

Step 3: Measure the torque required to induce slippage.

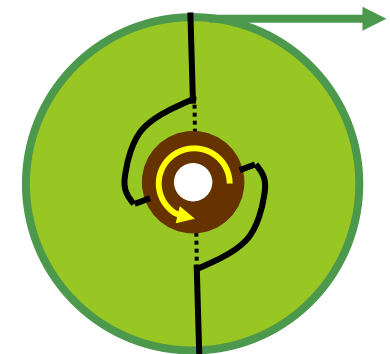


Tension
Torque

Low
Low



Medium
Medium

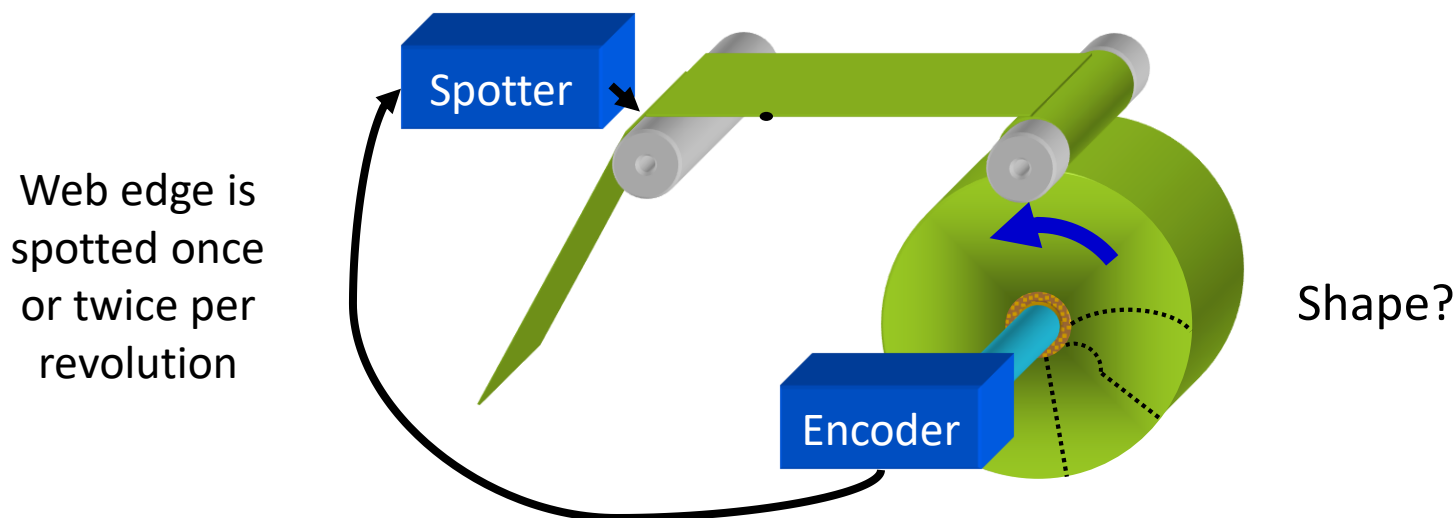


High
High

$\text{Torque} = \text{Force of Tension} \times \text{Radius}$

D-Line Inkjet Marking

Inkjet marks the moving web or edge of winding roll synched by an encoder to core's rotation (called an online J-line or a D-line system).

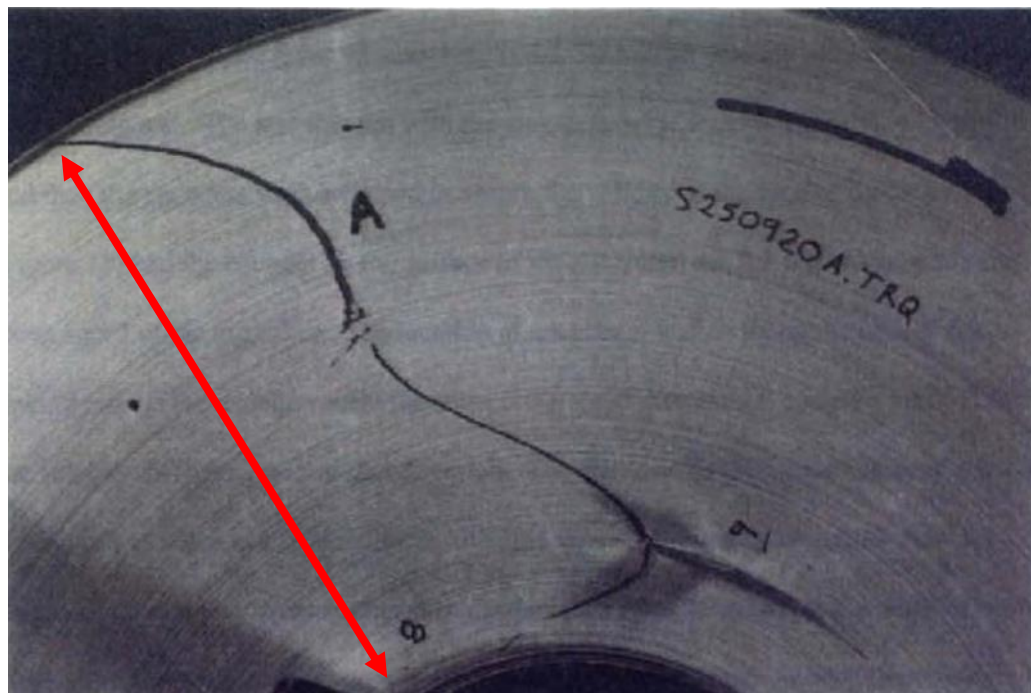


Outside roll mark should be in synch with core mark.

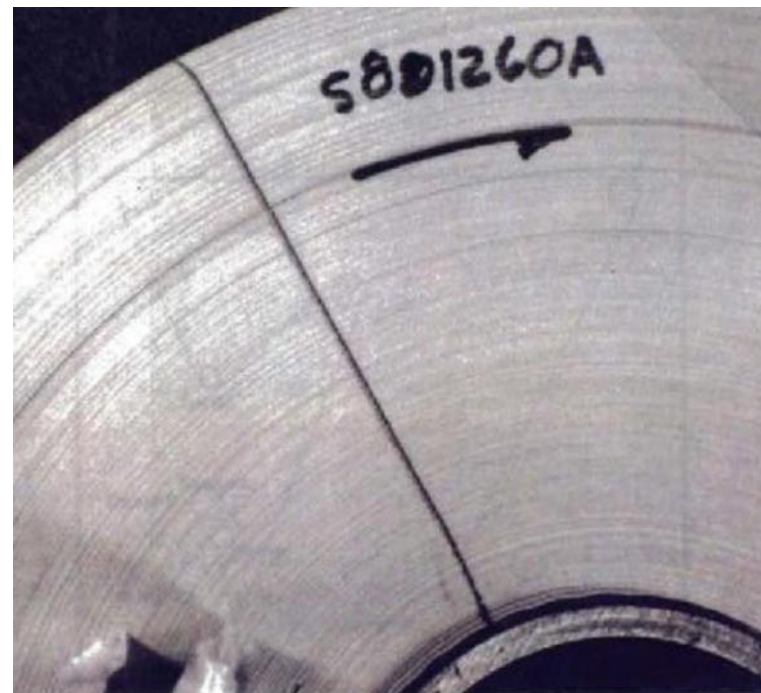
- Straight spoke = no slippage.
- D-shape = continuous slippage
- Partial D = slippage region (often near core)

Example D-Line Inkjet

D-Line shows significant inter-layer slippage throughout the roll, especially in the near core layers.



D-Line shows no significant inter-layer slippage.



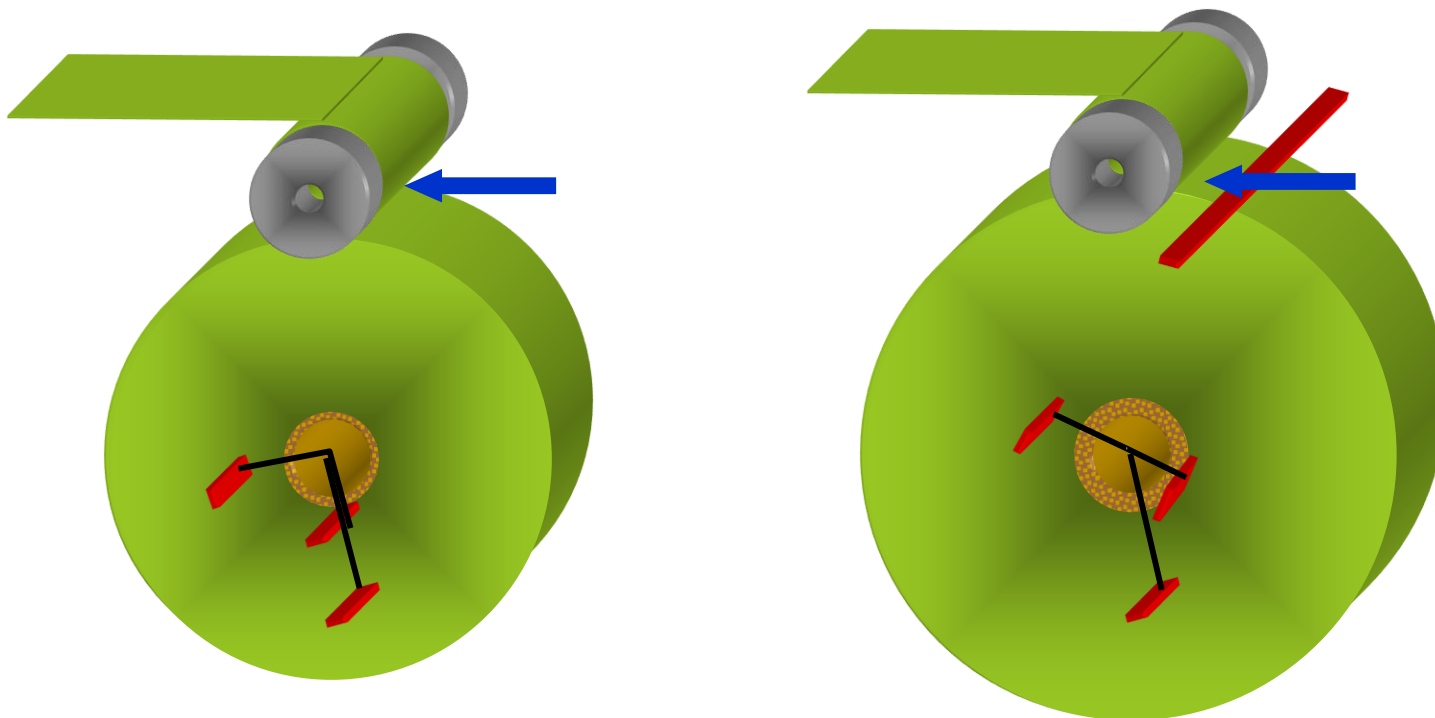
THE EFFECT OF NIP LOAD ON WOUND-ON-TENSION IN SURFACE WINDING

ROBERT EDWARD STEVES

Bachelor of Science
Mechanical Engineering
Oklahoma State University
Stillwater, OK.
May 1995

Tape Flag Video Analysis

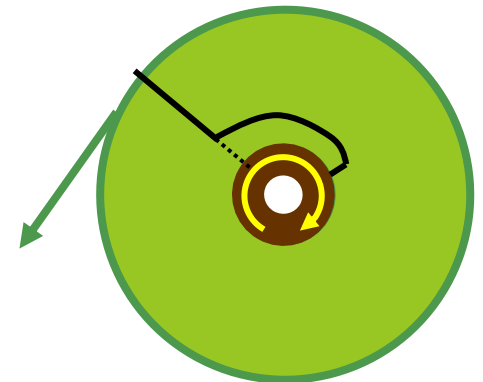
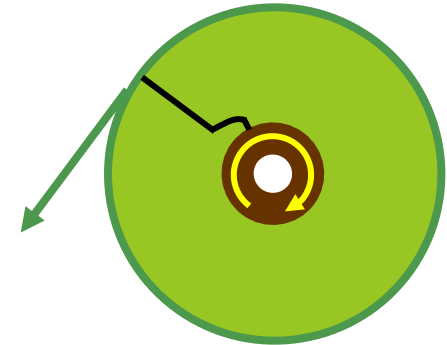
Wind random 'flags' into the side of the winding roll. Record a continuous high-speed video of the winding process and look for motion in the position of the flags relative to each other and the core. Flags will shift in cinching layers.



Unwind Torque Cinching of Near Core Layers

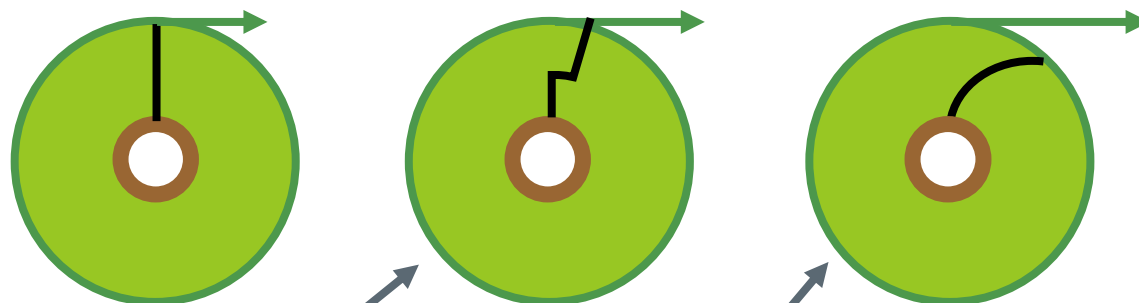


Cinching: Center Torque > Torque Capacity



Cinching Directions

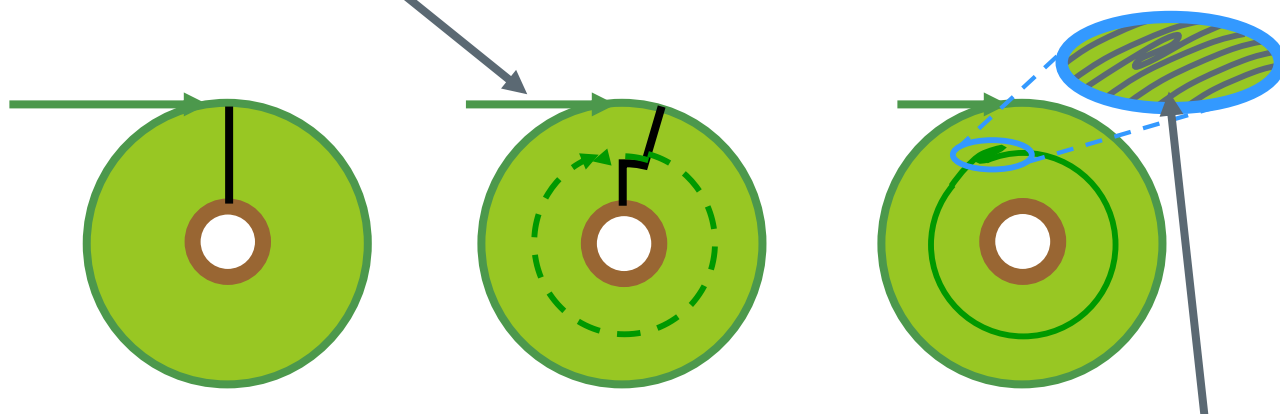
Cinching in
the tightening
direction.



Cinching layers can shift as a
stepped macro-slip...

...or continuous series
of micro-slip.

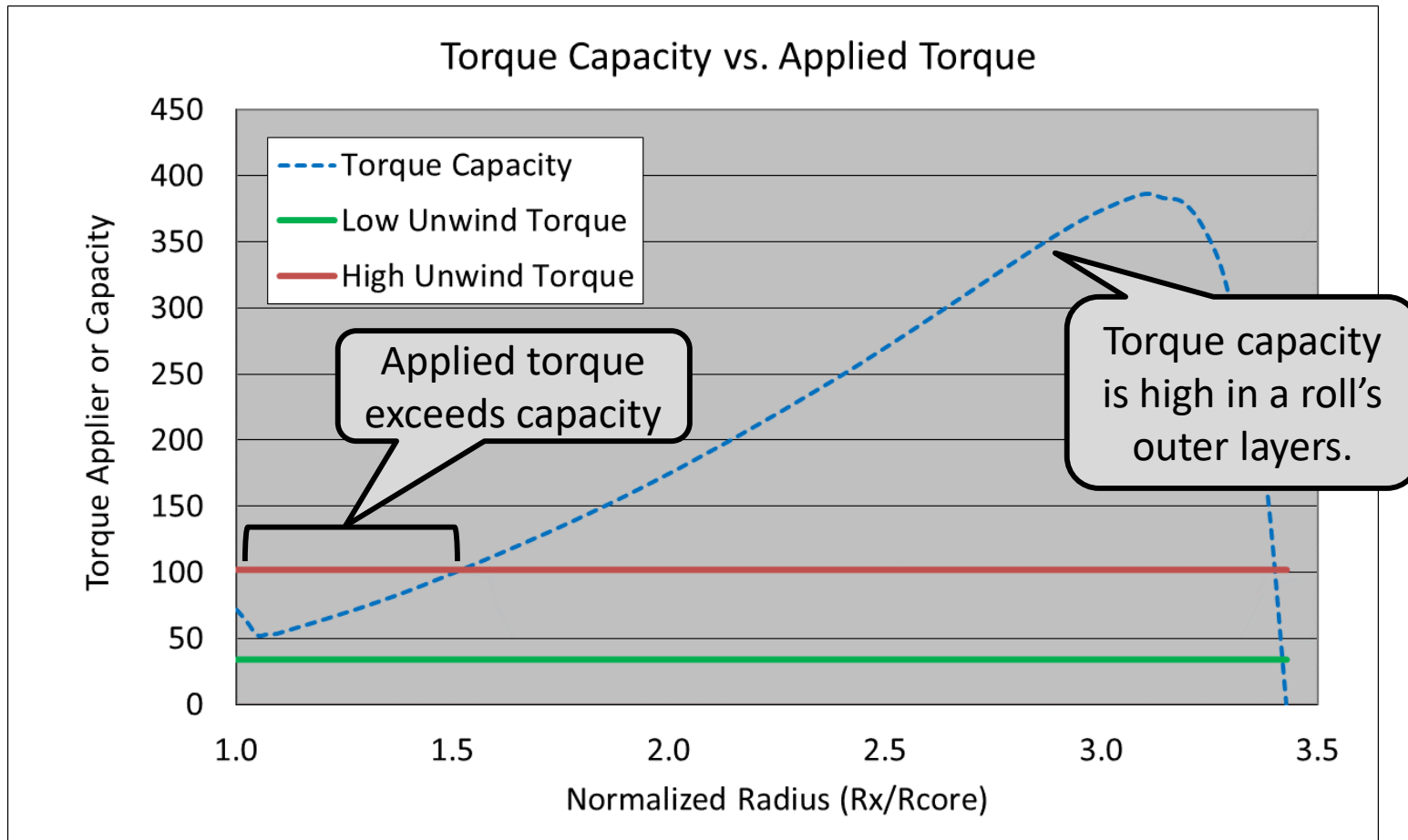
Cinching in
the loosening
direction.



Cinching can cause internal fold-over.

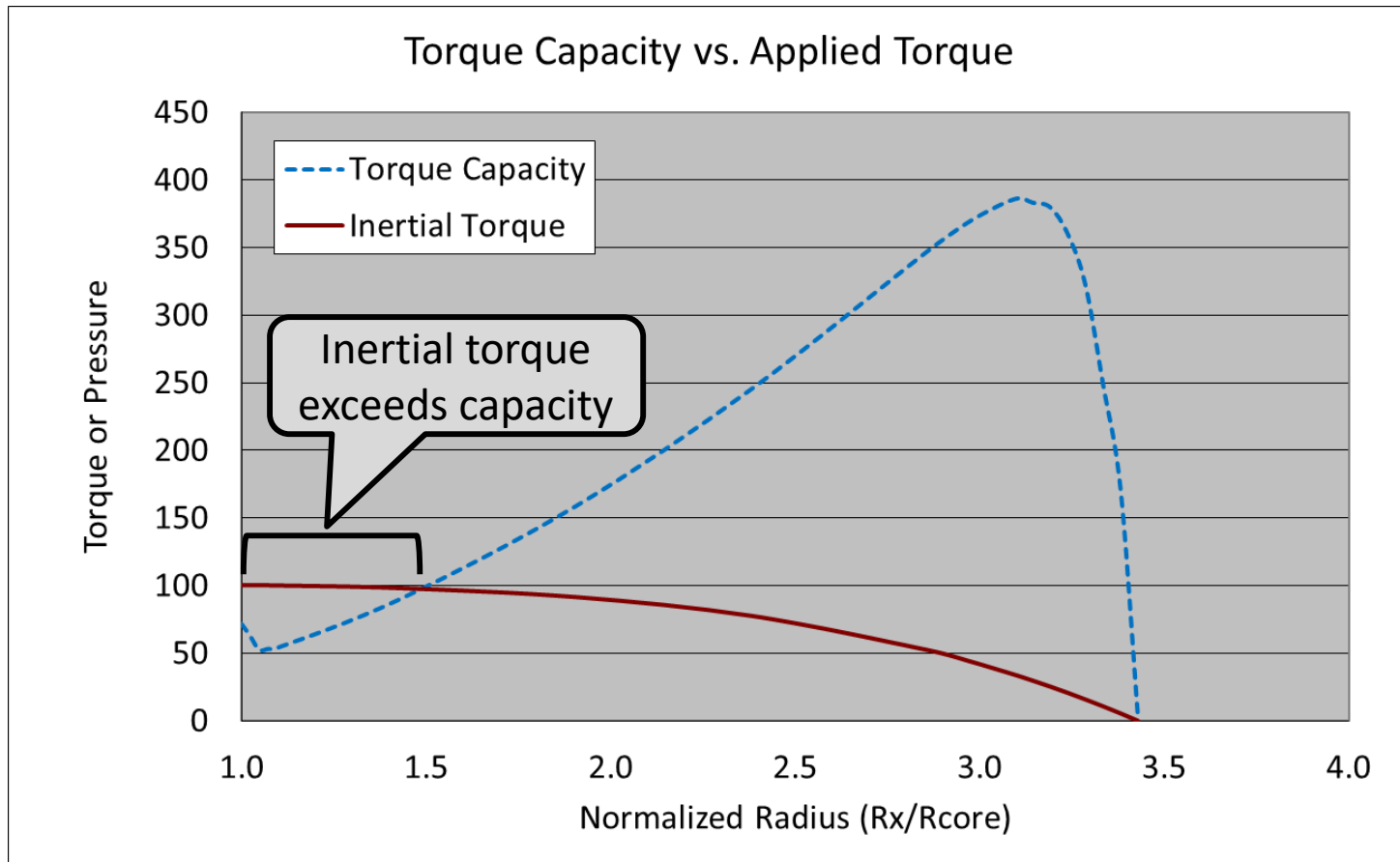
Slippage from Center Tensioning

Center torque of winding or unwinding can exceed a roll's torque transmission capacity, most likely near the core.



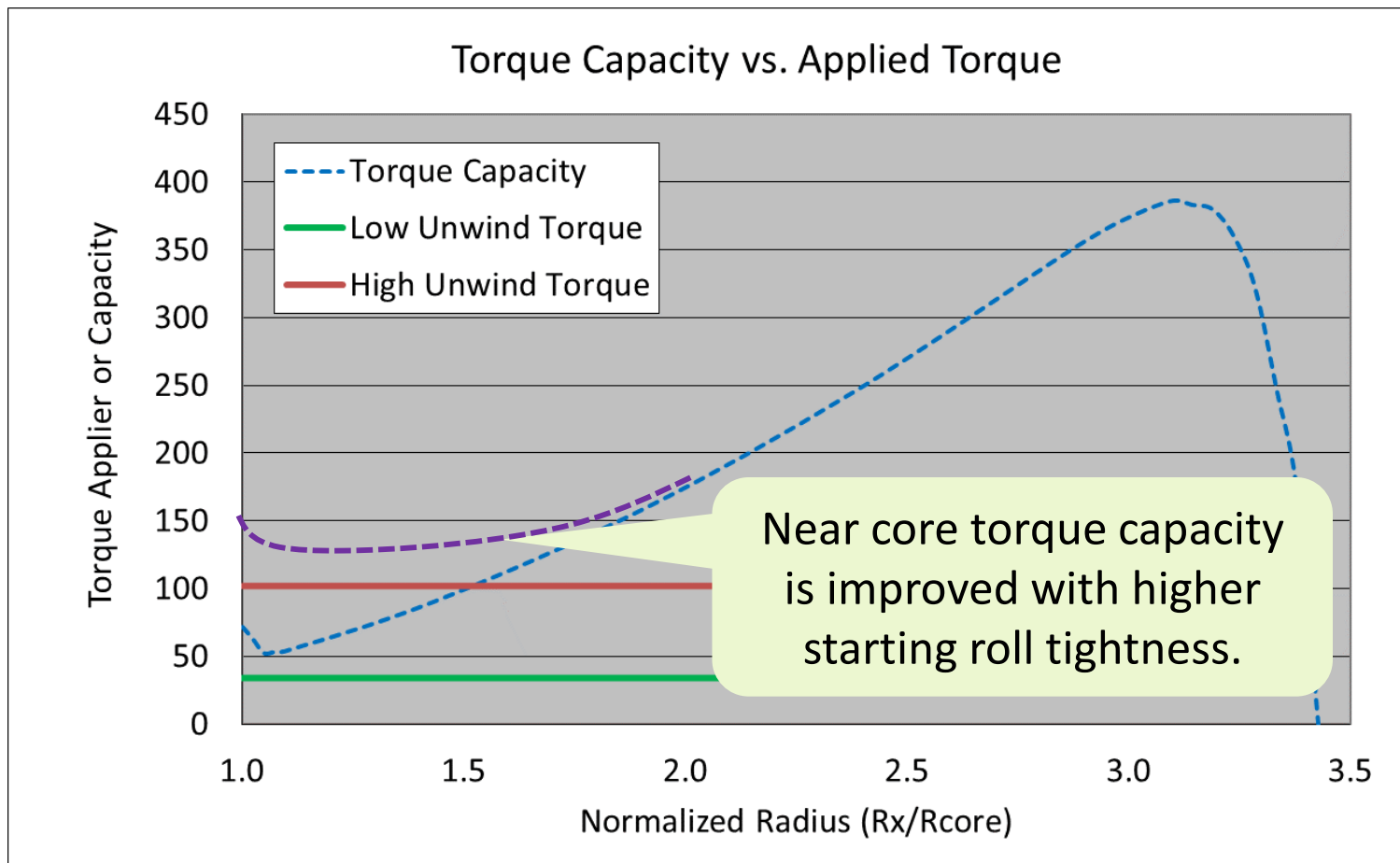
Slippage from Inertia

Excessive acceleration or deceleration requires torque from inner layers to outer layers greater than inertia, which can exceed capacity, usually near the core.



Increase Near-Core Torque Capacity

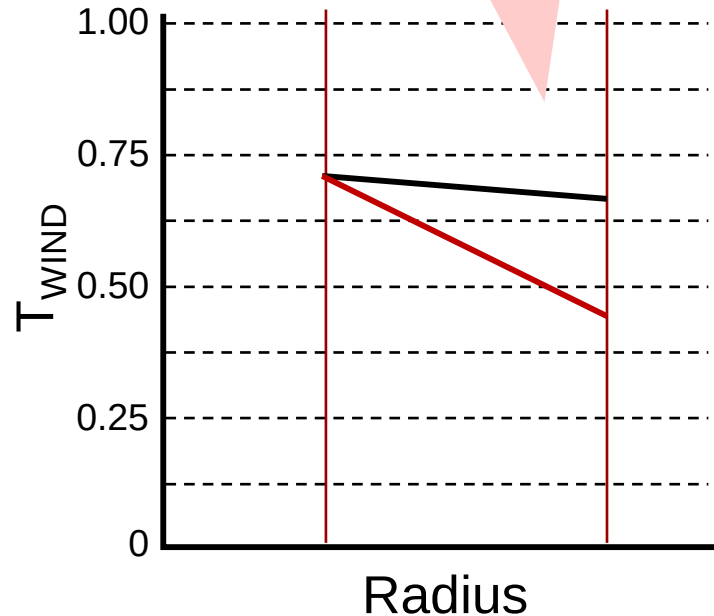
To avoid near core slippage, the roll must start with higher tension, nipping, and wrinkle-free.



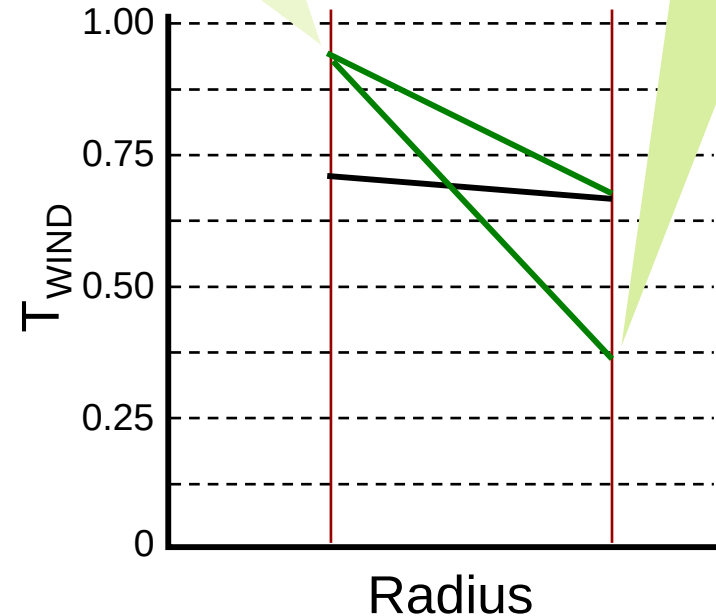
Near core torque capacity is improved with higher starting roll tightness.

Taper Tension Experiments

Changing taper tension does not increase near core tightness.



Starting with more tension (from torque and nipping) will improve near core torque capacity.

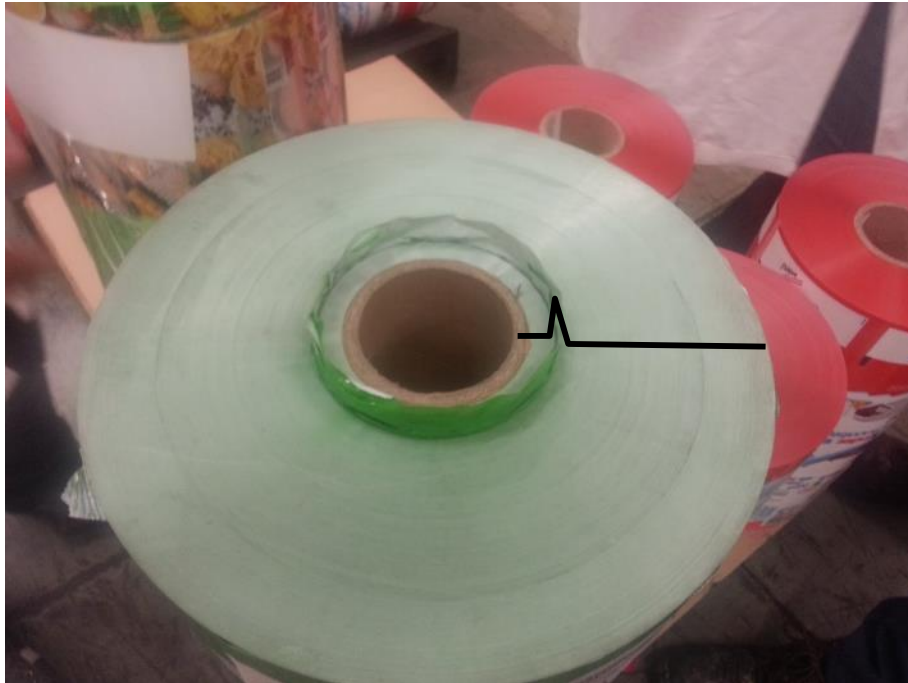


Ending at lower tension will reduce the applied torque.

Roll Handling Telescoping

In lifting a roll vertically from its core, each layer must have enough friction to support the weight of all the roll layers outside of it's position.

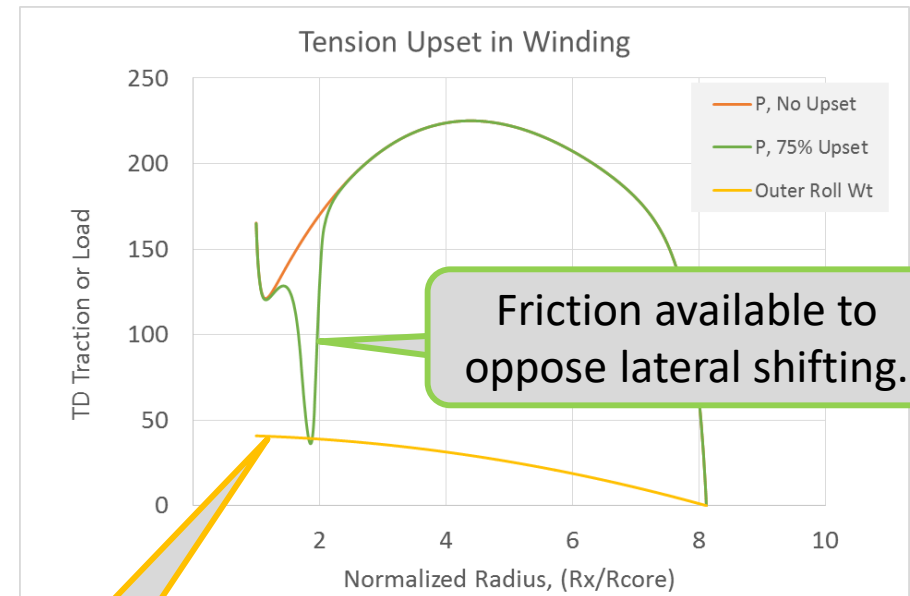
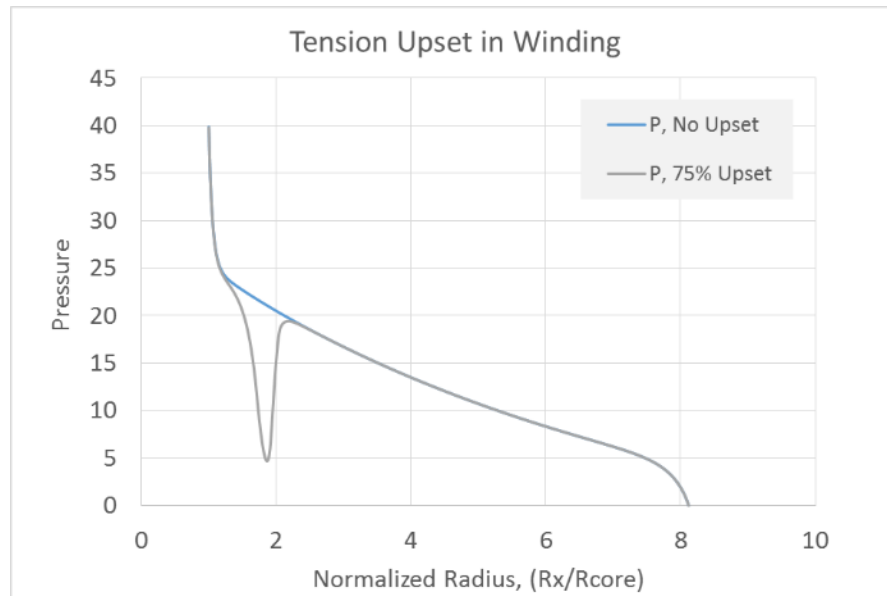
In lifting a roll vertically from its outside, each layer must have enough friction to support the weight of all the layer inside of it's position.



Roll Structure and Discontinuity of Tension

If winding has an upset in tension, such as from stopping-restarting or wrinkles, the roll will have a local pressure loss.

The lower pressure layers have lower friction, creating a likely point of slippage in vertical roll handling or torque of unwinding.



Cinching-Induced Telescoping Solutions

- ▶ Increase the winding roll's torque capacity:
 - ▶ Increase tension, especially near the core (consider 'hard start')
 - ▶ Increase friction coefficient
 - ▶ Increase area and reduce the mechanical disadvantage of layers near the core **with larger core diameter**.
 - ▶ Increase CD thickness variations (including induced thickness variations from printing, coating, or knurling).

Cinching-Induced Telescoping Solutions

- ▶ Increase the winding roll's torque capacity:
 - ▶ Minimize core compression or shrinkage, including manage paper core moisture content to ensure cores are dry, use a core with sufficient stiffness for the internal roll pressures of your product (consider changing to stiffer core)
 - ▶ Minimize shrinkage of layers, including substrates, coatings (including lateral flow of viscoelastic coatings, like adhesives), roughness or texture, and entrained air.
 - ▶ Minimize soft layers near the core from wrinkling or misaligned core starts. If surface winding, add center torque assist.

Cinching-Induced Telescoping Solutions

- ▶ Decrease the applied center tensioning torque by reducing final diameter or final tension (more taper tension) or switch to surface winding.
- ▶ Decrease the applied inertial torque by reducing final diameter or accel/decel rates.
- ▶ Stop even the smallest amount of cinching to prevent cinching-induced cinching.
- ▶ Physically restrain the web from shifting laterally with flanges, spool walls, or rotating elements. (This is certainly a bandage for the problem, but bandages are quite helpful.)

Better Rolls Through Chemistry

Non-telescoping polyethylene film roll and method for producing same

It has been discovered that telescoping of film rolls comprising a layer polyethylene film can be substantially eliminated by blending small amounts of a polypropylene material with the polyethylene prior to or during extrusion of the film. The resulting composition has been found to substantially eliminate telescoping both in single layer polyethylene films and in multilayer films having polyethylene as an outer layer.

Publication number	US5023142 A
Publication type	Grant
Application number	US 07/407,415
Publication date	Jun 11, 1991
Filing date	Sep 14, 1989
Priority date [?]	Aug 4, 1988
Fee status [?]	Paid
Inventors	Kevin G. Hetzler, Billie C. Munger, Wayne M. Wegner
Original Assignee	Reynolds Consumer Products, Inc.

FIG-1-

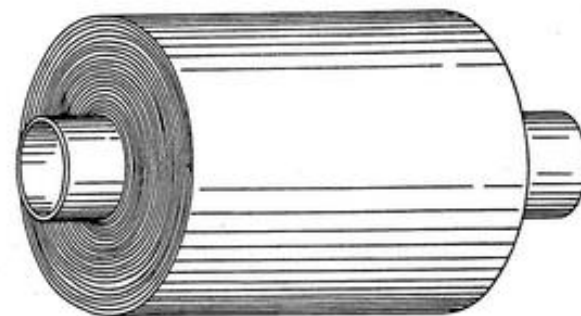
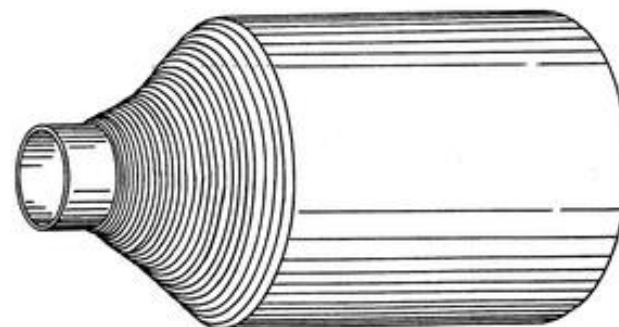
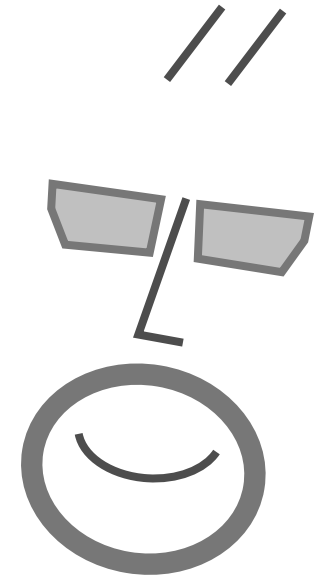


FIG-2-

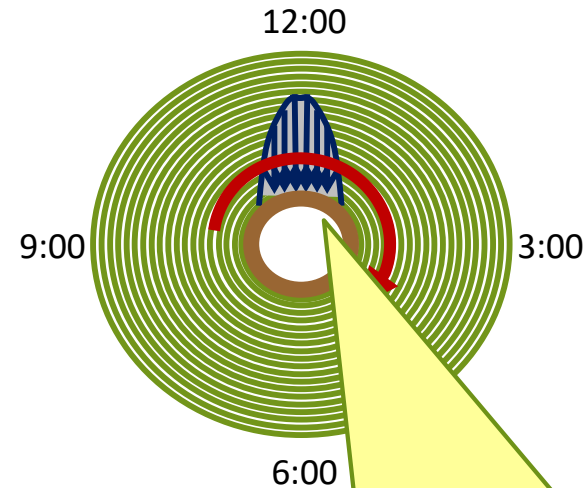
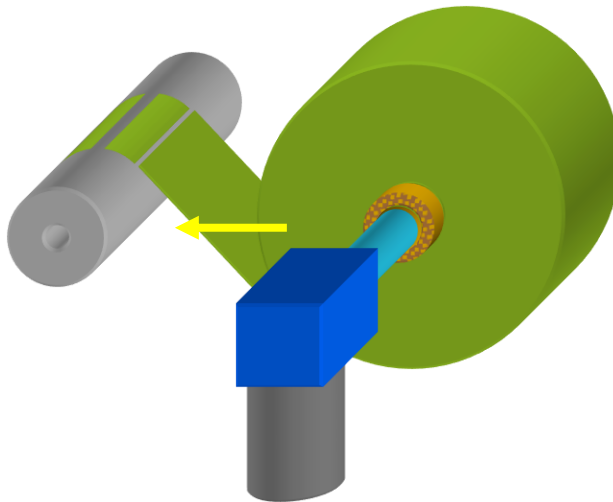


Winding Solutions Qs

What is nip-induced cinching?

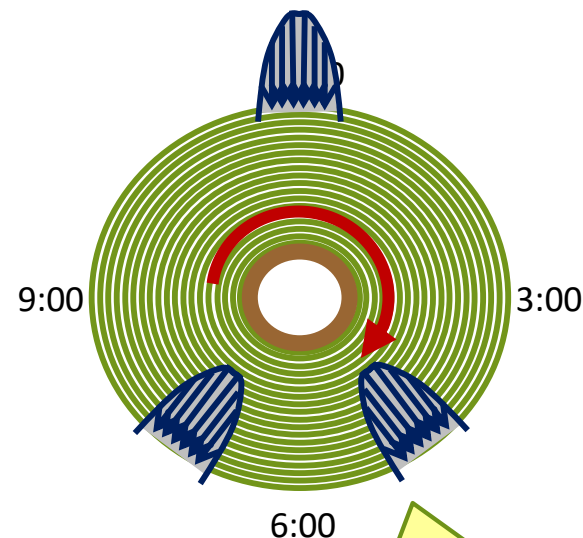
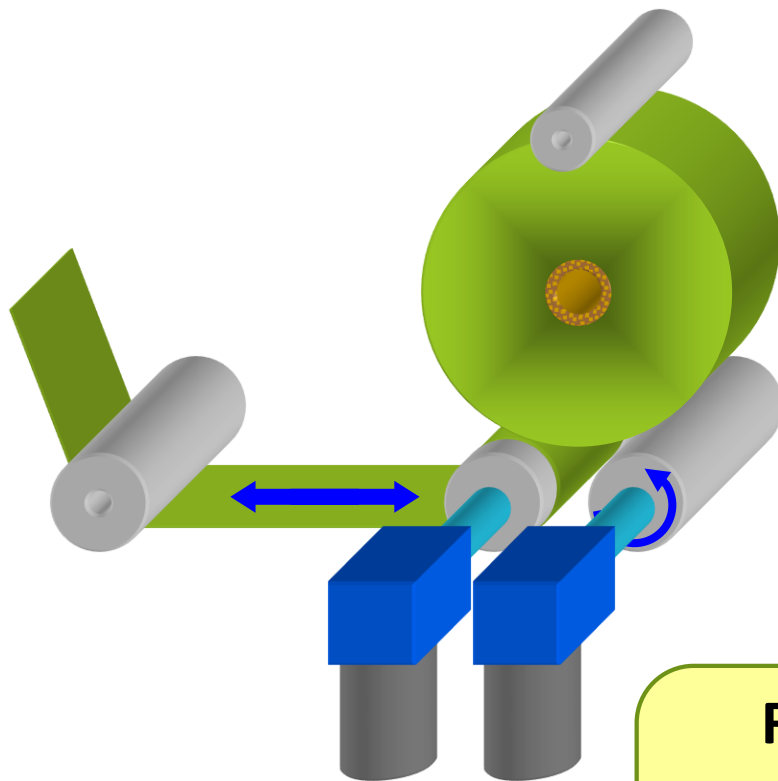


Internal 'Nip' of Center Supported Roll



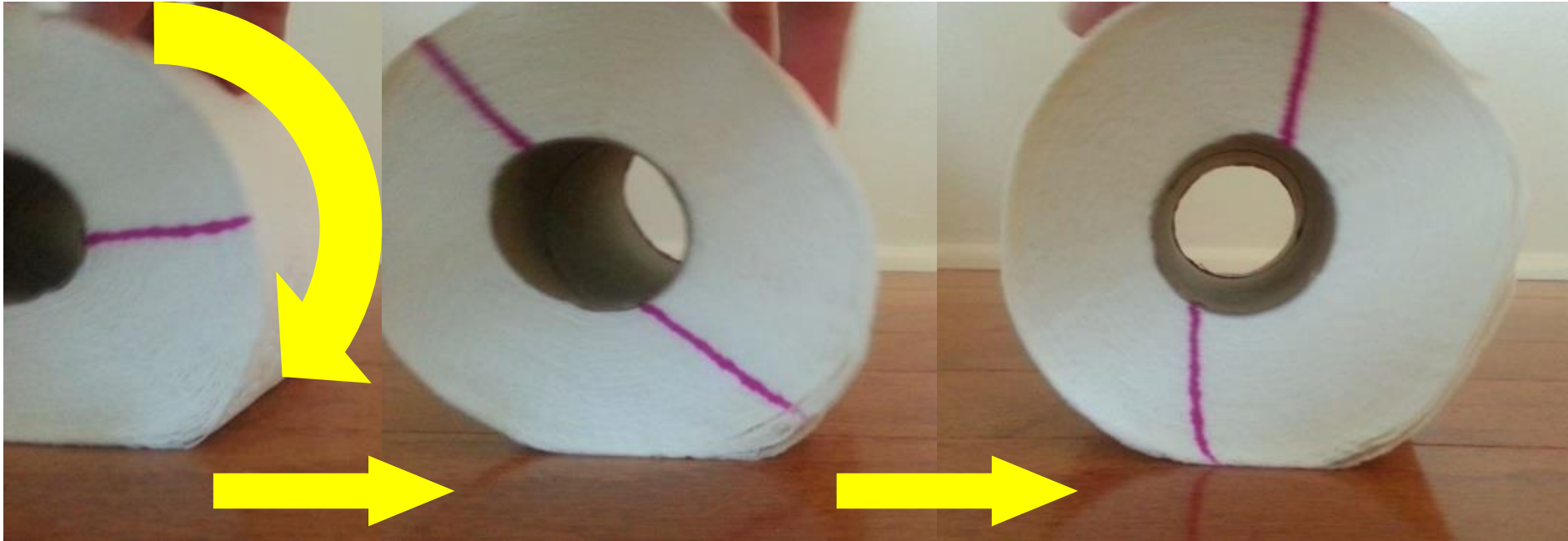
Roll weight hangs over core, creating high pressure zone in layers on core top side (and low on core bottom).

Internal 'Nip' of Surface Supported Roll



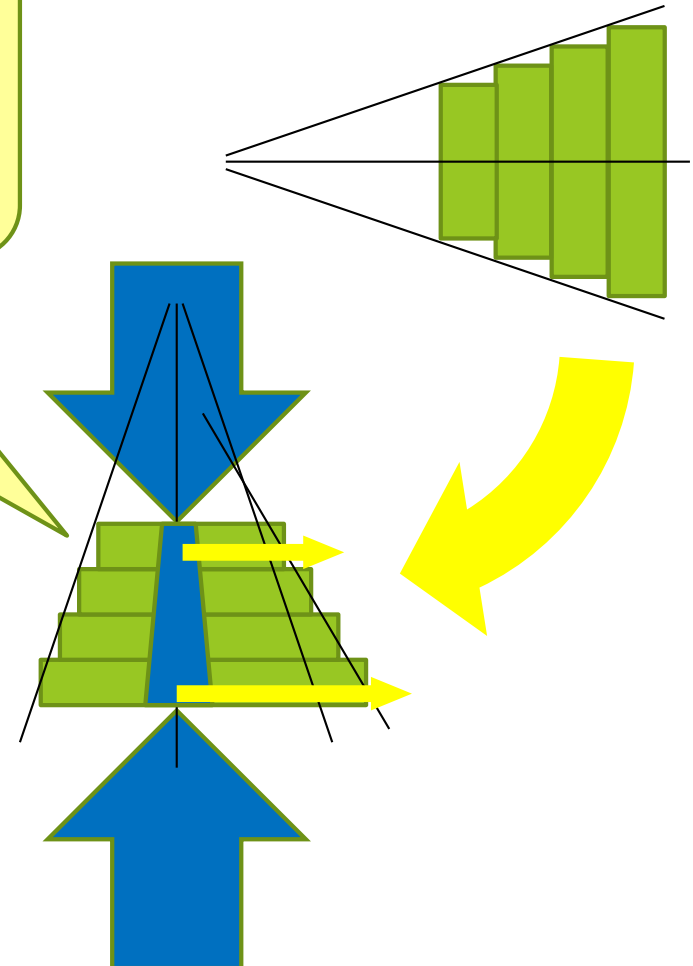
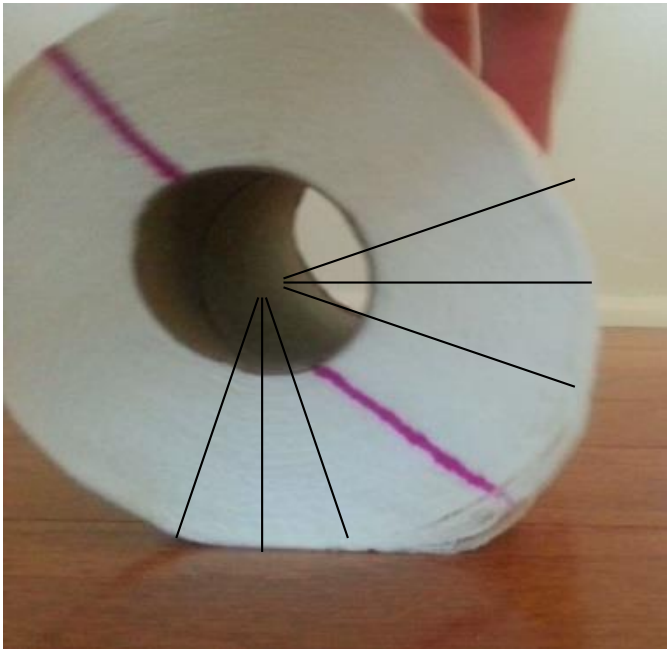
Roll weight and rider roller load create three nip points on the roll's outer surface, but can induce slippage down to the core.

Nip-Induced MD Slippage

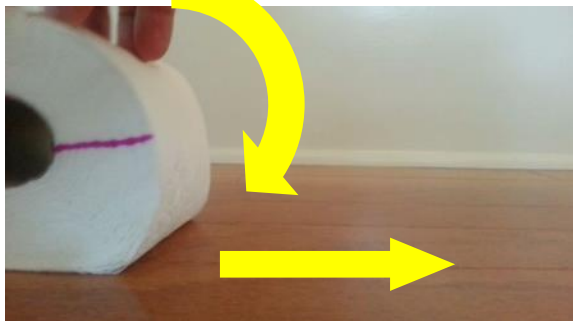
**1****2****3**

Nip-Induced MD Slippage

The nip elongates the roll's layers. In rotation, the longer, outer layers slide relative to shorter, inner layers.



1



2



3



4



5



6



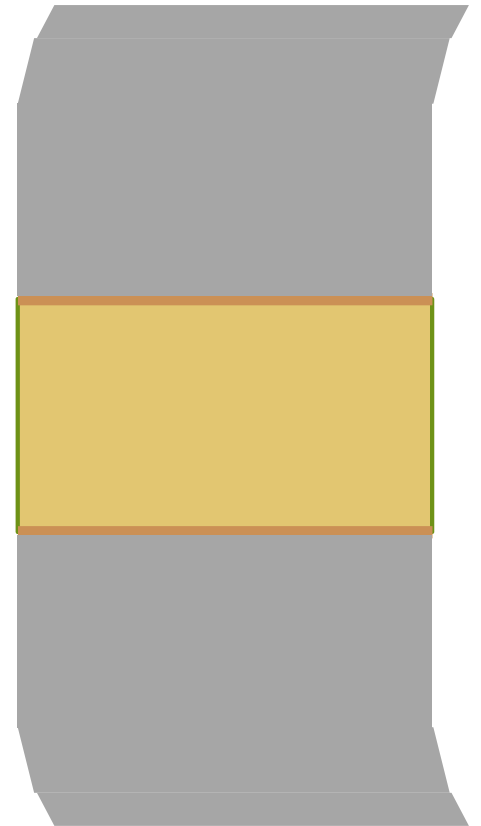
7



8



Nip-Induced Telescoping

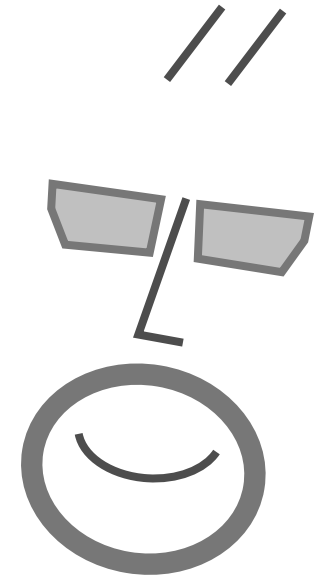


Nip-Induced Cinching/Telescoping

- ▶ Worse with higher nip loads
- ▶ Worse with lower roll tightness
- ▶ Worse with compressible materials
- ▶ Worse with low friction materials
- ▶ May occur only in outer layers, not entire roll.

Winding Solutions Qs

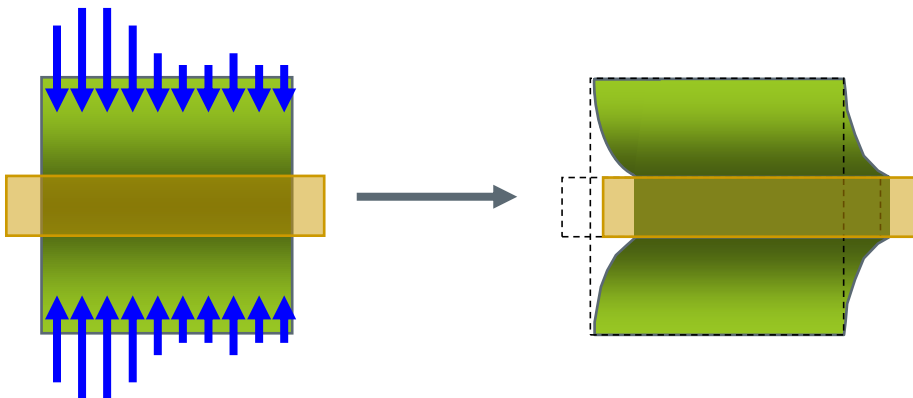
What is creep-related telescoping?



Pressure Sensitive Adhesive (PSA) Telescoping

PSA telescoping is high tension, high pressure defect. Rolls will be straight immediately after winding, but will slowly creep over time. The adhesive layers creep to relieve high stresses, shifting away from the tight side of uneven roll stresses.

- PSA telescoping may occur in 24 hours or may take weeks or months to develop.
- More likely with thick, soft adhesives, higher temperatures, narrow widths, and large roll buildups.

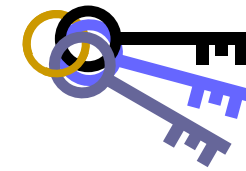


Winding Solutions Qs

What is handling-related telescoping?

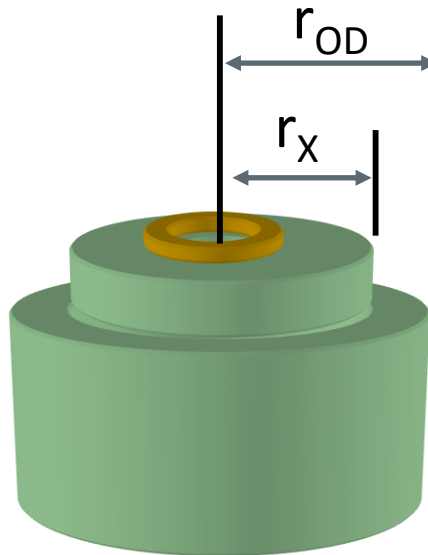
What is the remedy?





Lateral Traction Capacity

To lift a roll turned on end from the core, each layer must have traction forces greater than the load created by the weight of the outer layers.



To prevent slip while lifting:

Tractional Forces > Gravitational Forces

$$\mu P_X A_X > \pi w \rho (r_{OD}^2 - r_X^2)$$

Minimum pressure at radius x
to prevent slip:

$$P_X > \frac{\rho (r_{OD}^2 - r_X^2)}{2 \mu r_X}$$

r_{OD} = rolls outer radius, in.

P = roll density, lbs/in³

μ = friction coefficient

P_X = internal pressure at radius x, psi

A_X = area at radius x, in²

r_X = any given radius x, in.

w = roll width, in

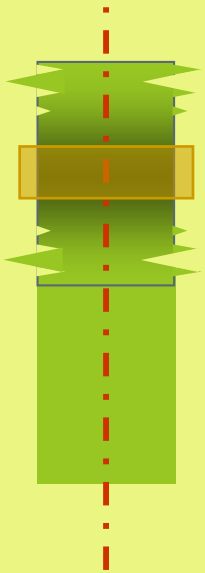
T = tension, lbf/in

In-Roll Lateral Errors

In-Roll Error

Roll layers shift during winding
(after entering winding roll on core centerline)

Air-Related Telescoping



Onset based on web roughness and air layer
 $[f(V,T,R)]$

Number of “shooting”
layers increases with roll
radius.

Individual events will not
worsen over time.

Cinching-Related Telescoping

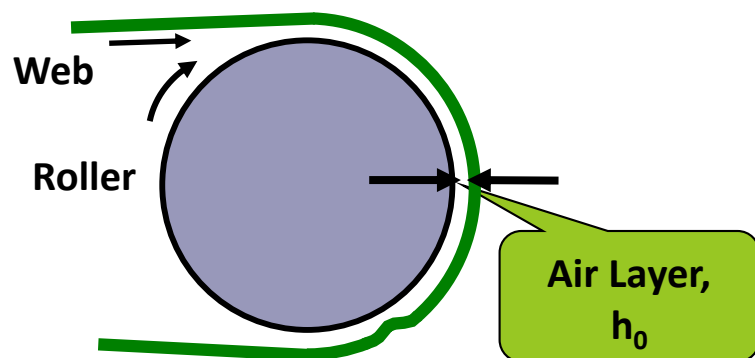


Onset begins when roll
torque capacity is exceeded.

Direction of telescoping
dependent on roll and web
CD variations.

Lateral errors may continue
to get worse with more
cinching.

Air Lubrication



$$h_0 = 0.643r \left(\frac{6\eta(V_W + V_R)}{T} \right)^{2/3}$$

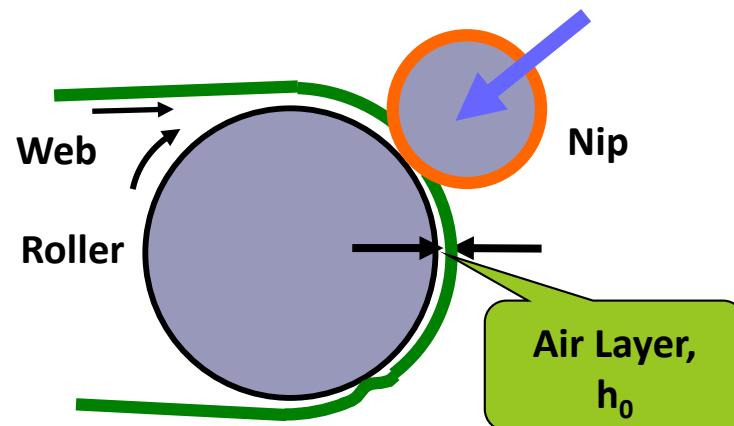
h_0 = Steady-state float height, in.

r = Roller radius, in.

V_W = Web speed, in/s

V_R = Roller speed, in/s

T = Tension, pli



$$h_0 = 4\eta V \frac{r_{eq}}{N}$$

r_{eq} = Equivalent roller radius, in.

η = Air Viscosity,

V = Process speed, in/s

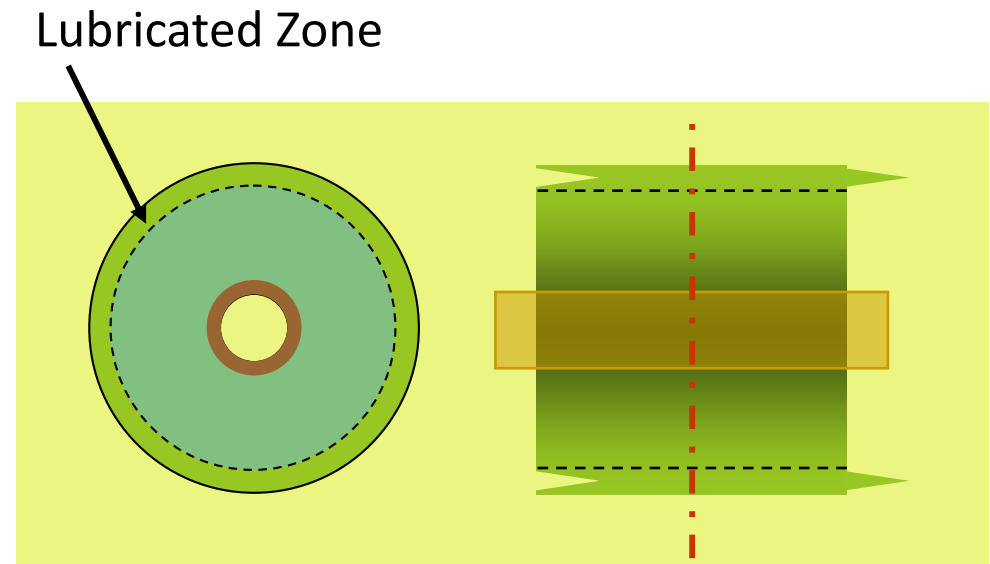
N = Nip load, pli

Air Lubrication in Winding

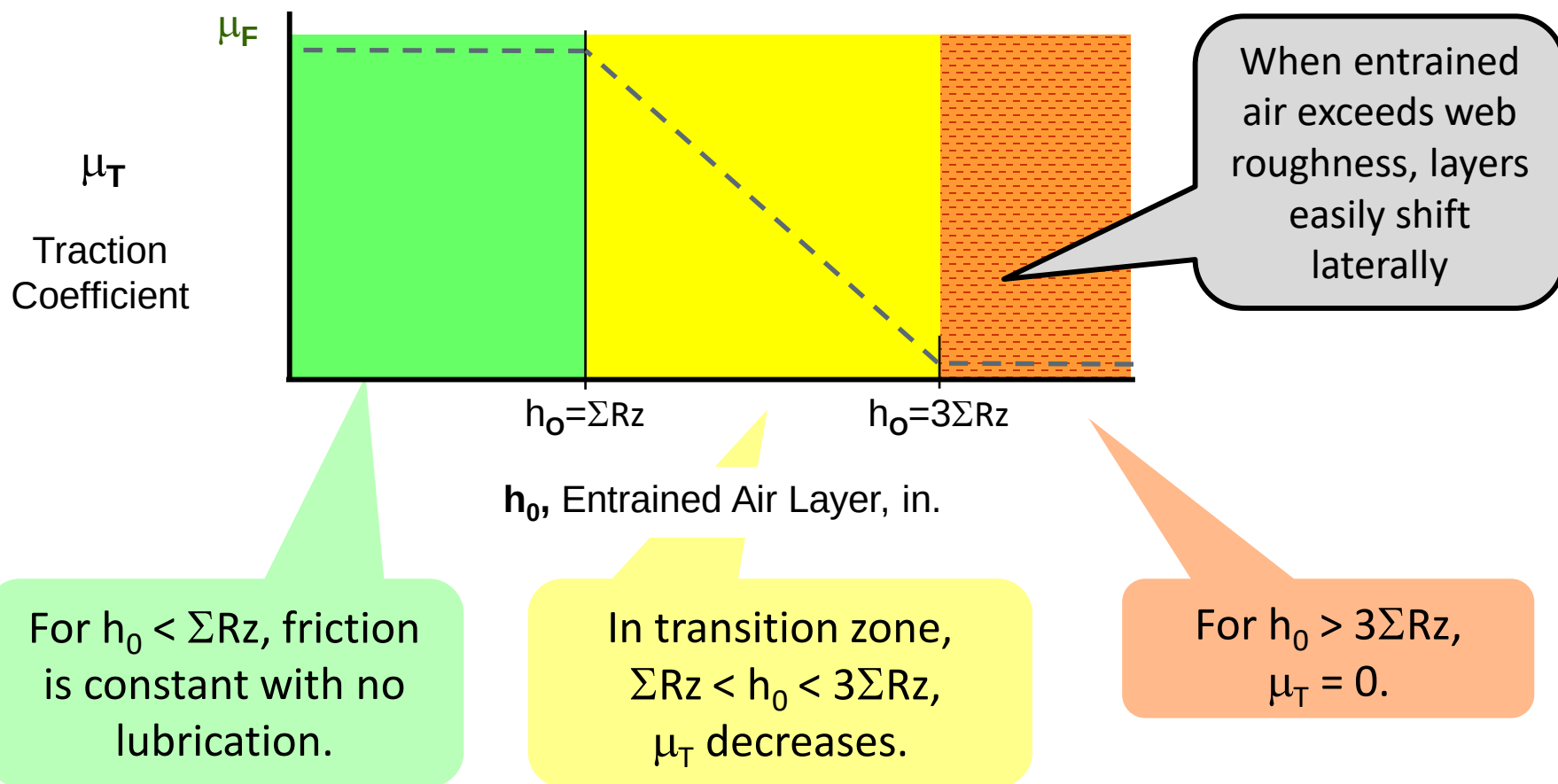
The entrained air takes time to escape, creating a lubricated zone in the winding roll's outer layers.

The lubricated zone increases with speed and diameter.

Smooth webs are especially sensitive to air lubrication and shifting layers at the outside of the winding roll.



Traction vs. Air Lubrication



ΣRz = sum of web and roller Ra roughness, in.

μ_F = friction coefficient, no lubrication

μ_{crit} = traction coefficient require to turn roller

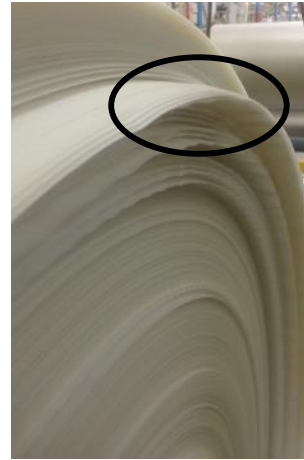
Air Lubrication Outer Wrap Telescoping

Layers shift in the outer wraps of the winding roll:

At small diameters, the tension creates sufficient pressure to prevent lubrication (air layer is less than roughness).

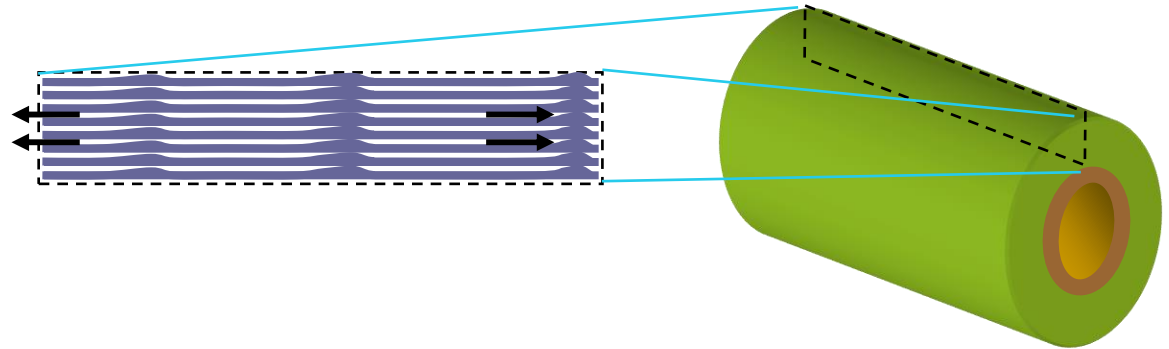
As diameter increase (and tension drops, if using taper tension), air layer is greater than roughness, for a number of layers at the outside of the winding roll.

These layers will 'shoot out' laterally in 'V' shaped telescoping, with the layer in the middle of the lubricate stack moving the furthest.



Entrapped Air: Good or Bad?

Some air will be trapped in pockets sealed by high thickness lanes.



Advantage of Entrapped Air:

For many products, entrapped air reduces defect, such as baggy web, caused by diameter and stress variations

Disadvantage of Entrapped Air:

In storage, the air may flow out the roll's end, causing the roll to loosen.

Air Lubrication Telescoping Solutions

- ▶ **Use a winding nip** - a roller nipping against the outside of the roll during winding. The winding nip, found in all surface winding and many center winders, has two air management benefits. First, the nip load reduces, but does not eliminate, the amount of air that enters the winding roll. Second, under the footprint of the nip, the entrained air is compressed, creating at least a once per revolution traction point, enough to greatly reduce lateral shifting (a.k.a. air lubrication telescoping).
- ▶ Physically restrain the web from shifting laterally with flanges, spool walls, or rotating elements. (This is certainly a bandage for the problem, but bandages are quite helpful.)

Air Lubrication Telescoping Solutions

- ▶ Reduce the air layer thickness by limiting roll diameter, reducing speed, and keeping tension high.
- ▶ Winding in a vacuum chamber eliminates the problem, but usually raises other problems as the benefits of air on winding friction and roll cylindricity are lost.
- ▶ If possible in product design, consider a pattern or texture on one side of the web to manage air.

Your Solution to Telescoping?

Plan A Apply a remedy based on cause.

Plan B



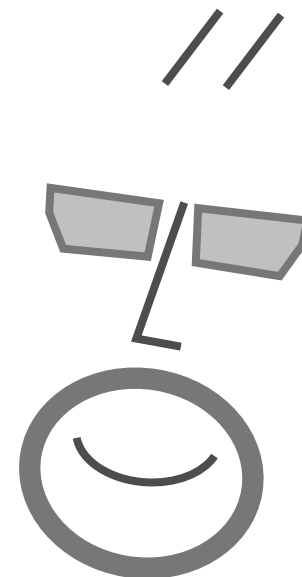
6.4 Roll Structure and Web Quality

- Wound-In Stresses
- Lateral Errors
- In-Roll Buckling 
- Transverse Variations
(Gauge Bands and Bagginess)

Winding Solutions Qs

*What is... Starring? Spoking?
Roll sagging? PSA gapping?*

*What do these
defects have in
common?*



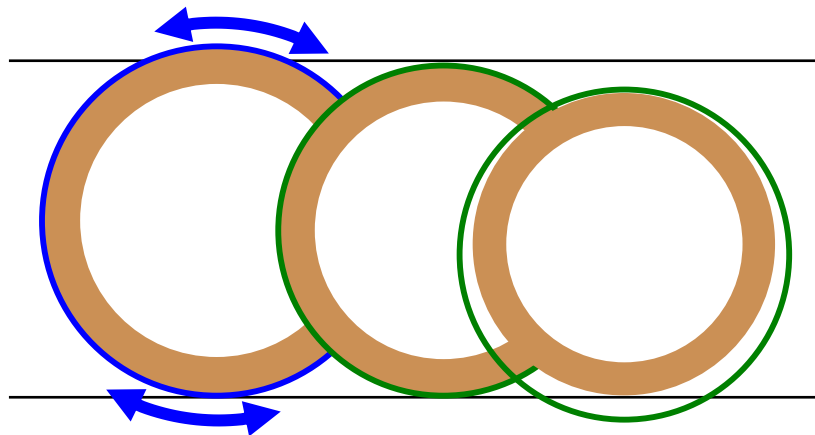
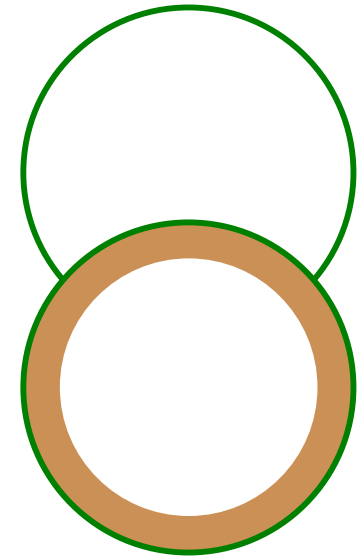
Hoop Tension, Hoop Compression

Start with an untensioned ring of radius, r_0 .

If the ring is placed on a core of the same radius, there is no tension.

If the ring is placed on a core of large radius, it will have tension proportional the strain.

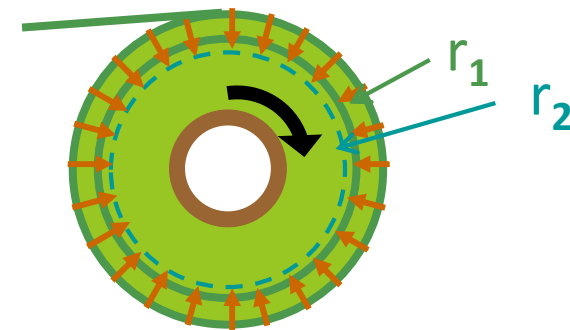
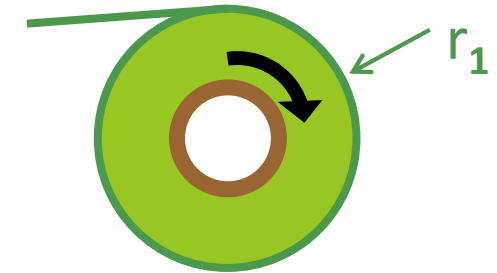
If the ring is placed on a core of smaller radius, it will be loose.



Hoop Tension, Hoop Compression

Consider a tensioned layer that joins the a winding roll at a given radius, r_1 .

As additional layers are wound on top of this layer, it will shift to a smaller radius, r_2 , as the core and layer below it are compressed.

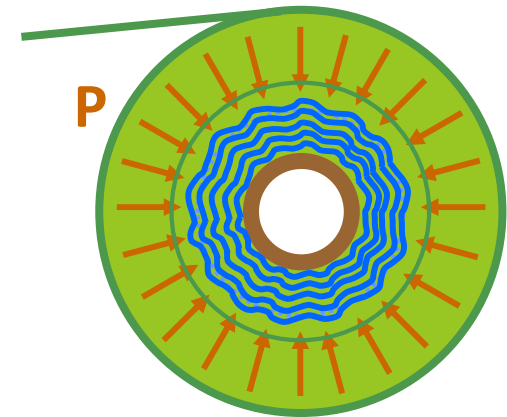
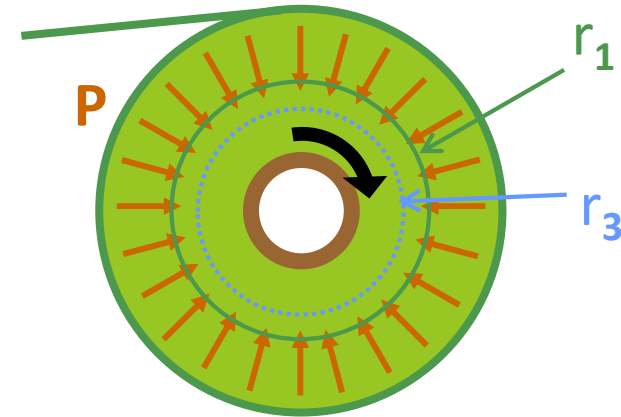


Hoop Tension, Hoop Compression

As the roll continues to wind, the layer will be compressed to another smaller radius, r_3 , from the rising pressure of additional layers.

In moving to smaller radial positions, the strain and tension of a given layer can drop to zero tension or into hoop compression.

Beyond critical compressive stresses and strain the layers will buckle.



Spoking, Starring, Gapping, Soft Edge

Spoking in film by winding on top of soft layers.

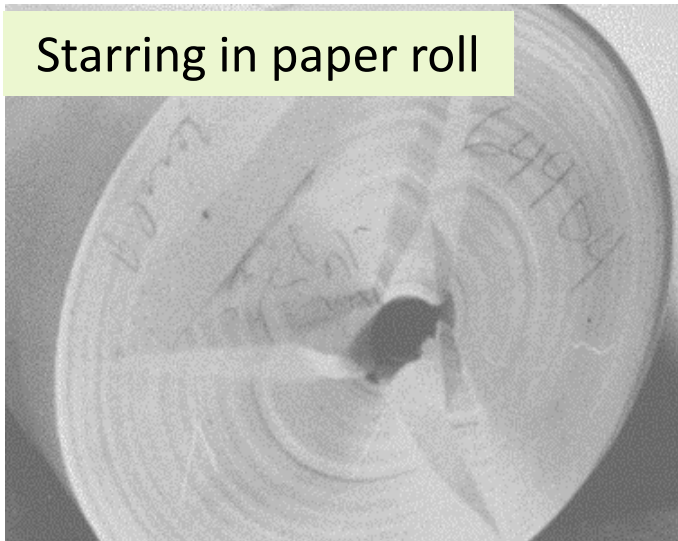


Starring in film roll



Buckling of thinner uncoated edge
(not a defect)

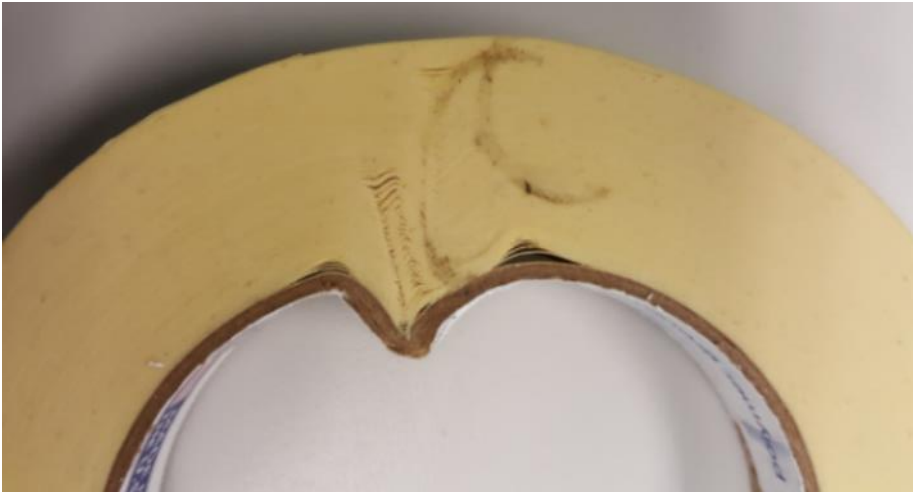
Starring in paper roll



Gapping in PSA roll

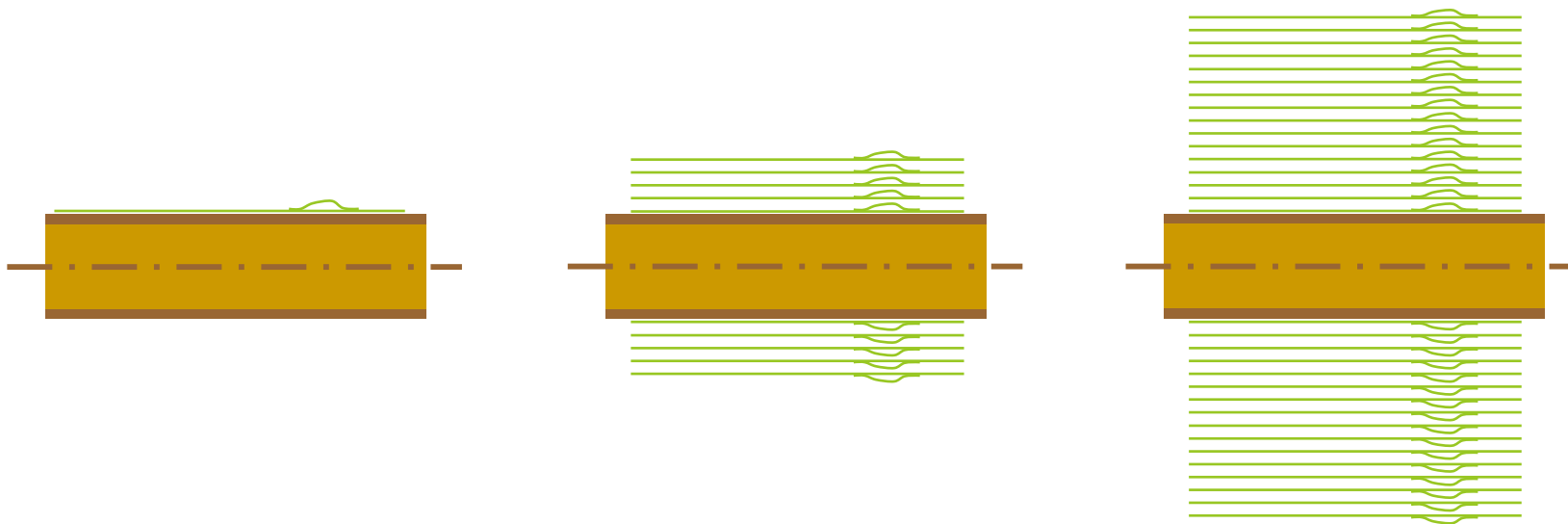


CD Buckles



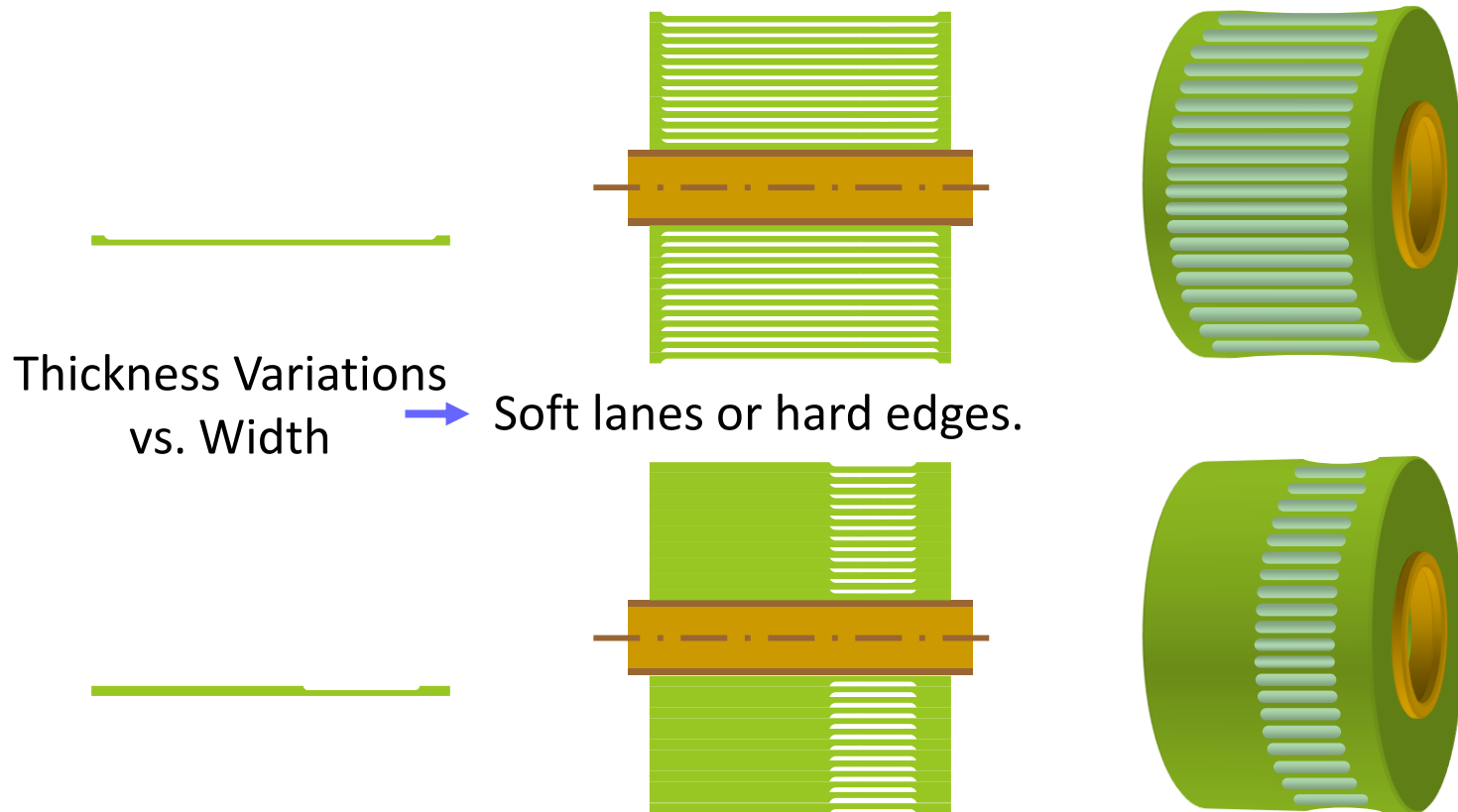
Oscillation and Radial Variations

If a roll builds without oscillation, the thick lanes will create diameter variations. The percent diameter variations will usually be a lower than the percent thickness variations that caused them.



CD Buckles and Softbands

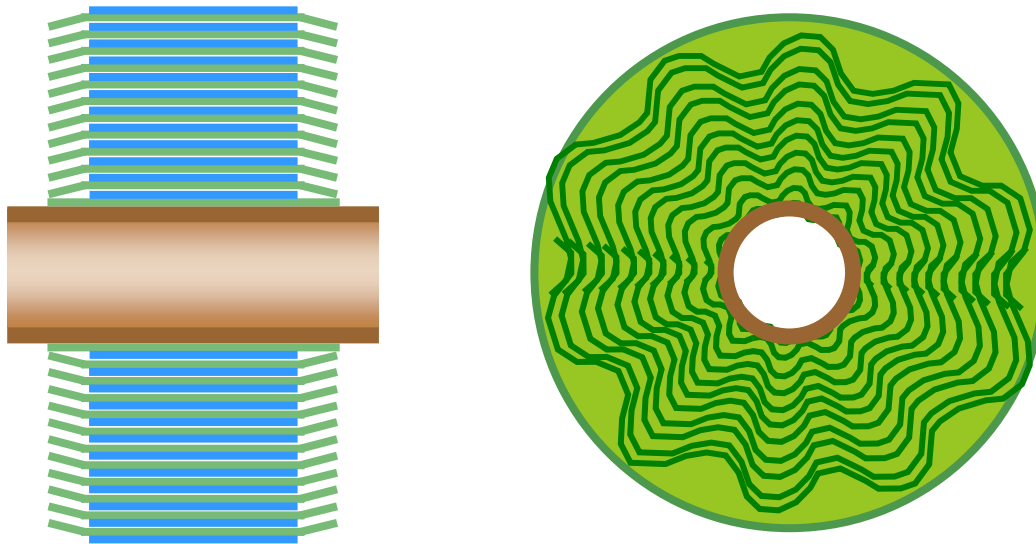
Crossbuckles or softbands are magnifications of web or coating thickness variations.



Radial Error: Starring

Not a Defect?

- Don't confuse starring from core or interlayer compressive buckling with a similar end-of-roll appearance commonly seen in coated or laminated products.
- When the coating or laminated layer is relatively thick and narrower than the web the excess web on either side will fall loosely toward the core, resulting in a starring appearance.

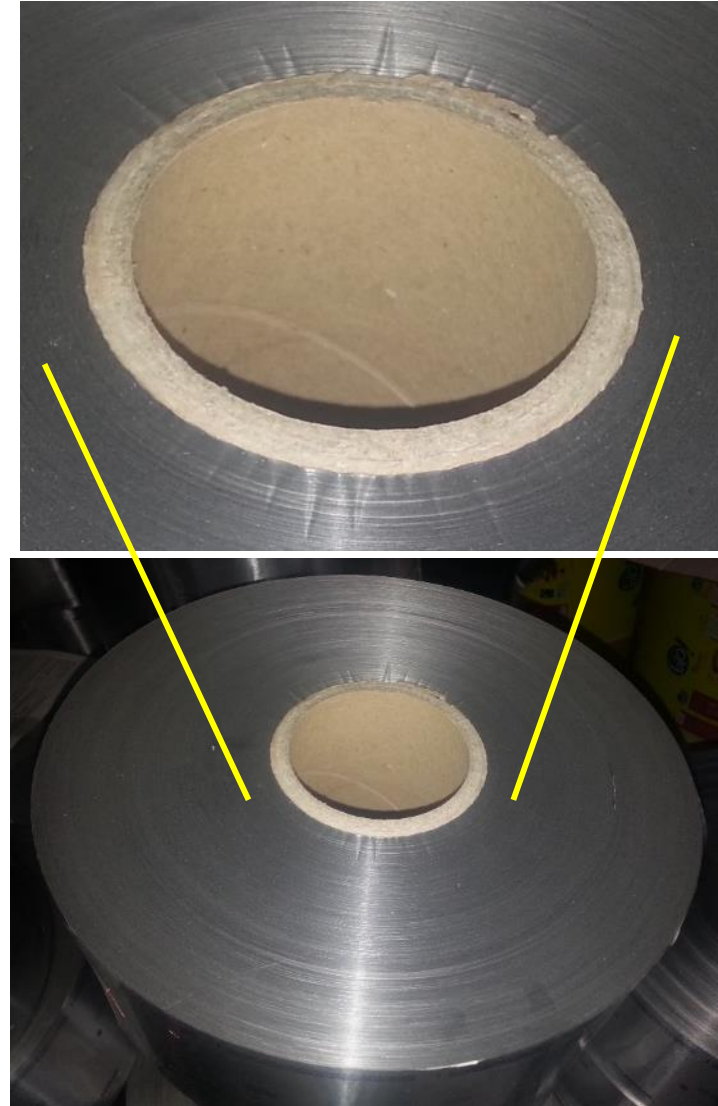


Roll Near-Core Buckling – Starring

Starring is a buckling defect.

Any layer in hoop compression may potentially buckle and star.

- Starring is usually a high tension defect. High tension creates high pressure on the core.
- When the percent change in the core's outer diameter is greater than the web strain of winding, layers go from tension to compression.
- When compressive stresses are high enough, the layers buckle.



Roll Radial Compression Factors

- ▶ Core compression from winding pressure
- ▶ Core shrinkage (winding on 'wet' core that later dry and shrink).
- ▶ Winding at high temperature than storage, shrinking with cooling.
- ▶ Compressing wrinkles layers.
- ▶ Compressing web surface roughness or internal voids.
- ▶ Escape of excessive wound-in air.
- ▶ Aligning of thin lanes in web thickness profile.

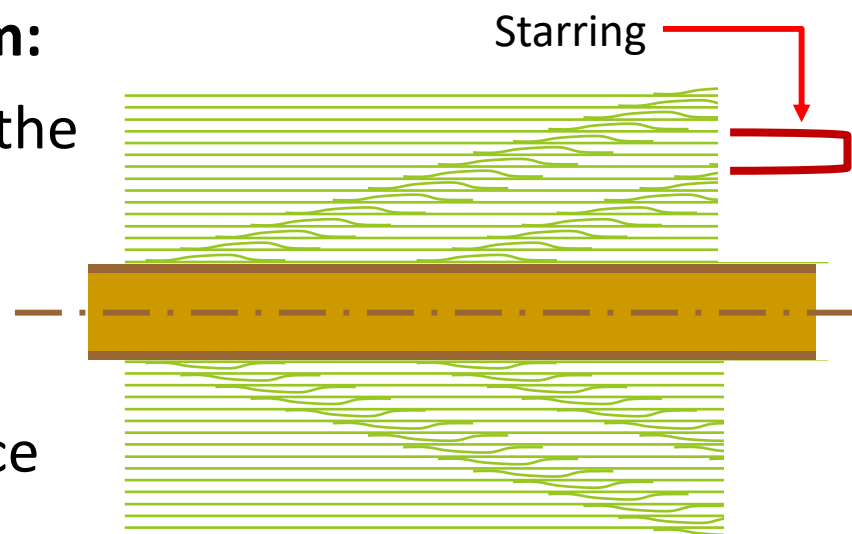
Oscillation and Radial Variations

Discussion of this starring mechanism:

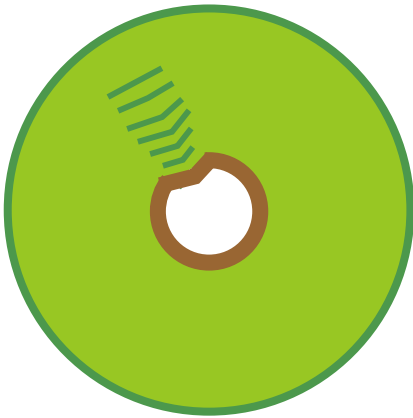
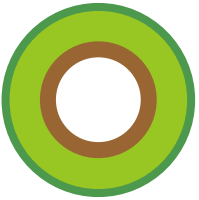
Q: Based on this mechanism, what is the remedy?

A: As with any starring defect, the goal is to either start the layer on the winding roll with more strain or reduce the loss of strain within the roll that leads to buckling.

- Increasing starting strain through higher tension (created by center torque).
- Decrease profile variations.
- Avoid oscillating largest profile variations to roll edges.



Radial Error: Crushed Cores



Mechanism:

- Material properties likely promote tourniquet effect, increasing core pressure with every additional wrap.
- Eventually, pressure on core exceeds cylindrical shell buckling criteria.
- Post-wind dimensional changes can also contribute to core crushing, such as viscoelastic strain recovery or thermal expansion (in heat-aged rolls).

Remedies:

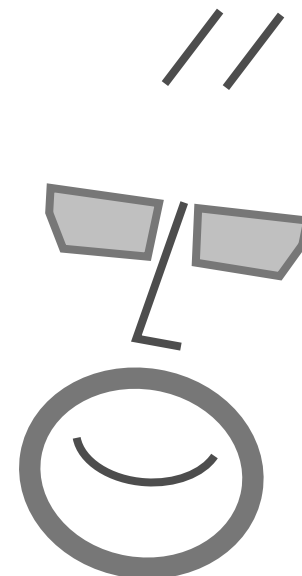
- Increase core stiffness. Increasing diameter is the strongest factor (D^4), but increasing core modulus or thickness will also help.
- Decrease pressure buildup by winding shorter rolls or starting with a larger core.
- Change process or product to improve dimensional stability.

Winding Solutions Qs

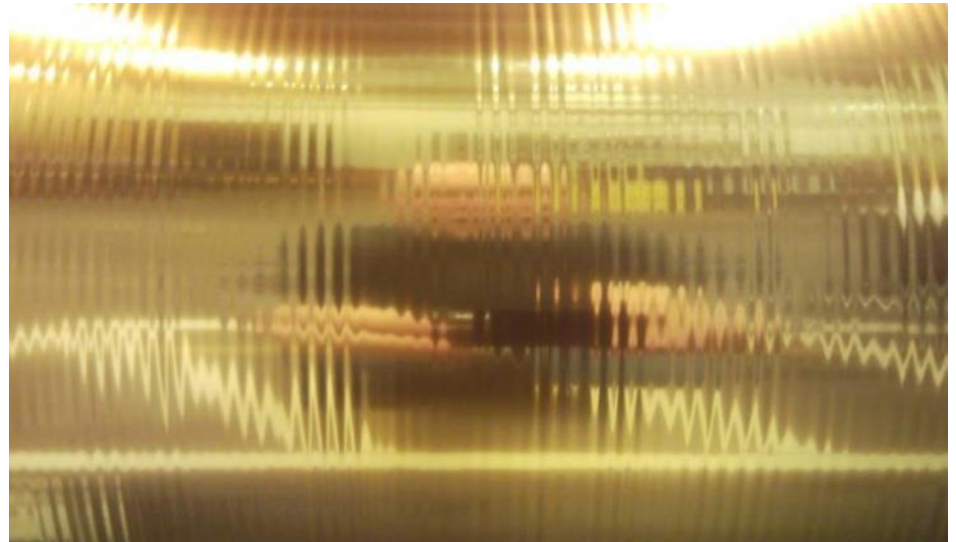
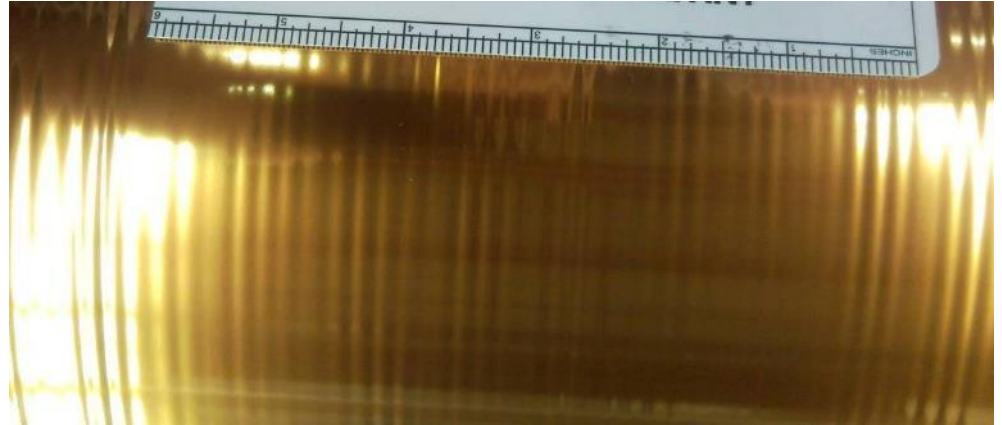
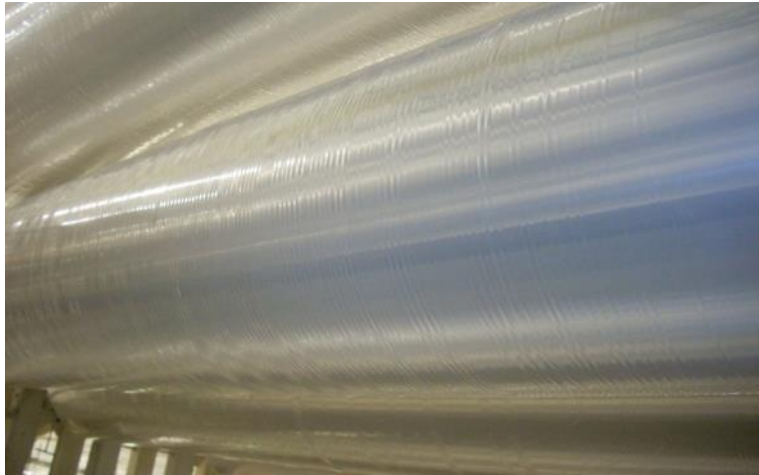
*What are MD lines, MD ridges,
or tin-canning?*



*What is the
remedy?*



MD Buckles (Tin-Canning)

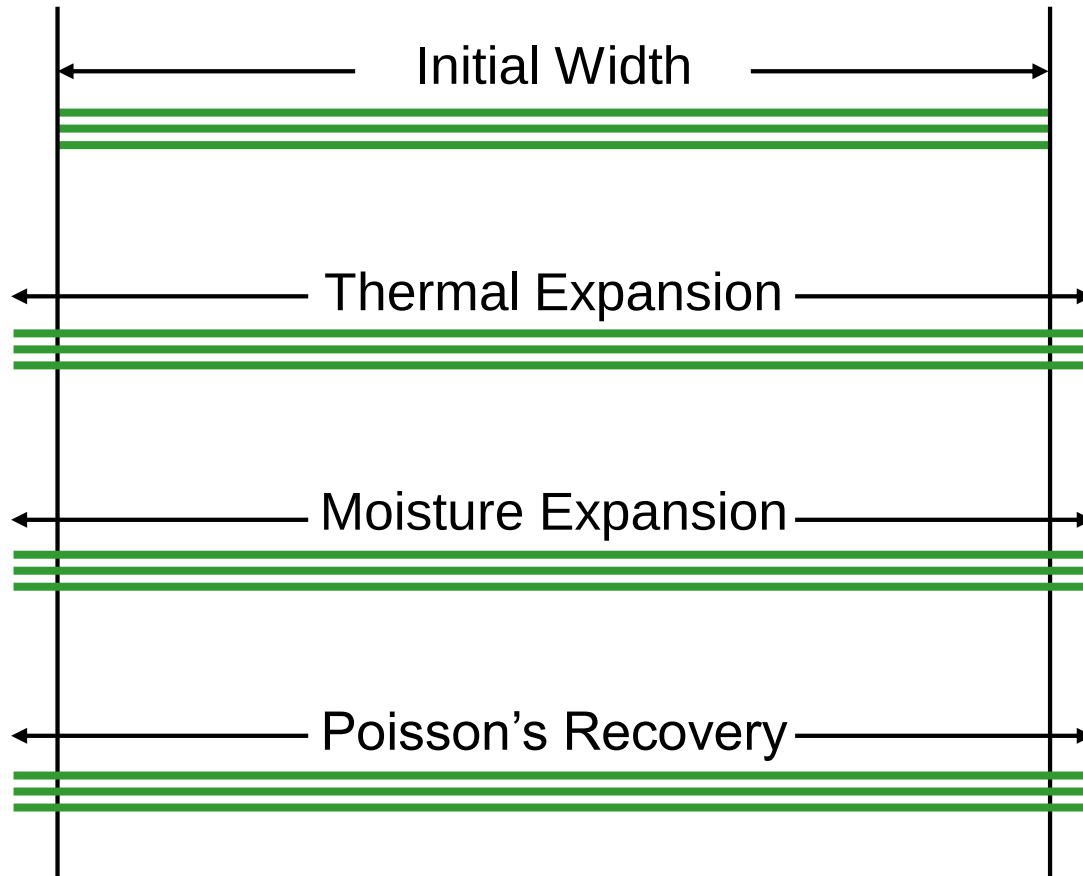


How Do MD Lines Form?

MD buckles form when the layers within roll expand their width while constrained from sliding laterally.



What Causes Width Expansion?



$$\frac{\delta w}{w} = C_{\alpha} (\Delta Temp)$$

$$\frac{\delta w}{w} = C_h (\Delta \% Moisture)$$

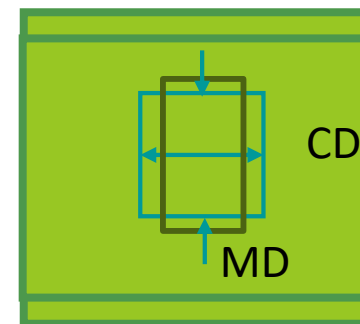
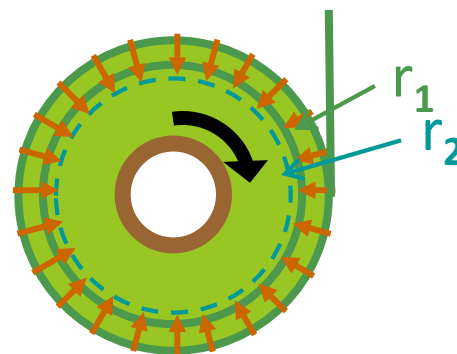
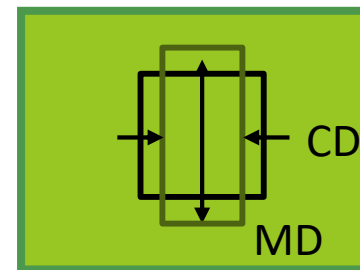
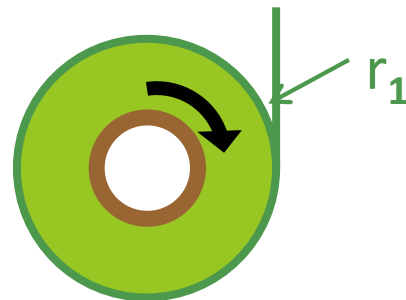
$$\frac{\delta w}{w} = \frac{\nu}{tE} (\Delta Tension)$$

Mechanics of MD Buckles (Tin-Canning)

Consider a tensioned layer that joins the winding roll at a given radius, r_1 with $\varepsilon_{MD} > 0$ and $\varepsilon_{CD} < 0$.

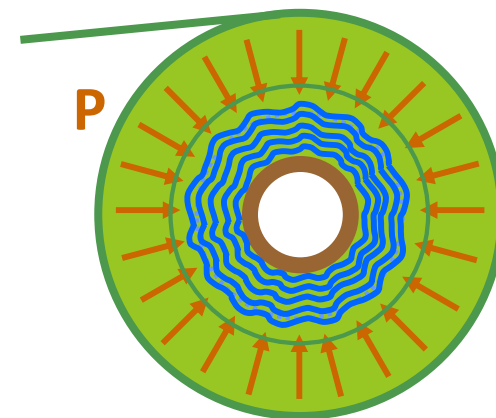
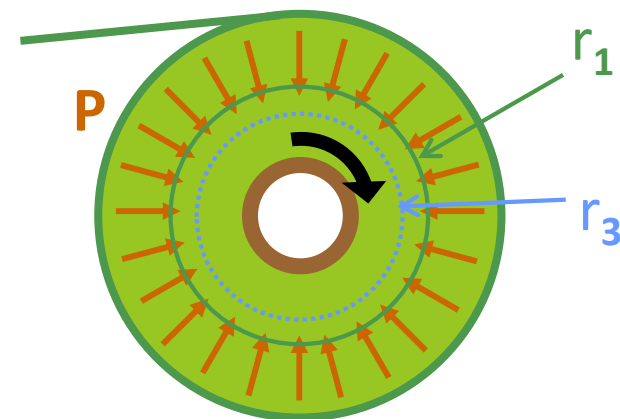
With additional layers, it will shift to a smaller radius, r_2 , losing MD strain and regaining width.

If width recover is constrained, MD buckles form.



Mechanics of MD Buckles (Tin-Canning)

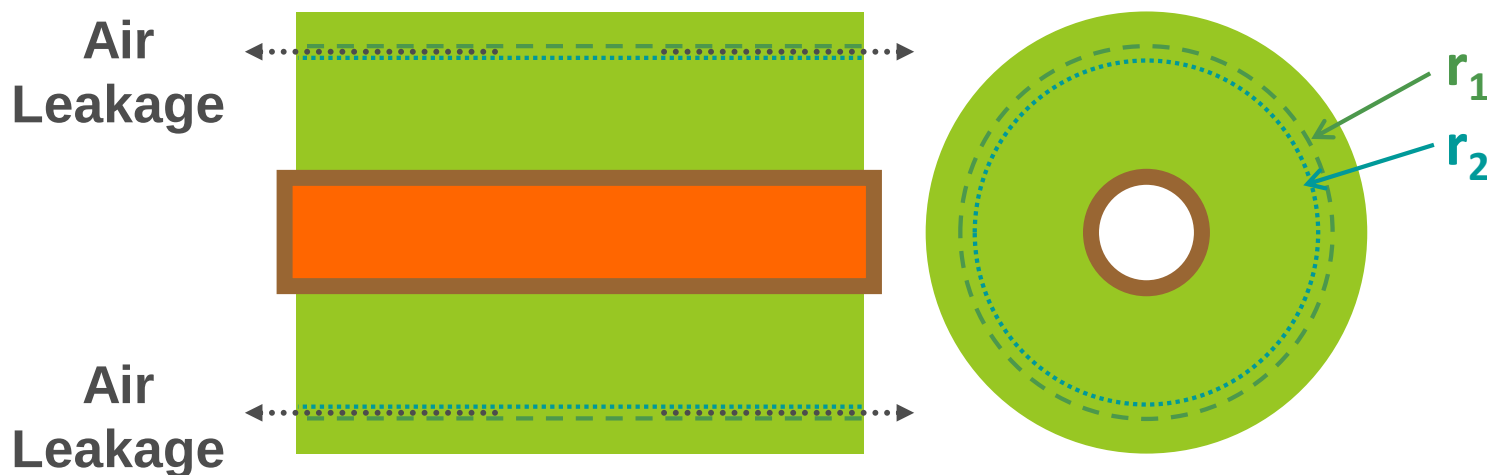
If the percent radial compression exceeds the initial wound-on strain, layers in MD hoop compression can buckle and form CD buckles (crossbuckles).



Air Side Leakage & Strain Change

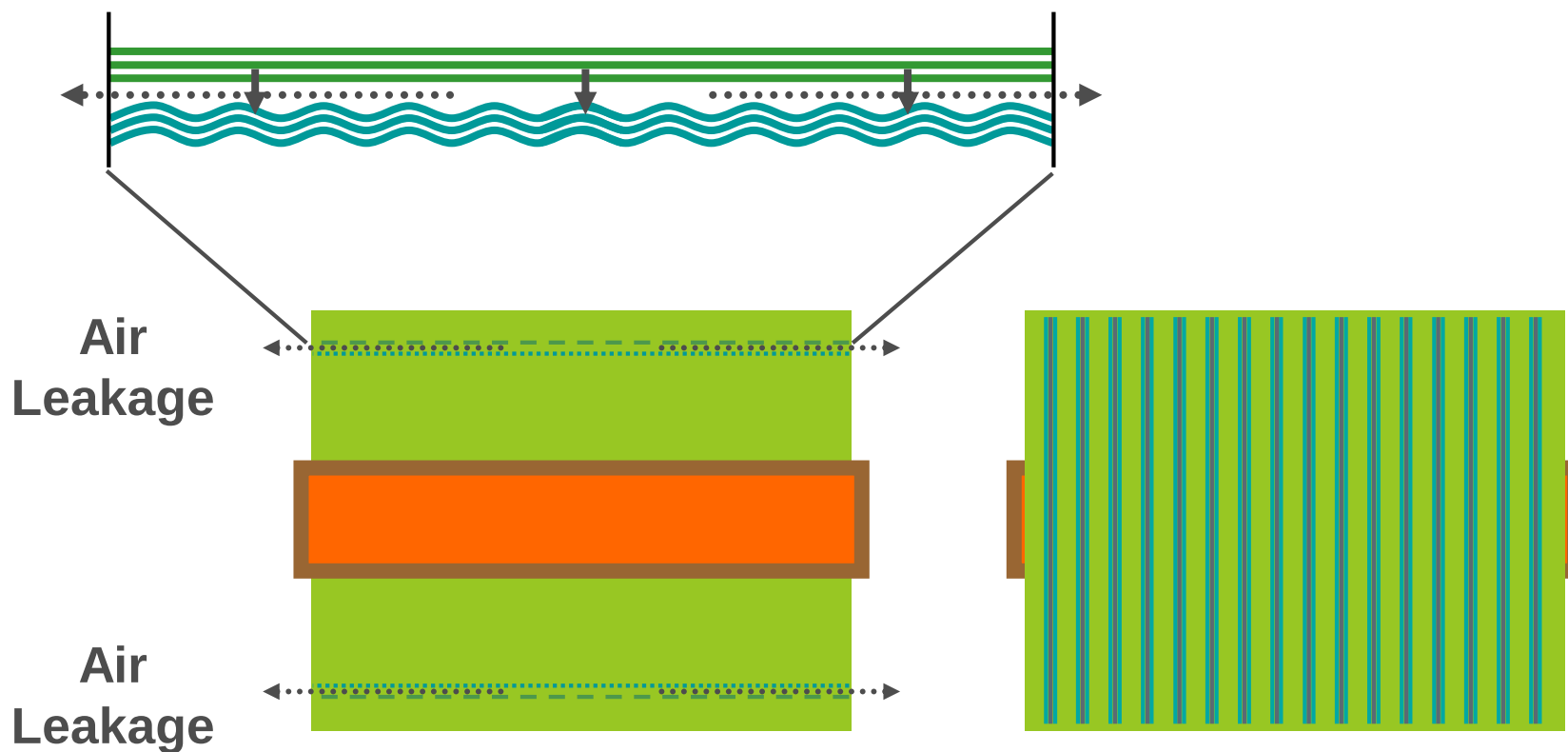
Roll wound at high speed have entrained air within the roll that will escape over time.

The air leakage will cause layers to shift from initial radius (r_1) to a smaller radius (r_2), reducing MD strain. As MD strain drops, the Poisson's effect width contraction attempts to recover.



Air Side Leakage & Strain Change

Where CD strain is constrained, CD stress will develop, leading to CD buckling and MD lines.



Mechanisms of MD Buckles (Tin-Canning)



Average Air Entrainment

- Speed
- Tension or Nip Load
- Radii (Roll and Nip)
- Air Viscosity (constant)

Other Air Entrainment Factors

- Core Deflection
- Nip Roller Deflection
- Roll Transfer Wrinkles
- Winding Wrinkles
- CD Diameter Variations

In-Roll Air Retention Volume

- Softbands
- Web Roughness

Air Escape

- Air Compression
- Roll Pressure
- Time
- Roll Width

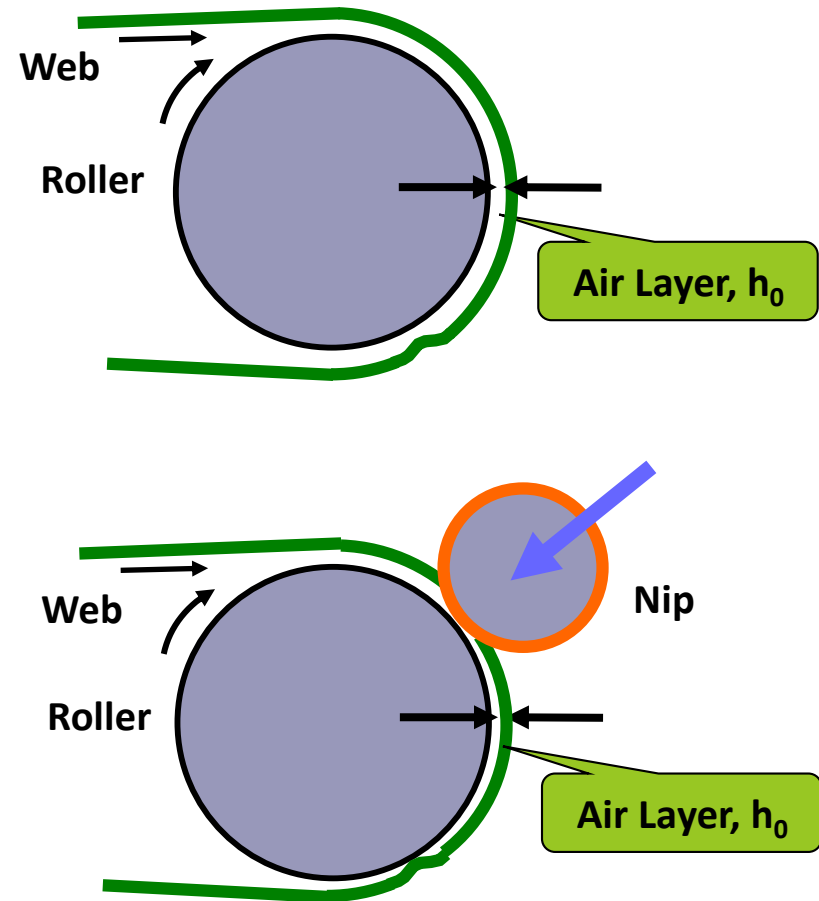
Understanding Tin-Canning Mechanics

Average Air Entrainment Increases with:

- Speed
- Roll Radius
- Nip Roller Radius

Decreases with:

- Tension
- Nip Load
- Air Viscosity*



* Air viscosity is mostly constant, except in vacuum.

Understanding Tin-Canning Mechanics

Other Air Entrainment Factors

CD Tension and Nip Load Variations

- Deflection: Core and Nip Roller
Allows more air in center of winding roll.
- CD Diameter Variations
Allows more air in low diameter lanes of winding roll.
- Wrinkles: Roll Transfer and Wound-In
Creates air pockets in open space of wrinkles pleats.

Understanding Tin-Canning Mechanics

In-Roll Air Retention Volume

- Web Roughness or Texture
Minimum air retention is in the valleys of surface asperities or textures.
- Softbands
Air pockets can form in the valleys between hardbands.
- Wrinkles
Air pockets can reside in pleats of wrinkled layers.

Understanding Tin-Canning Mechanics

Air Escape (How much air will escape roll?)

- Roll Pressure and Air Compression

If air compresses under in-roll pressure, it takes less volume to retain it.

- Time

Pressure will cause air flow, but the tortuous path out the narrow channels between layers can take time.

- Roll Width

Wider rolls have longer distances and require more time for air to escape.

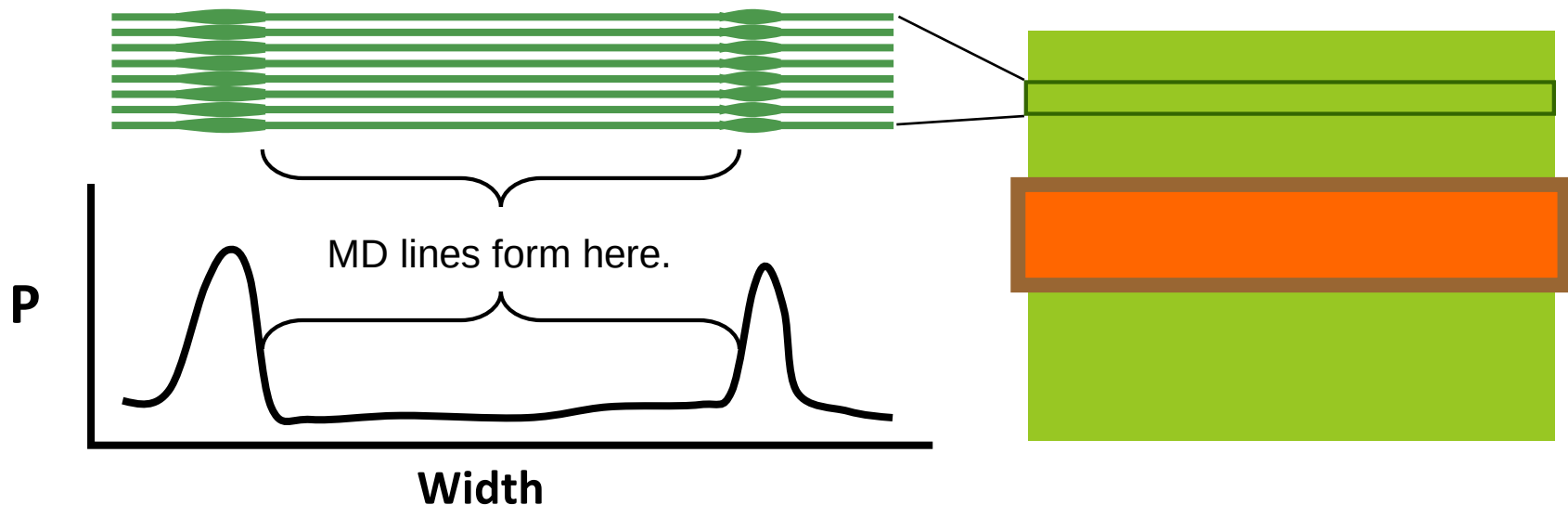
Wider rolls are more likely to have significant effects of deflection, wrinkling, and gauge bands.

What Constrains Necking Recovery?

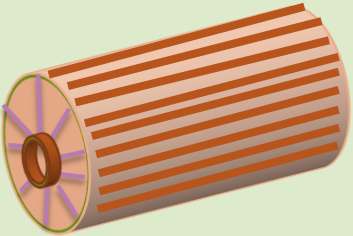
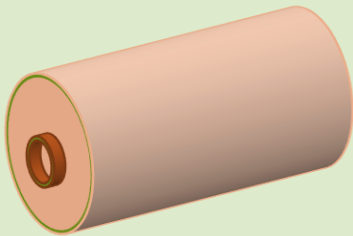
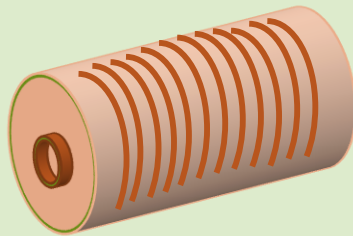
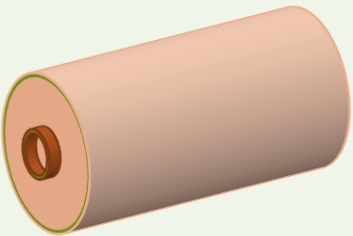
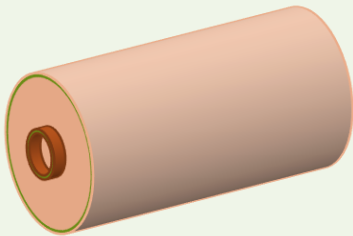
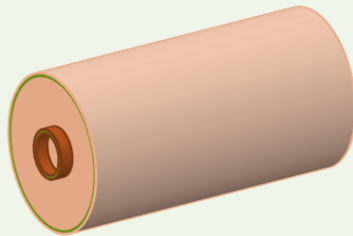
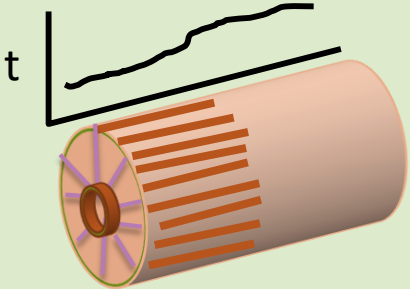
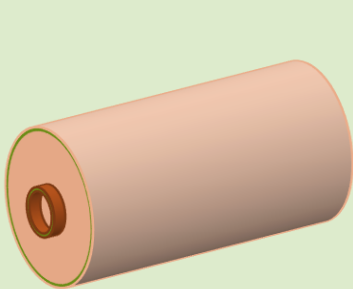
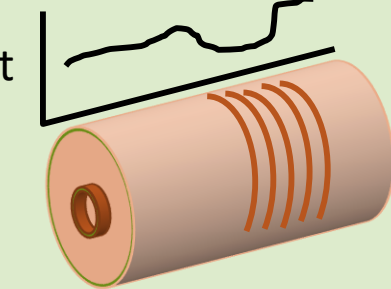
If the entire roll would expand, the web would freely recovery from necking. *What constrains necking recovery?*

Answer 1: Air escapes from roll edges first, locking the outer regions of the from lateral slippage.

Answer 2: Thickness variations create high pressure bands that may constrain lateral motion.



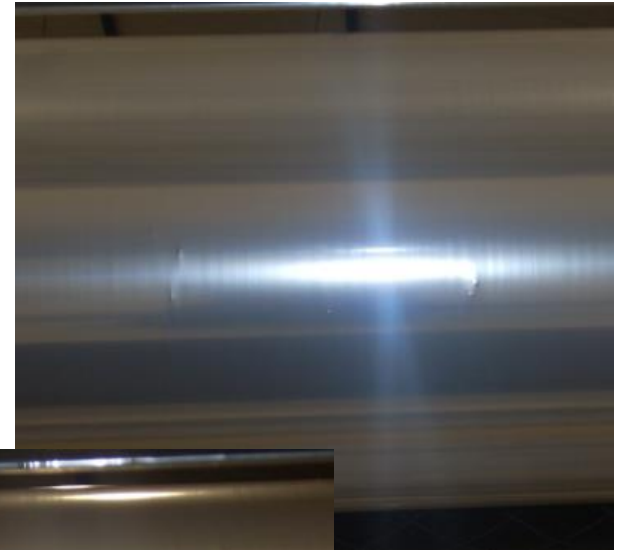
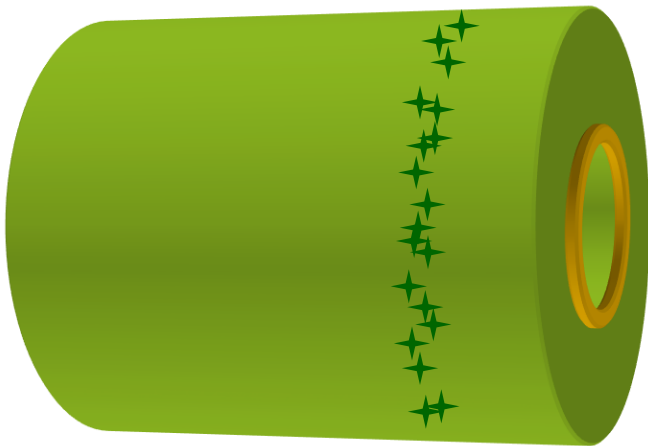
In-Roll Buckling vs. WOT, Air, Thickness Profile

	Low Winding Tension	Medium Winding Tension	High Winding Tension
Too Much Air, Uniform Thickness			
Well Managed Air, Uniform Thickness			
High Thickness Variations			

Slip Knot and Convolution Buckling

Slip knots are buckled regions of a winding roll where the shear between sticking and slipping contact buckles the web.

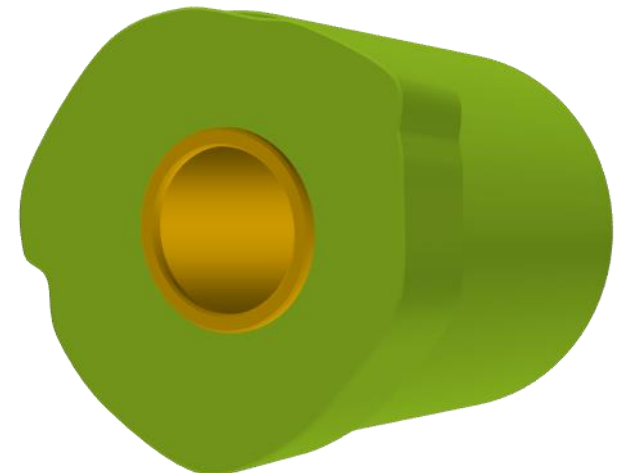
Slip knots can form from slitting debris or surface imperfections (such as poorly milled slip additives).



Slip Knot and Convolution Buckling Mechanism

Convolution buckling (buckling associated with the revolving roll)

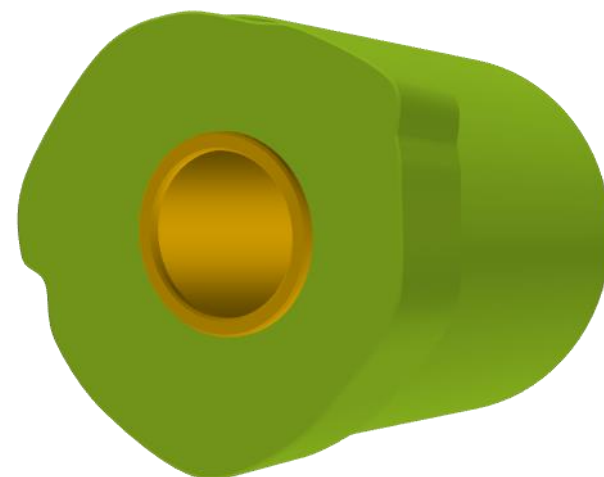
- Most common in thick optical films with ultra low roughness and little or no slip additives.
- These films have 'friction as a function of pressure.'
- The product side A to B contact slides well at low pressure, but sticks at high pressures.
- Any local high pressure on the outside of a roll (including small particles) cause local sticking in a finite local position, but slip elsewhere, leading to shear and local buckling.
- A small buckle will magnify if the next layer 'sticks' at this bump, snowballing to a larger size.



Slip Knot and Convolution Buckling Solutions

Most films are designed to avoid the problematic 'friction as a function of pressure.' Options include:

- Most polyester films have finely milled mineral additives that change the surface roughness during orientation.
- Many polymer films have chemistry in the polymer blends to change surface physics.
- Some films are embossed or cast on a roller with engineered surface roughness, changing the real area of contact and friction.
- Thinner films conform around bumps easier and are less likely to move from 'knots' to 'buckles.'
- Even small upsets can trigger or magnify this defect, including core and chuck eccentricity, tension upsets, and thickness variations.

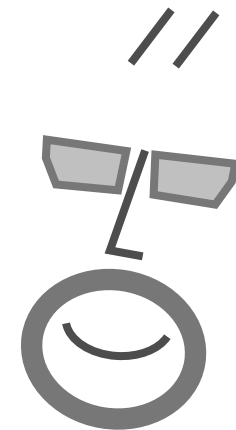


6.4 Roll Structure and Web Quality

- Wound-In Stresses
- Lateral Errors
- In-Roll Buckling
- Transverse Variations 
(Gauge Bands and Bagginess)

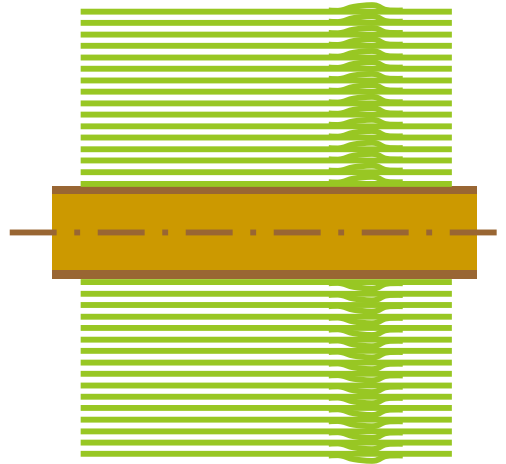
Winding Solutions Qs

How do crossweb thickness variations affect winding and wound roll structure?

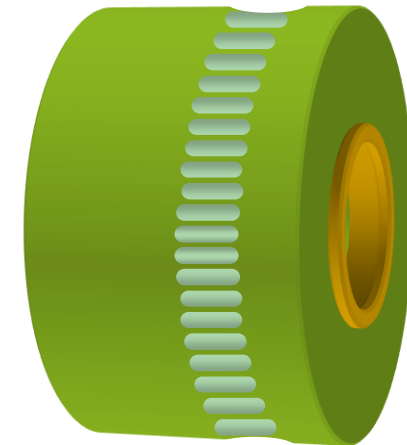
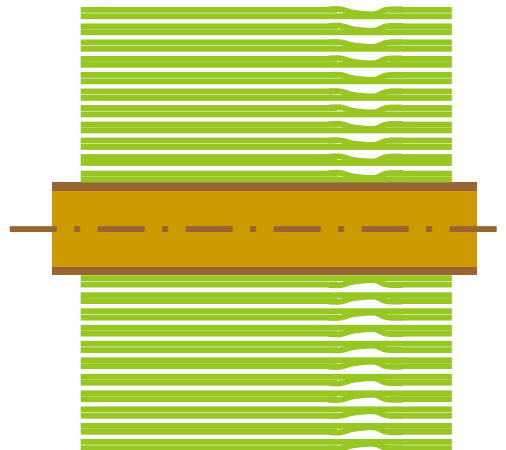


Softbands and Hardbands

Hardbands are magnifications of thick web or coating thickness variations.

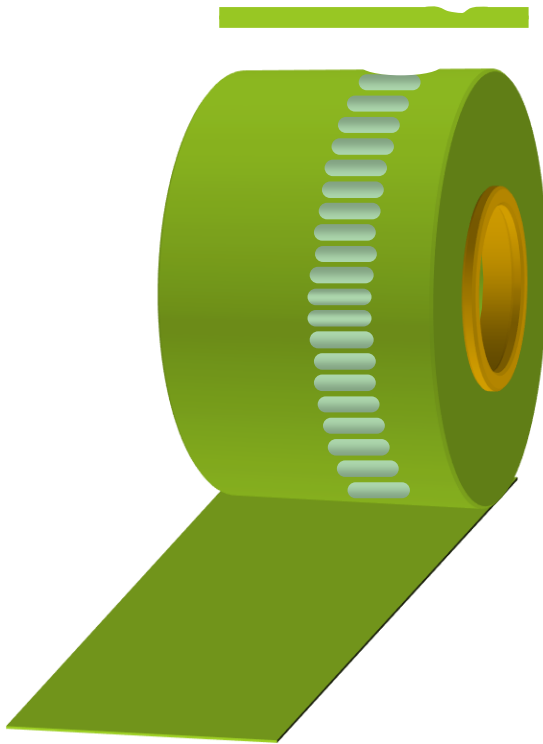


Softbands are magnifications of thin web or coating thickness variations.

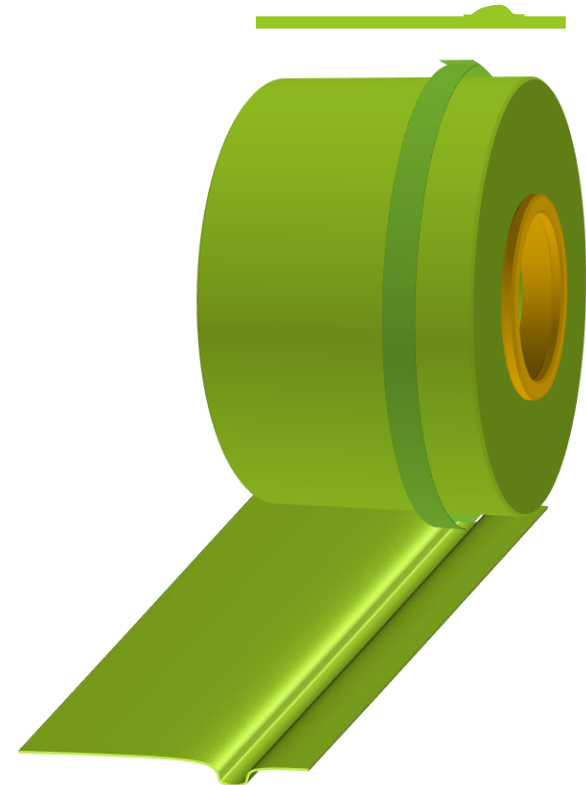


Thickness Variations and Bagginess

Thin lanes are less likely to create permanent deformation.



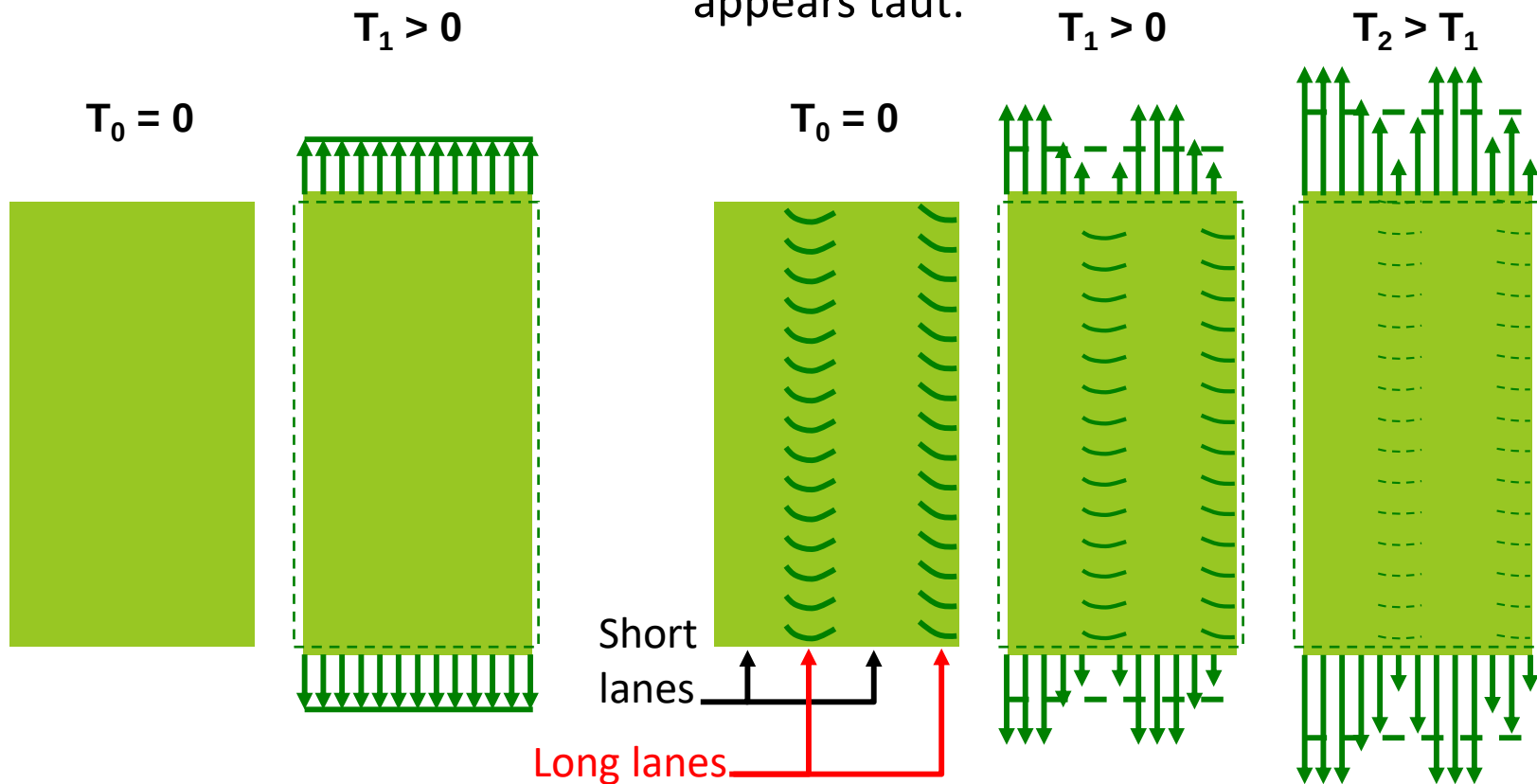
Thick lanes are more likely to deform the web, creating bagginess.



Tensioning Baggy Webs

The ideal web carries tension uniformly across the web width.

For an baggy web, tension loads and elongates short lanes first. When short lanes are stretched to equal the long lanes, the web appears taut.



Baggy Web

Asymmetrical baggy web will have a curve or skew viewed over long lengths.



If asymmetrical baggy web is forced to follow a straight line, the long side will buckle up due to its extra length.

How to Make Baggy Web:



1. Make a web with crossweb thickness variations and stiff properties in the thickness direction.
2. Magnify the thickness variations by winding with no lateral oscillation and ensure the final roll is many times the diameter of the core.
3. Wind the roll with high tension, no taper, and at low speed to avoid entrained air softening the roll tightness.

How to Make Baggy Web:



4. Ensure the product will be easy to yield or flow visco-elastically by promoting high moisture in papers and elevated temperatures in films.
5. Store the roll for a long time. For films, make sure to store them in a hot environment like a summer warehouse or semi trailer.
6. Unwind the roll is enjoy the bagginess. Make sure you don't measure it.



Remedies for Baggy Webs

1. Reduce Thickness Variations
2. Reduce Roll Tightness
 - a. Lower tension, increase tension taper
 - b. Reduce stack modulus (increase roughness, wind with soft interleave)
 - c. Reduce roll buildup ratio (increase core size)
3. Control Roll Aging – Relax polymer on-line, relax polymer in roll form and rewind, convert quickly.
4. Knurl Edges, Shifting Stress and Distortion to Edges
5. Thickness oscillation at winding
6. Decrease product stack modulus

How to Define Thickness Profile?

Percent variation is the most common profile parameter.

$$\% \text{ variation} = (\text{maximum} - \text{minimum}) / (\text{average})$$

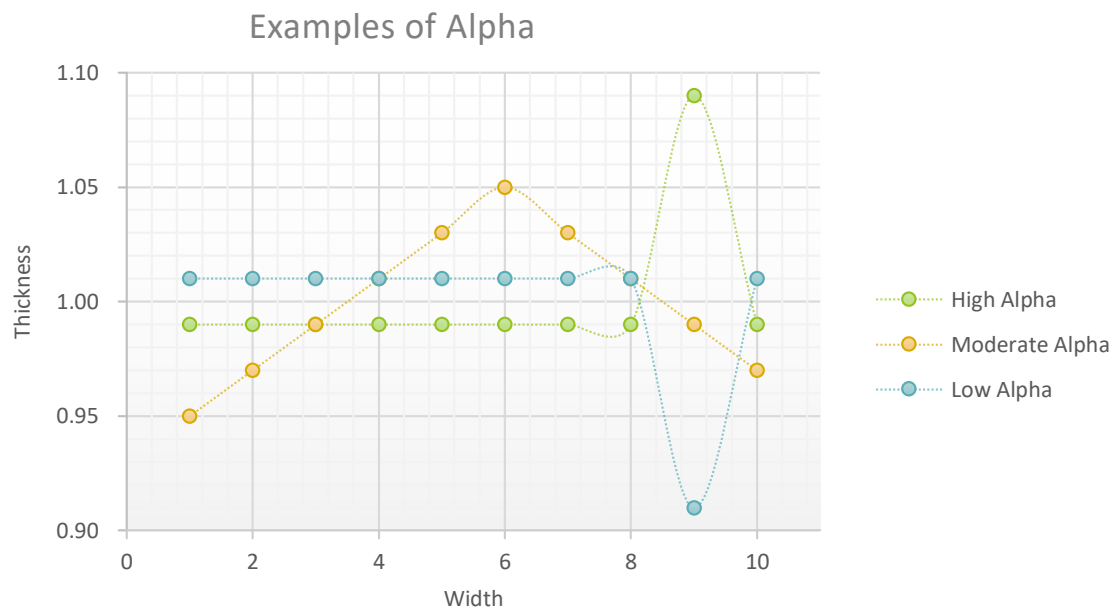
Other thickness profile parameters may be better predictors or defects.

- ▶ ‘Mountains’ like create more defects than equivalent variations in ‘valleys.’
- ▶ Profile features may cause more defects if they are:
 - ▶ Near the core
 - ▶ Near edges
 - ▶ Have greater slope or less width
 - ▶ Are not near other features (solo mountains)

Alpha vs. Range

Alpha is a thickness profile parameter useful in differentiating profiles more likely to create bagginess in winding. Alpha emphasizes 'up' features in thickness profile over 'down' features.

This graph shows three thickness profiles of equal average and range, but differing alphas. The first curve with the highest alpha is mostly like to create sever bagginess.



	%Range	%Max	alpha
High Alpha	0.1	0.09	0.10
Moderate Alpha	0.1	0.01	0.02
Low Alpha	0.1	0.05	0.08

Radial Variations from Thickness Profile

Dr. Lien Struik of TNO (Netherlands) calculated the percent radial variations that are create from a given percent thickness variation.

'a' is the ratio of thickness variation to average thickness.

$$\left(1 + \frac{b}{\varepsilon}\right)^2 = e^{K_2(a - (1+v)b)}$$

'b' is the ratio of radial variations to average radius.

a = thickness variations / average thickness

b = radial variation / radius

ε = strain of winding

ν = web Poisson's ratio

K_2 = radial modulus compressibility constant

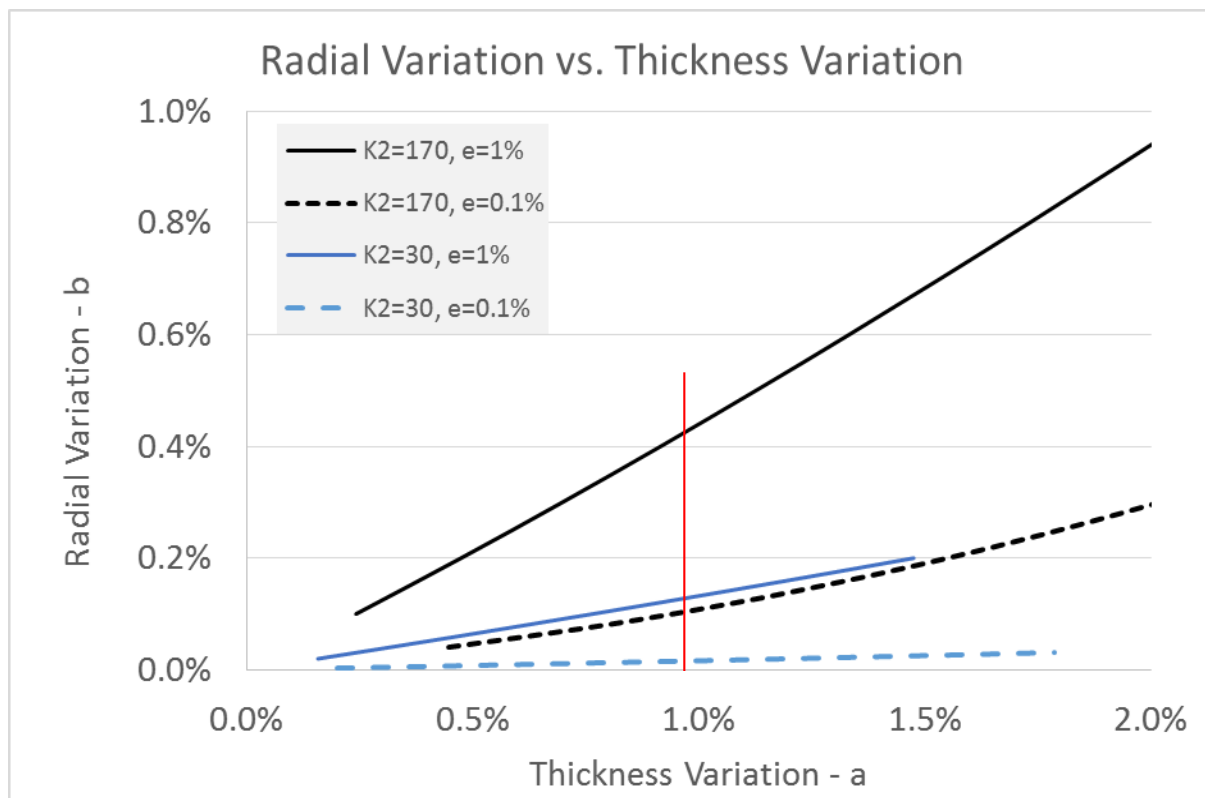
$$a = \frac{\ln \left[\left(1 + \frac{b}{\varepsilon}\right)^2 \right]}{K_2} + (1 + \nu)b$$

Radial Variations from Thickness Profile

K2 compression constant
= 170 for fine coated paper
and PE films
= 30 for newsprint paper and
PET films

A 2% thickness variation will
create a 1% or 0.3% radial
variation in fine coated paper
or PE film rolls wound with
high strain (1%) or low strain,
respectively.

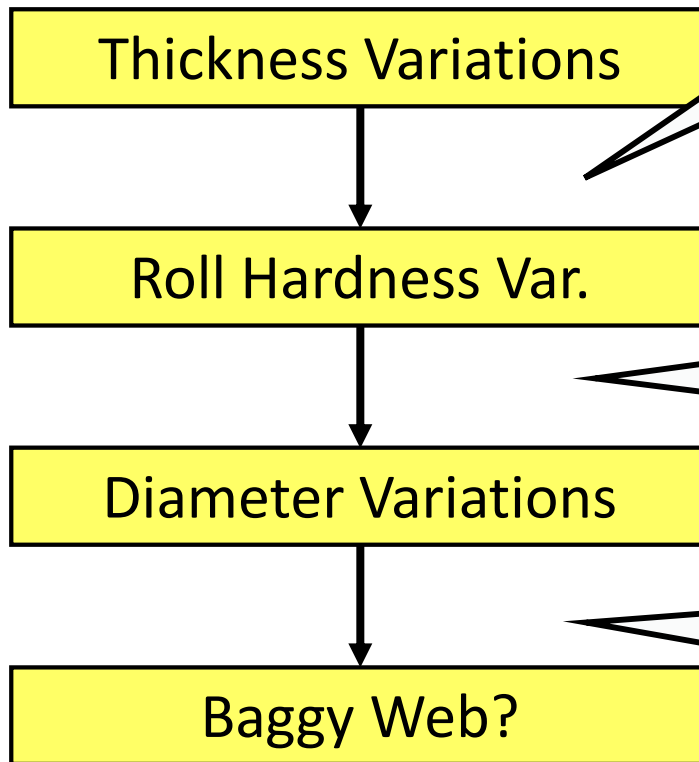
Much lower radial variation
will be seen in winding
newsprint paper or PET films
(0.3 to <0.05%) for high and
low winding strains.



Example: A 1% thickness variation will create a 0.4% radial variation in tightly wound PE, but less than 0.1% variation at lower tension or in newsprint paper.

Creating Baggy Web

Thickness variations do not automatically create baggy web.



Materials that compress in the stack direction will create less diameter variation for a given thickness variation.

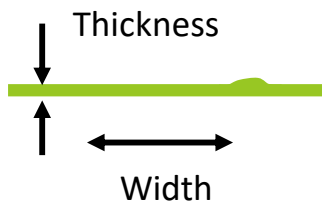
Even in small roll buildups, thick lanes will create hardness variations, but may not create significant diameter variations.

Through experiments, look to correlate the percent diameter variation and winding stress to where bagginess is seen.

Thickness Variations + Winding = Bagginess

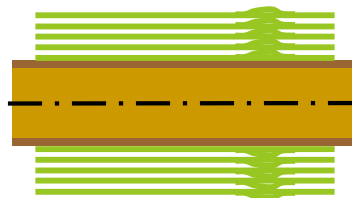
CD thickness variations before winding

- Many processes will have CD thickness variation, usually slowing changing relative to winding times.



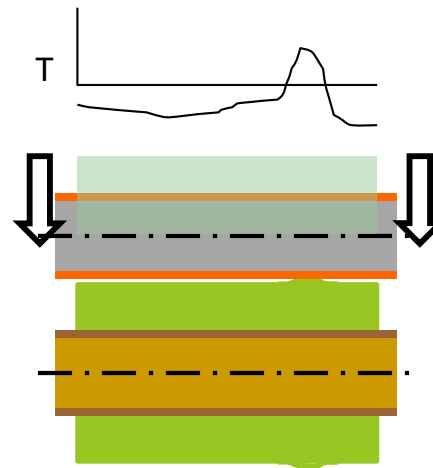
CD diameter variations in winding roll

- Starting with the core, layers accumulate to create diameter variations in the winding roll.
- Stack modulus, roughness, and air entrainment determine how radial variations become diameter variations.



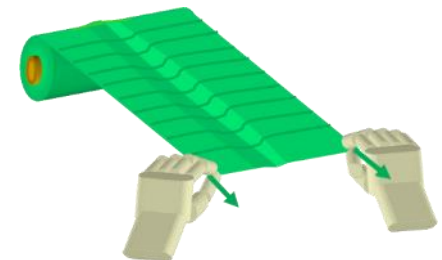
CD winding stress variations

- The growing CD diameter variations shift tension from low gauge, low diameter lanes to high gauge, high diameter lanes.



Creep and visco-elastic strains

- Creep or visco-elastic strain changes occur within wound rolls at strains much lower than typical short tension-elongation testing would predict due to the long times of roll aging.



Action Plan to Reduce Bagginess

1: Measure Defect Baseline

- Planarity gauge
- Measure defect magnitude (good vs. bad rolls, your product vs. competition, map defect vs. roll radial and CD position .

2: Collect Product Data

- Measure thickness vs. width and length
- Measure stack modulus, visco-elasticity, and roughness
- Measure core properties, roll geometry

3: Collect Process Data

- Details on winder design (tension control, gap/nip roller system)
- Details on tension and nip control.
- Details on roll start process, accel/decel, oscillation

4: Roll Storage and Aging

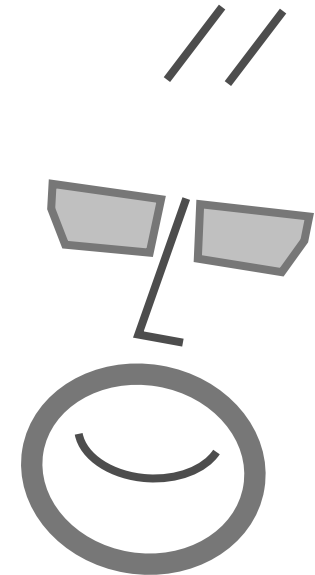
- Determine roll aging / storage times and conditions
- Identify, model, and empirically verify appropriate material and process dynamics (moisture, heat, visco-elasticity, polymer relaxation, air escape)

5: Reduce Process Sensitivity

- For Bagginess: Eliminate process nips, apply spreader rollers, increase process tension, use skewing roller or nips
- For Thickness: Improve uniformity, decrease tension, modify winding process (e.g., gap vs. nip)

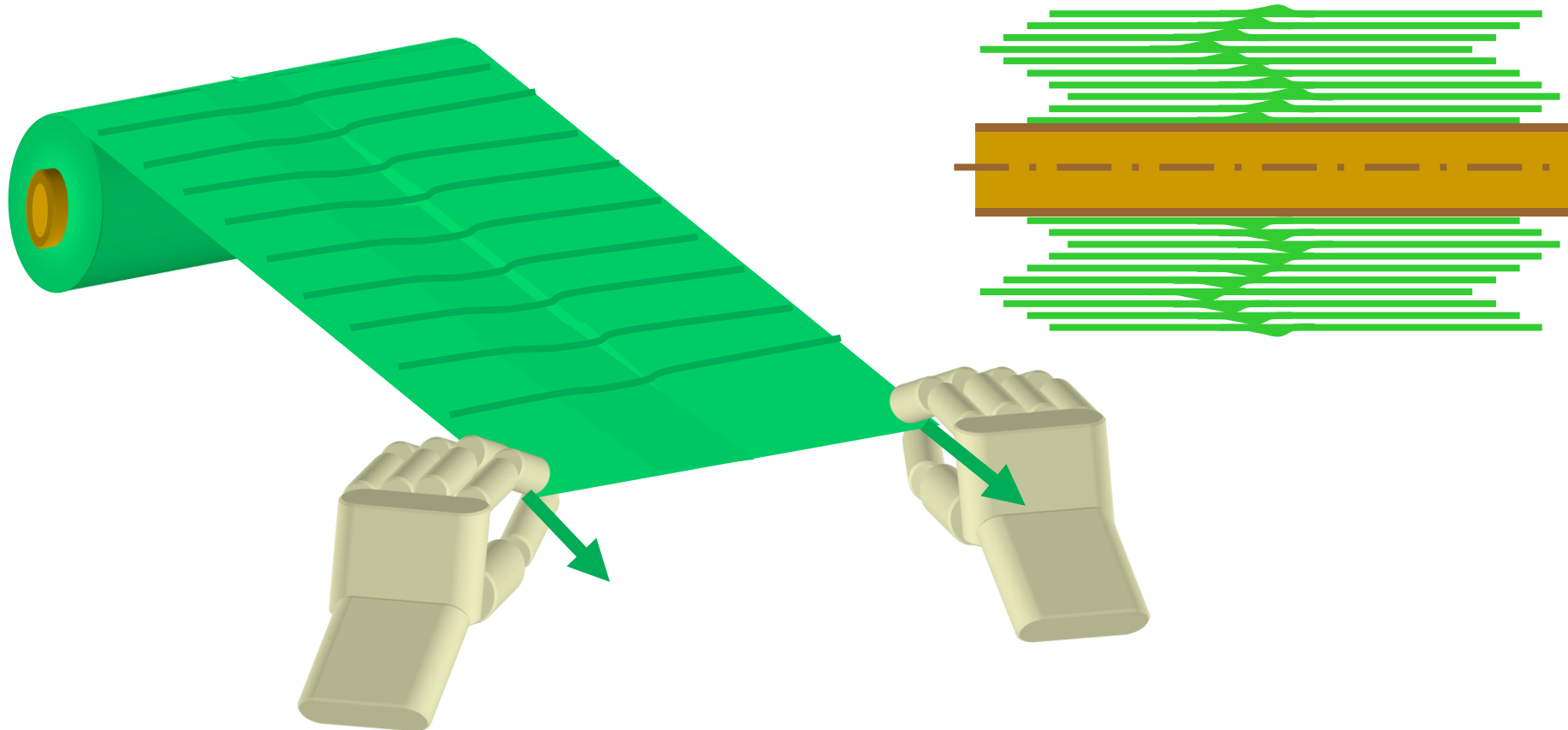
Winding Solutions Qs

What is oscillation winding
and why is it done?



Thickness Profile Oscillation

Winder oscillation moves thick lanes laterally on a triangle wave pattern, reducing wound roll stress and resulting bagginess.



Thickness Oscillation

Method 1: Staggered Edge Winding (used when trim is non-recyclable)

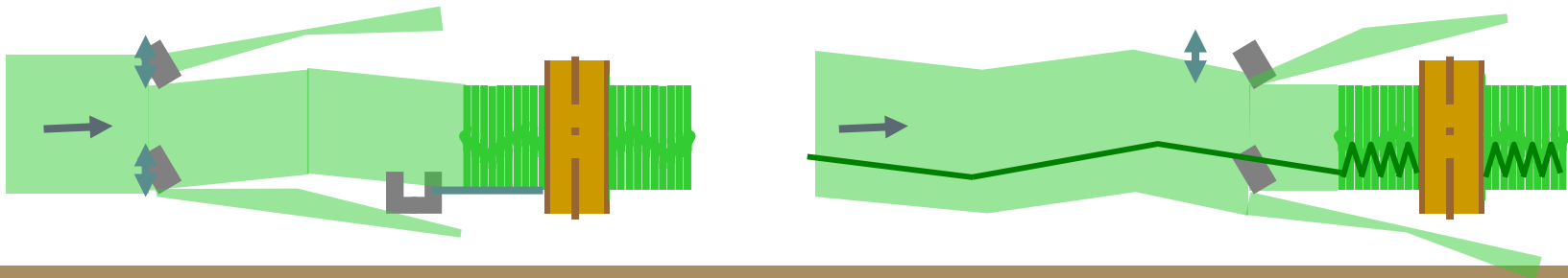
Oscillate rewind edge sensor relative to the core.

(a.k.a. wiggle winding.)



Method 2: Oscillating Trim Width (used when trim is recyclable)

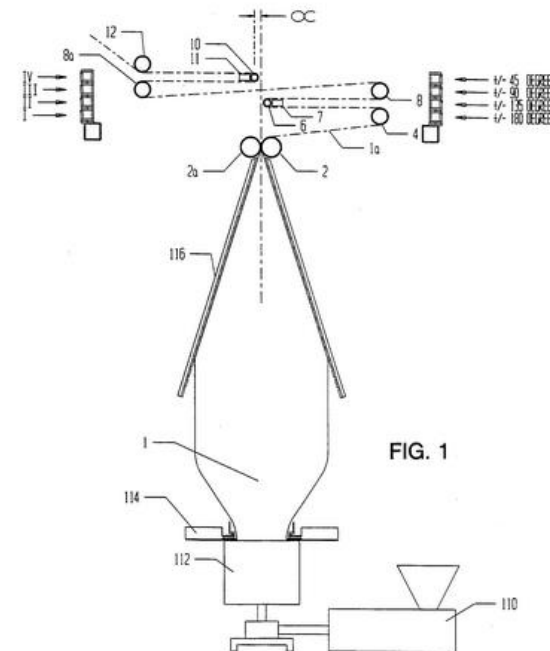
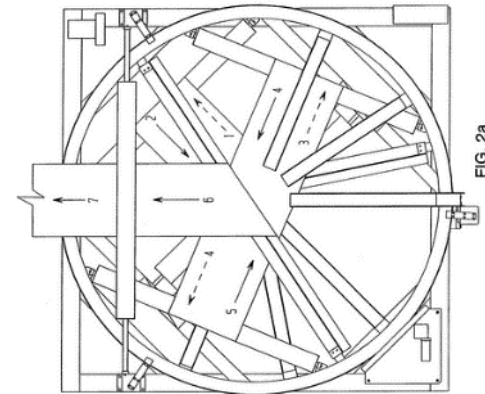
Oscillate trim knives then rewind guide or oscillate upstream of fixed trim knives for a straight roll edge (but more trim waste).



Blown Film Oscillation

Blown film lines are able to wind 'good' rolls with more thickness profile variation than flat processes by using nearly 360-degree oscillation.

Gauge bands can be fully oscillated across a winding roll, greatly reducing the roll diameter variations.

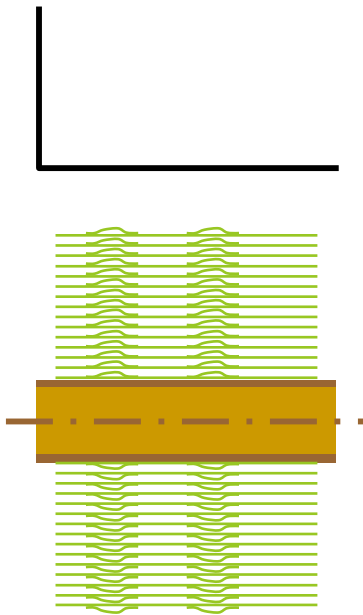


In an oscillatory hauloff mechanism for a blown film line

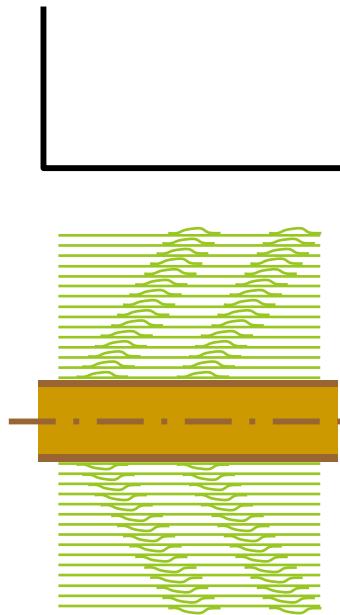
US 5360328 A

Oscillation and Radial Variations

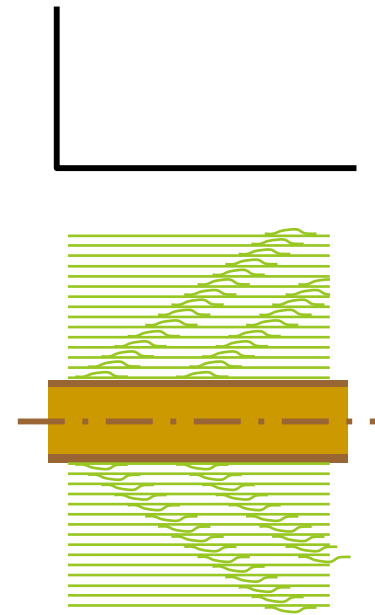
No oscillation will
create the largest
CD radial variations.



Slow oscillation will
reduce the
CD radial variations



Faster oscillation
will have the least
CD radial variations



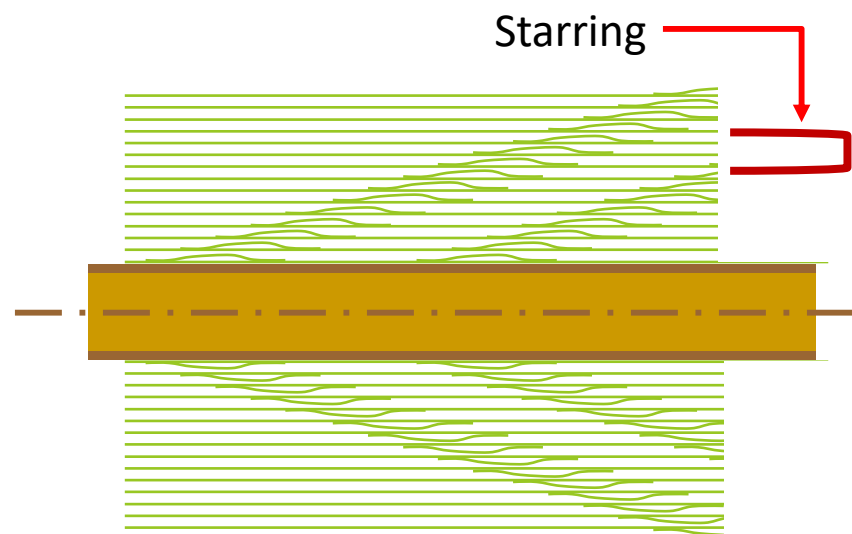
Oscillation and Radial Variations

How intermittent starring defect may form:

The thickness profile has high and low features.

Oscillation moves the 'up' thick and 'down' thin features across the winding roll.

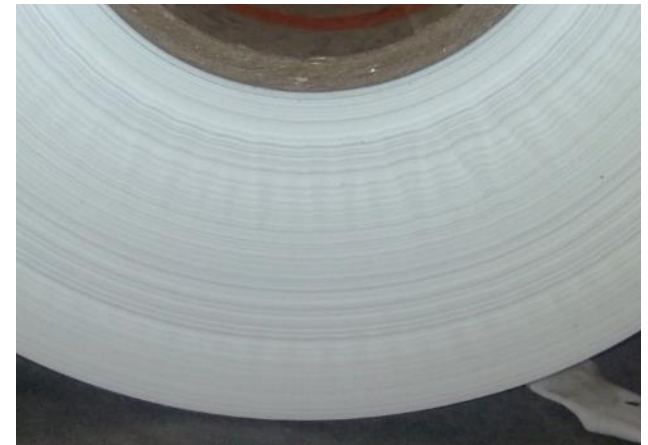
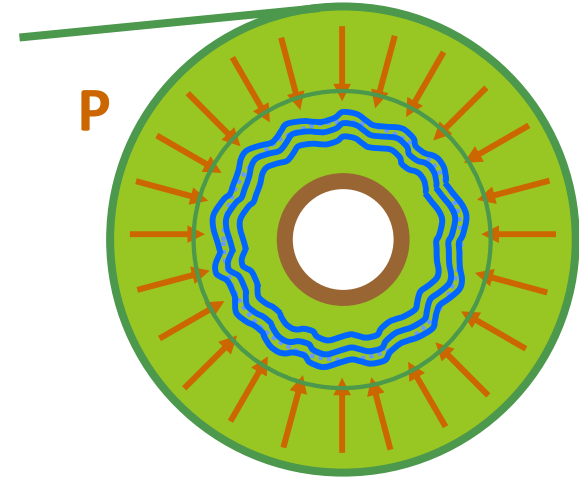
Starring forms when a larger than average low feature passes through the edge of the roll followed by a high feature.



Oscillation and Radial Variations

Wherever a low 'down' feature builds on itself, it forms a local CD band of lower than average diameter and lower than average winding stress and strain. Away from the edge of a roll, the low features can entrap air, but at the edges, the air can escape.

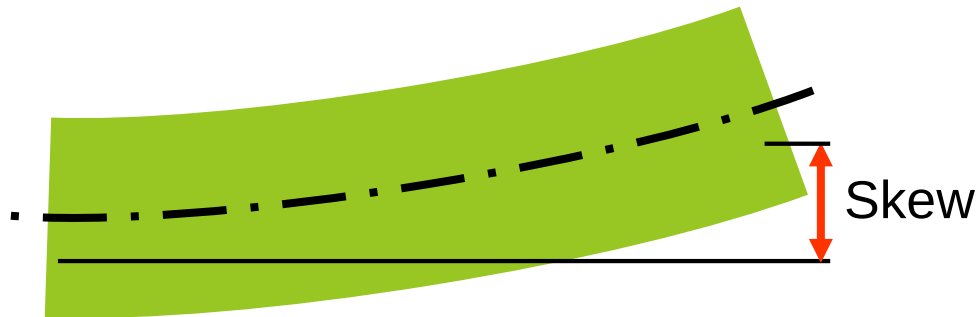
When the low thickness bands move off the roll and high thickness bands follow, the stress of the added layers will compress the thin layers into compression and buckling.



Measuring Baggy Webs (Offline)

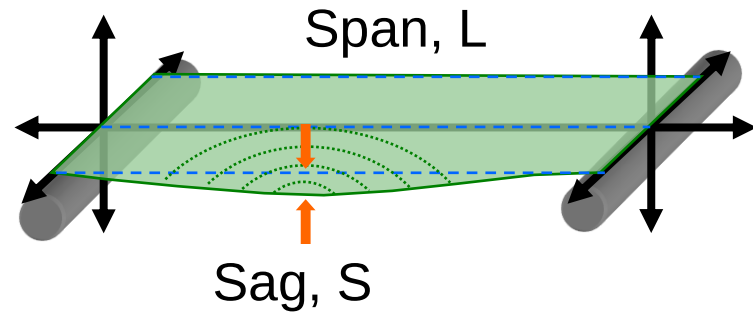
Sweep Out Skew Test:

A web with a dominant baggy edge will have a measurable long-term curvature.



Sag Test:

At a fixed tension, measure the crossweb sag variation between two parallel rollers.

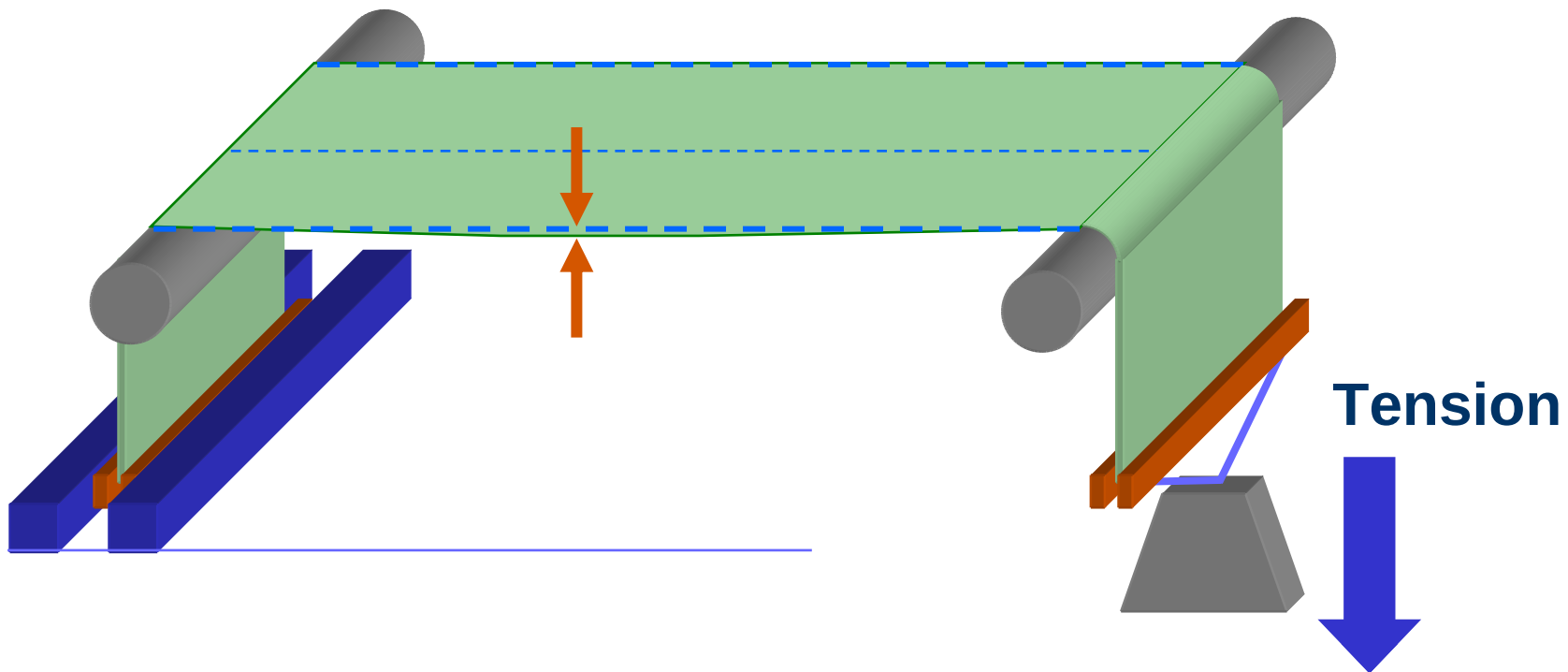


$$\Delta \varepsilon = \frac{S^2}{2L^2}$$

Measuring Baggy Webs (Offline)

Tension Pull Out Test:

Measure the tension to pull the web to near flatness (e.g. less than 1/4" droop).



Measuring Baggy Webs (Offline)

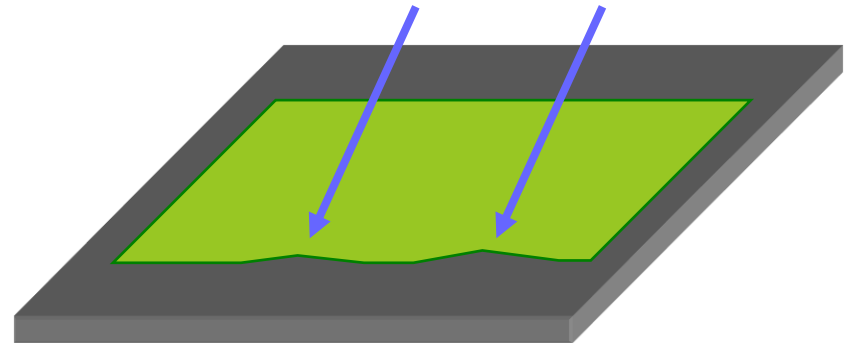
Slit Length Differential:

1. Draw parallel crossweb reference lines.
2. Slit the web into strips.
3. Measure length of strips.



Lay Flat Test:

Lay flat test, place sheet on flat plate and measure ripple or curl deviations from planar.





Measuring Flatness or Planarity



Scanning gantry of ultrasonic sensors

Web Planarity Gauge

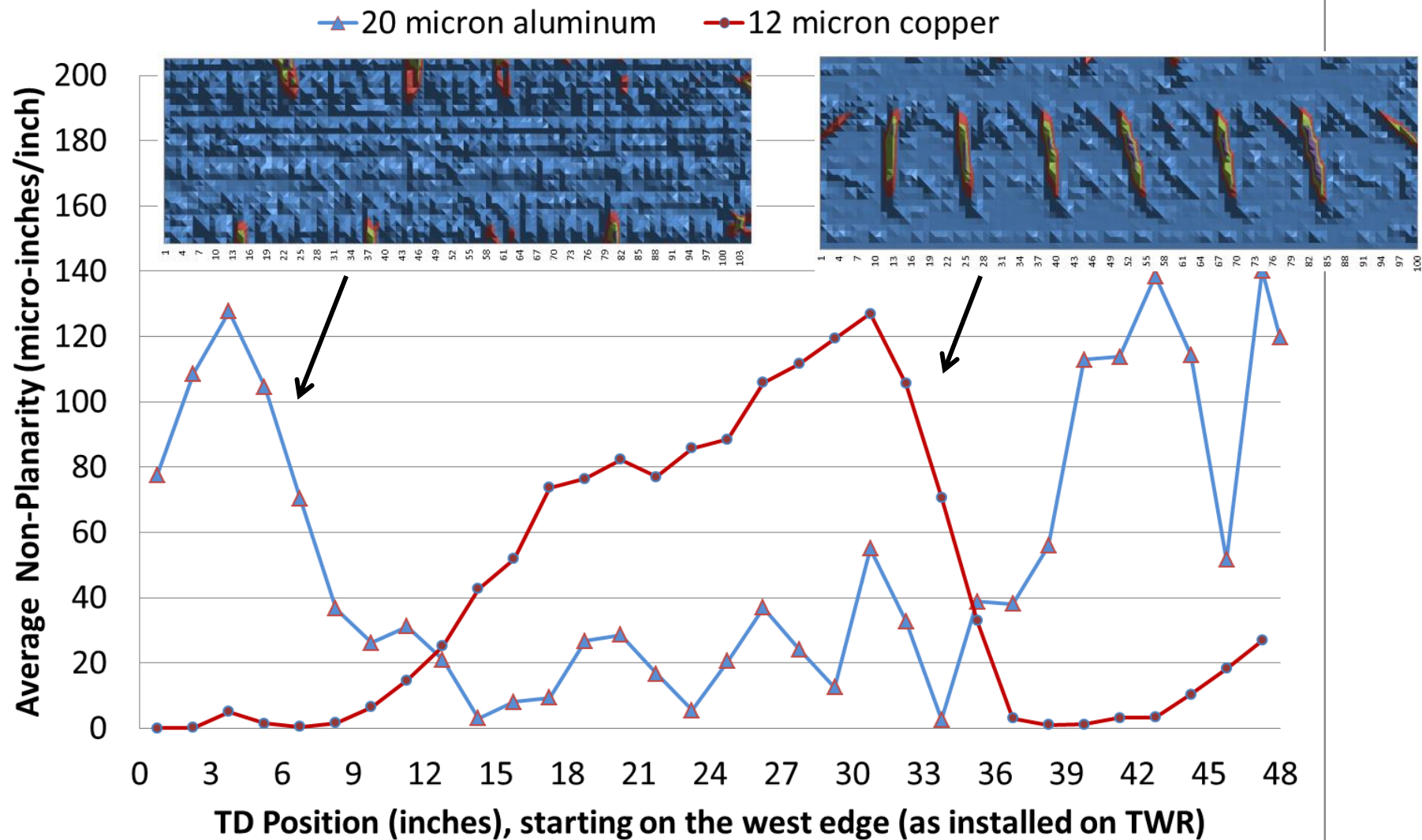
- Sample placed on an optically flat table.
- Measure topography of web in lanes, noting where it lays flat and dimensions of buckles up from the table.
- Web length computed vs. width and normalized to units of strain.

Information courtesy of Optimization Technologies, Inc.



Copper Nonplanarity - visual



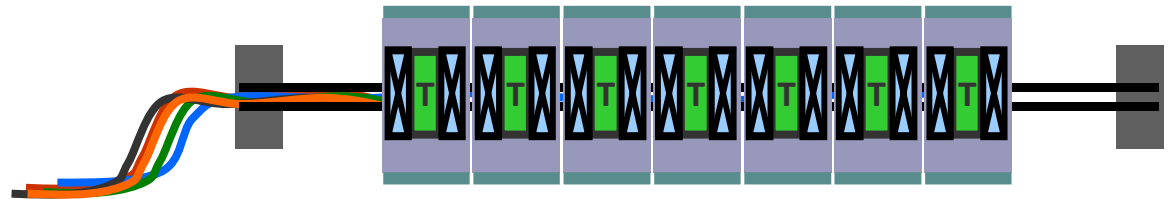


Measuring Baggy Webs (Online)

Segmented Tension Roller:

Individual narrow rollers on a common shaft.
(a.k.a. the shish-kabob roller)

Multi-Internal Transducers*



*Available from Dover-Flexo USA

This segmented tension roller
reads every 6", 10x over 60".
(150 mm over 1.5 m)

Cost: \$41,135

(or 5 segments for \$27,150)

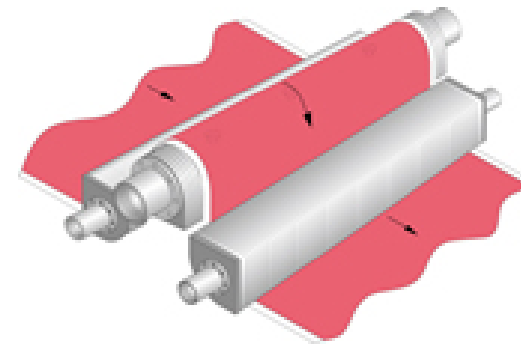
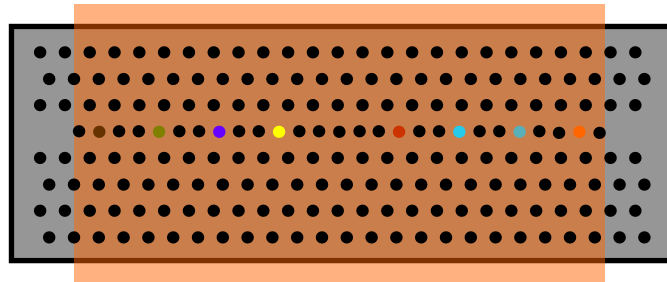
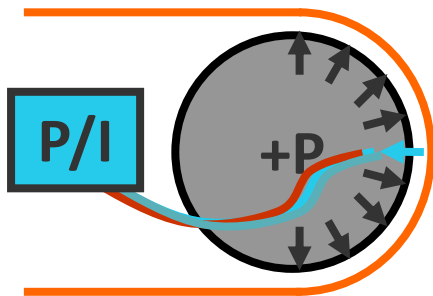


<http://www.dfe.com/solutions.html>

Measuring Baggy Webs (Online)

Air Turn Back Pressure:

By measuring crossweb back pressure on a air turn you can estimate tension variations.



$$P = \frac{T}{r}$$

$$T = rP$$

P = Pressure under web, psi

T = Tension, pli

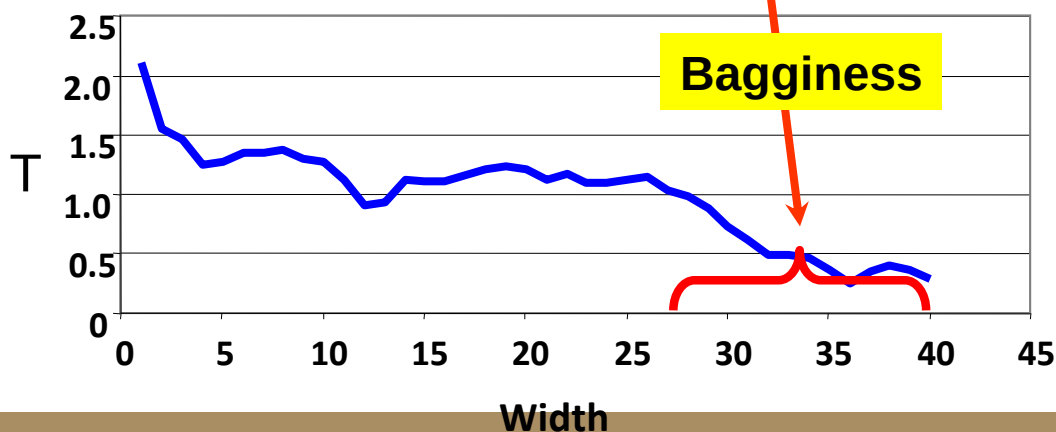
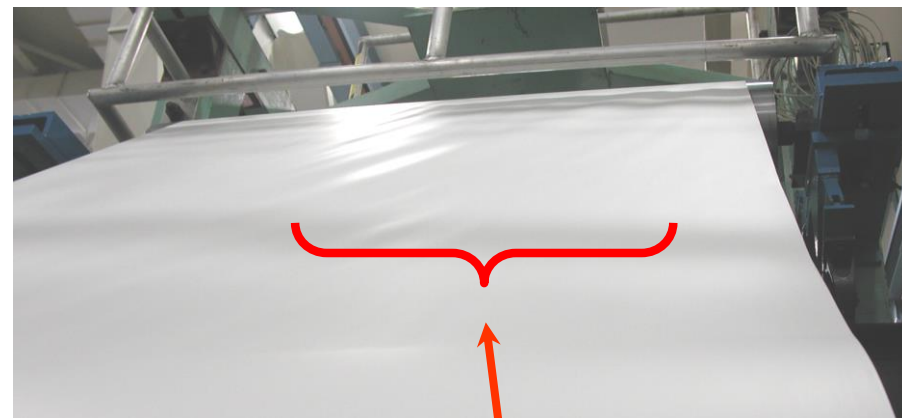
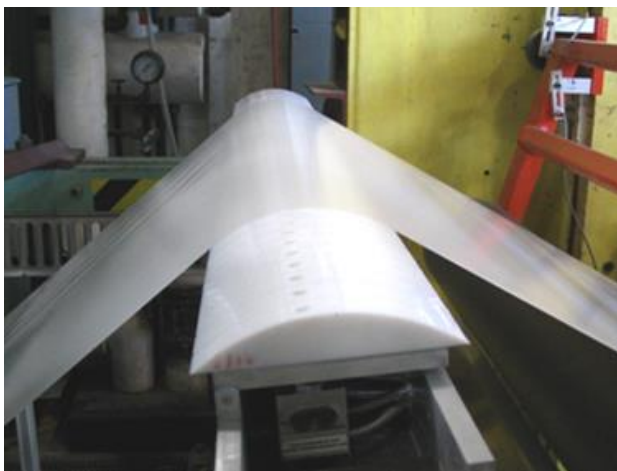
r = Radius, in.

Image courtesy of Bellmatic Ltd.
www.bellma.com

Measure Baggy Webs (Online)

Segmented Tension Bar:

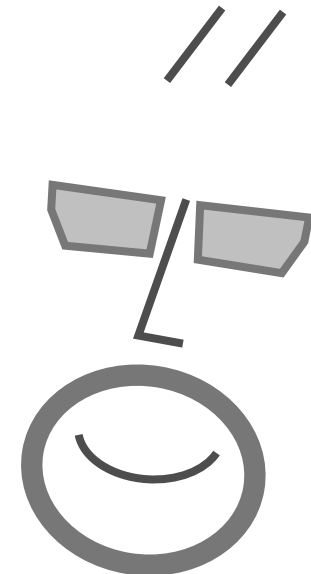
Developed at Paprican, this bar has tension beams reading tension every 1" (25 mm) over 50" (1.25 m).



Winding Solutions Qs

What is roll hardness?
How is roll hardness measured?
Why is roll hardness measured?

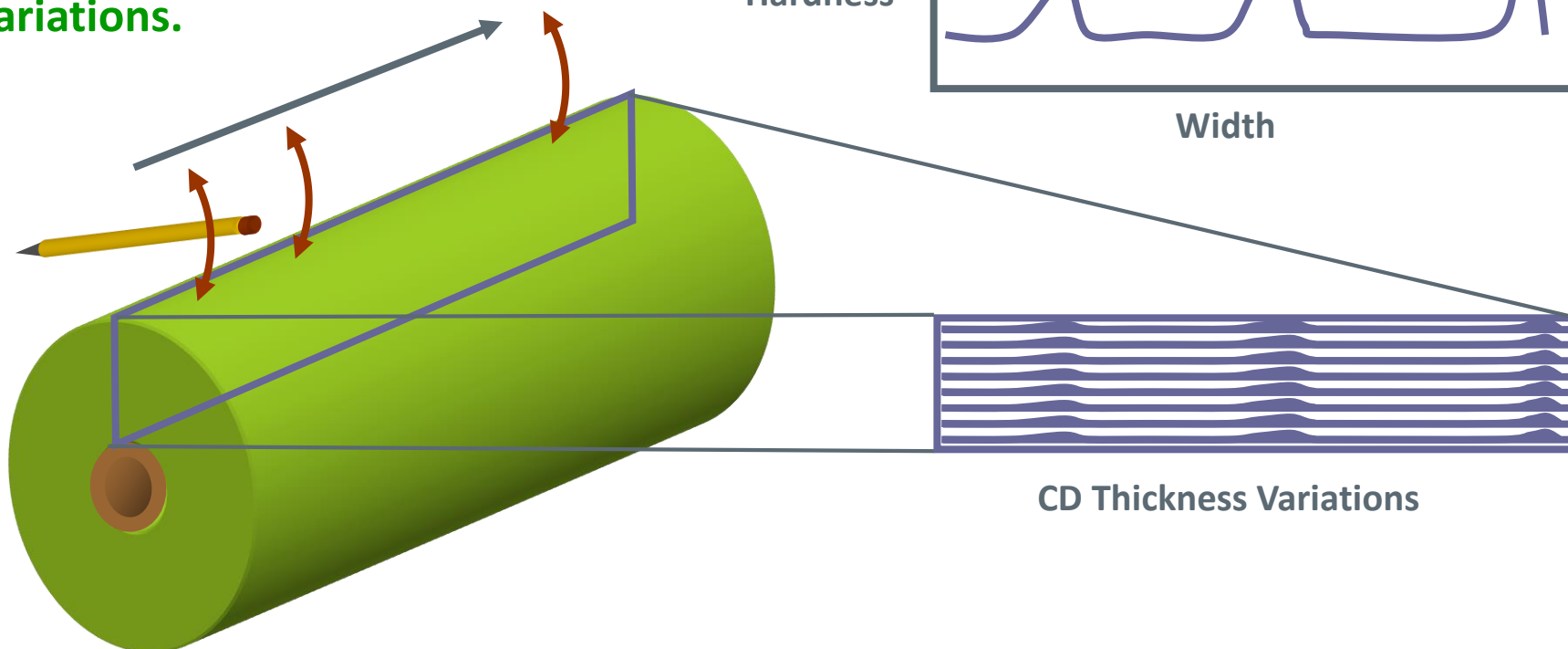
- Measuring bagginess is difficult and requires unwinding the roll.
- Roll hardness is an easy measurement that can be a good predictor of winding consistency or CD roll variations.



What is Roll Hardness?

“Hardness” is a term to describe roll density or tightness, especially roll-to-roll or CD roll variation.

Use a pencil or stick to rap on the roll's surface and feel the hardness variations.



Quantifying Roll Hardness

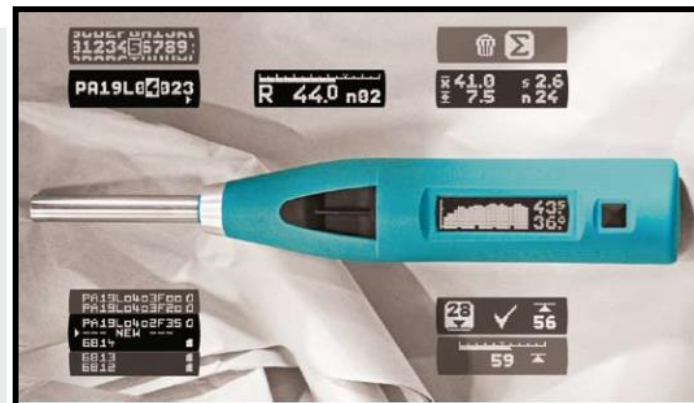
Rhometer



Paro-Tester



PaperSchmidt



Bel-2000-II



Leeb



Tapio RQP V2.0

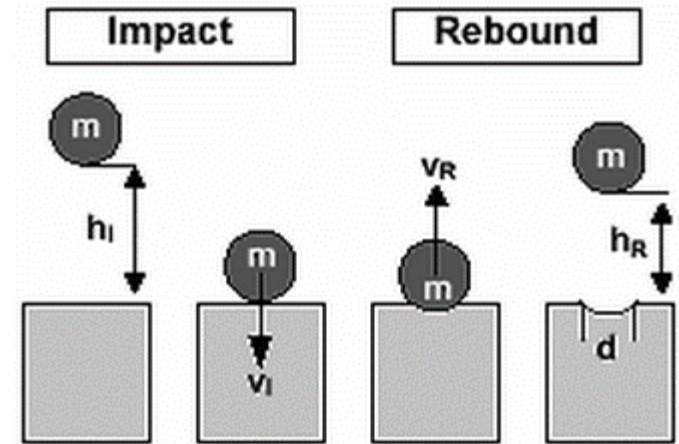


ACA RoQ

Leeb Hardness (from Leeb or Paro Tester)

The Leeb quotient (v_i/v_r) is taken as measure of the energy loss by plastic deformation.

The impact body velocity changes from pre- and post-impact as a function of energy absorption. Harder test samples will have higher rebound velocity than softer ones.

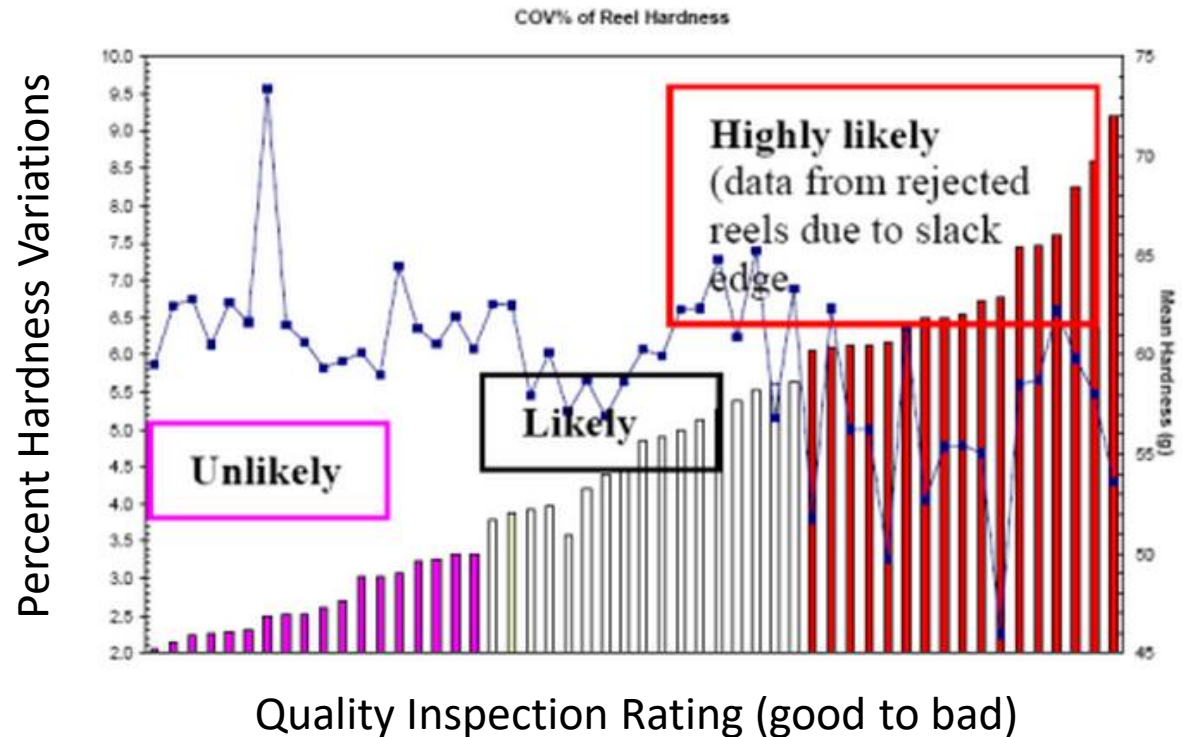


The value of $1000 V_{\text{IMPACT}}/V_{\text{REBOUND}}$ is the Leeb rebound hardness unit HL.

Tapio RQP (Roll Quality Profiler)



Tapio Technologies Oy
 Nuijalantie 13
 FIN-02630 Espoo
 FINLAND
 tel. +358 207 416 220
 fax. +358 207 416 239
 info @ tapiotechnologies.com
 www.tapiotechnologies.com



%Variation

0-3.5%

3.5-6.0%

> 6.0%

Quality Rating

Unlikely to cause waste

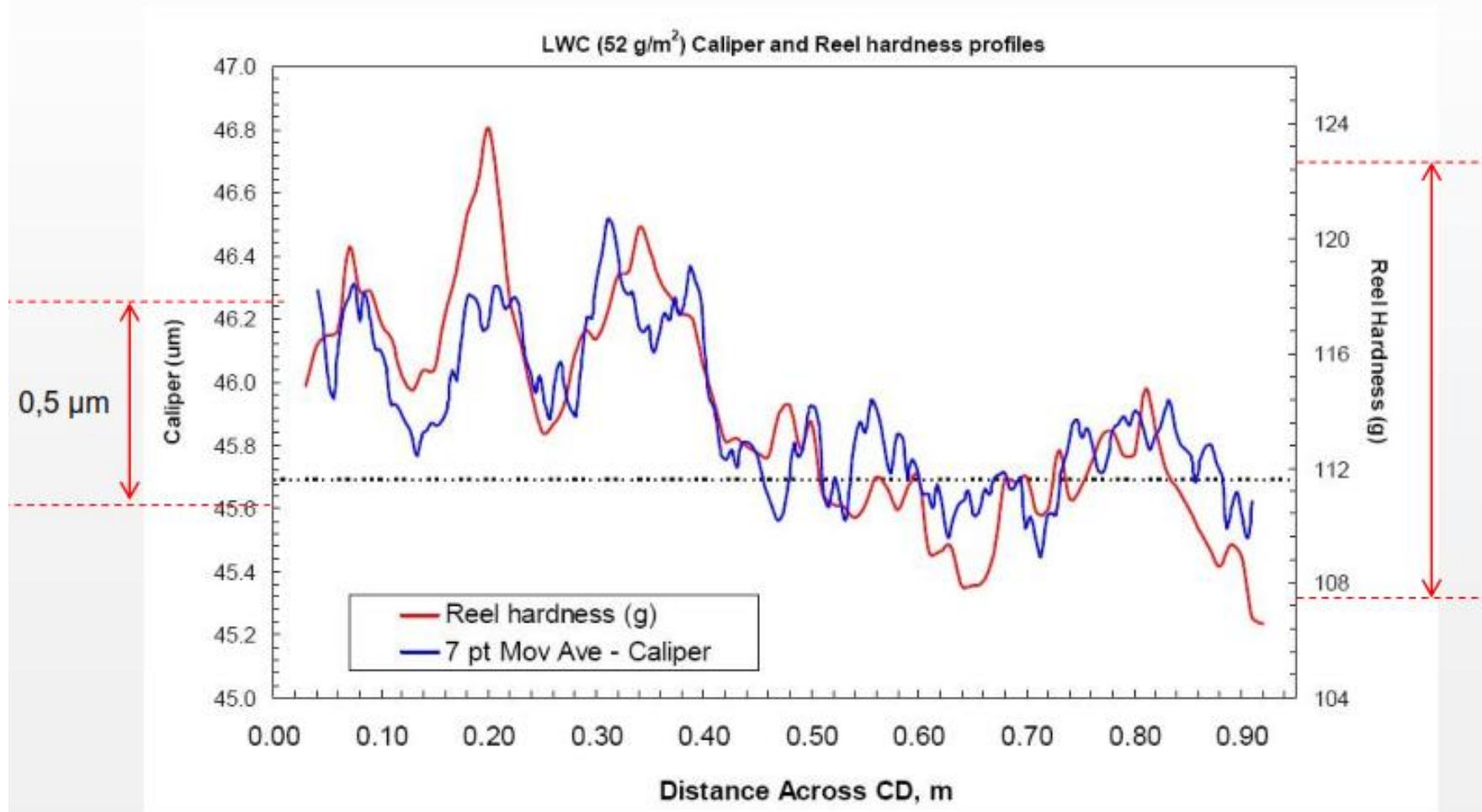
Likely to cause waste

Highly likely to cause waste

*D. Wanigaratne, Appita 2010

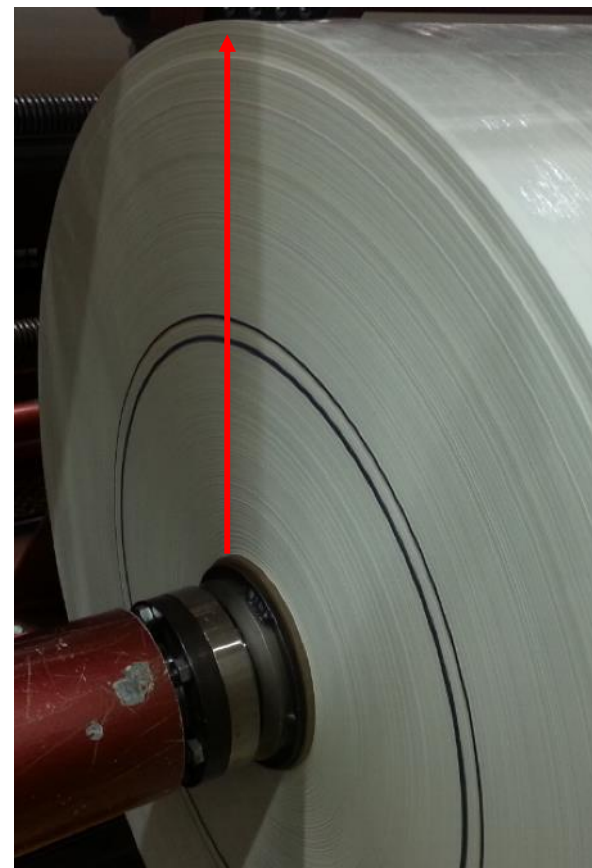
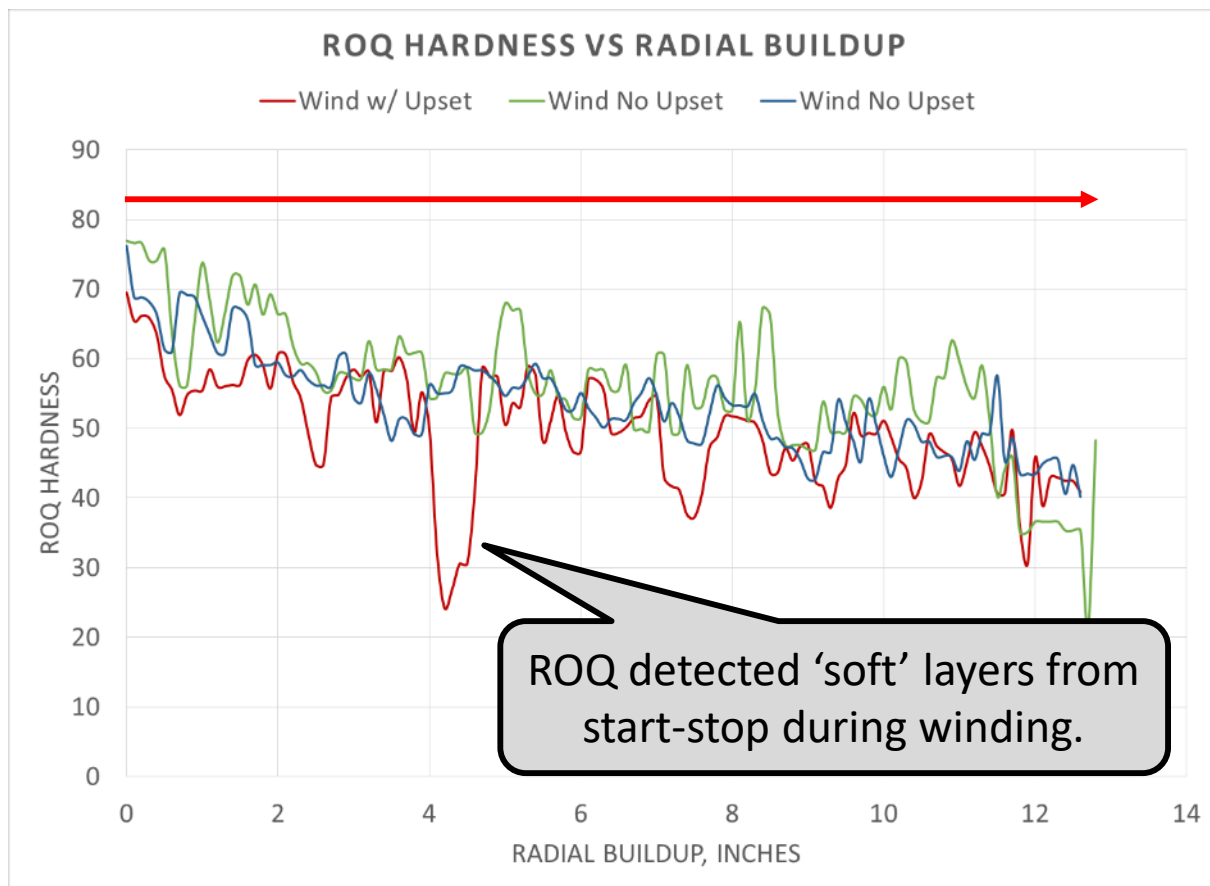
ACA RoQ

Example from ACA where 0.5 micron of thickness variations (about 1%) created a 15% hardness variations.



Radial Hardness Profile with ROQ

<https://youtu.be/lx9BP9Gcdzw>



Core

Outer Diameter

Roll Hardness and Defects

Roll hardness can be effective in differentiating rolls of similar material and roll geometry.

High Hardness Defects:

Blocking, core crushing, hardbands

Low Hardness Defects:

Sag, loose wind, lateral error in transport

Variable Hardness Defects:

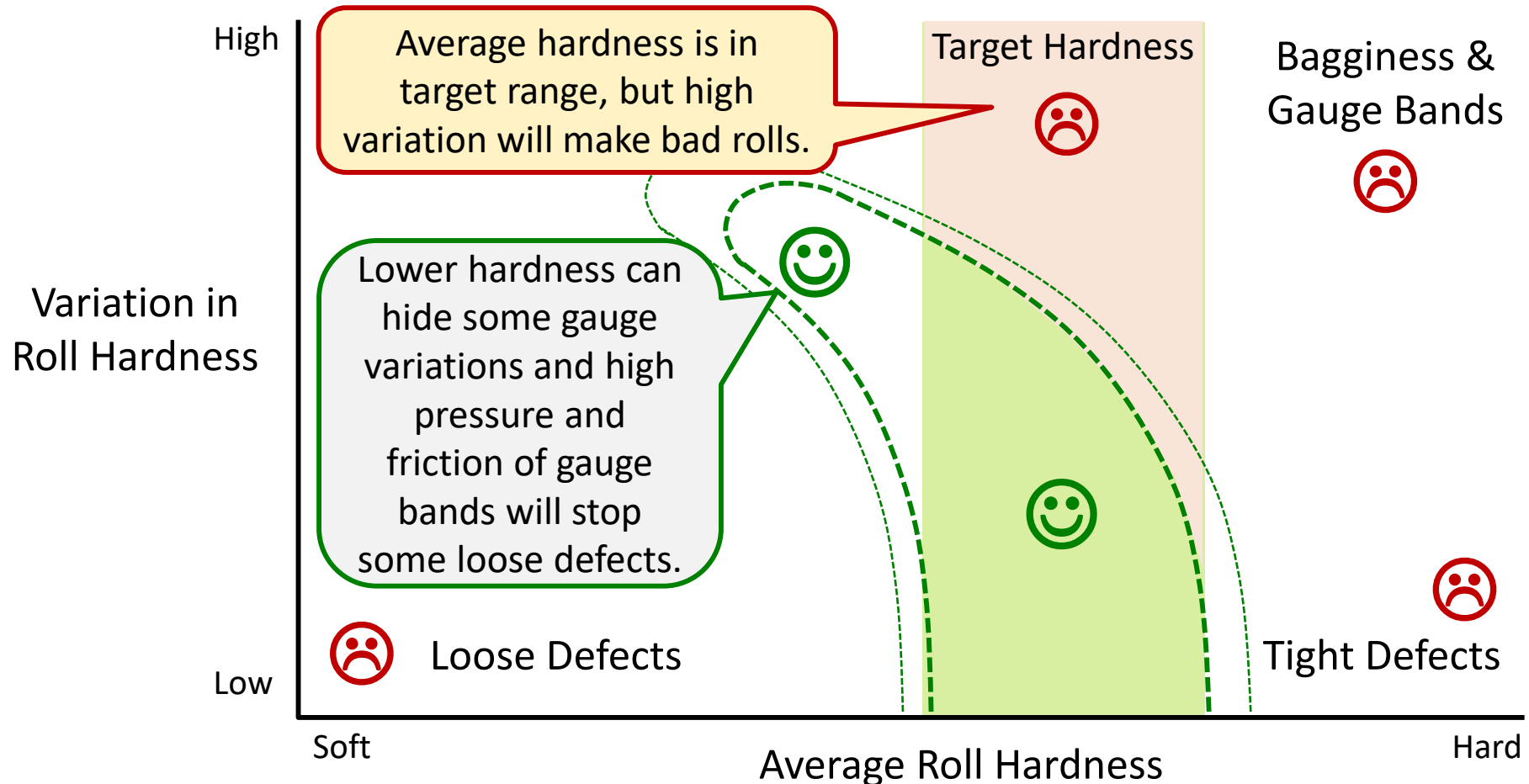
Baggy web, skewed web, loose edges

Roll hardness will be ineffective in finding defects:

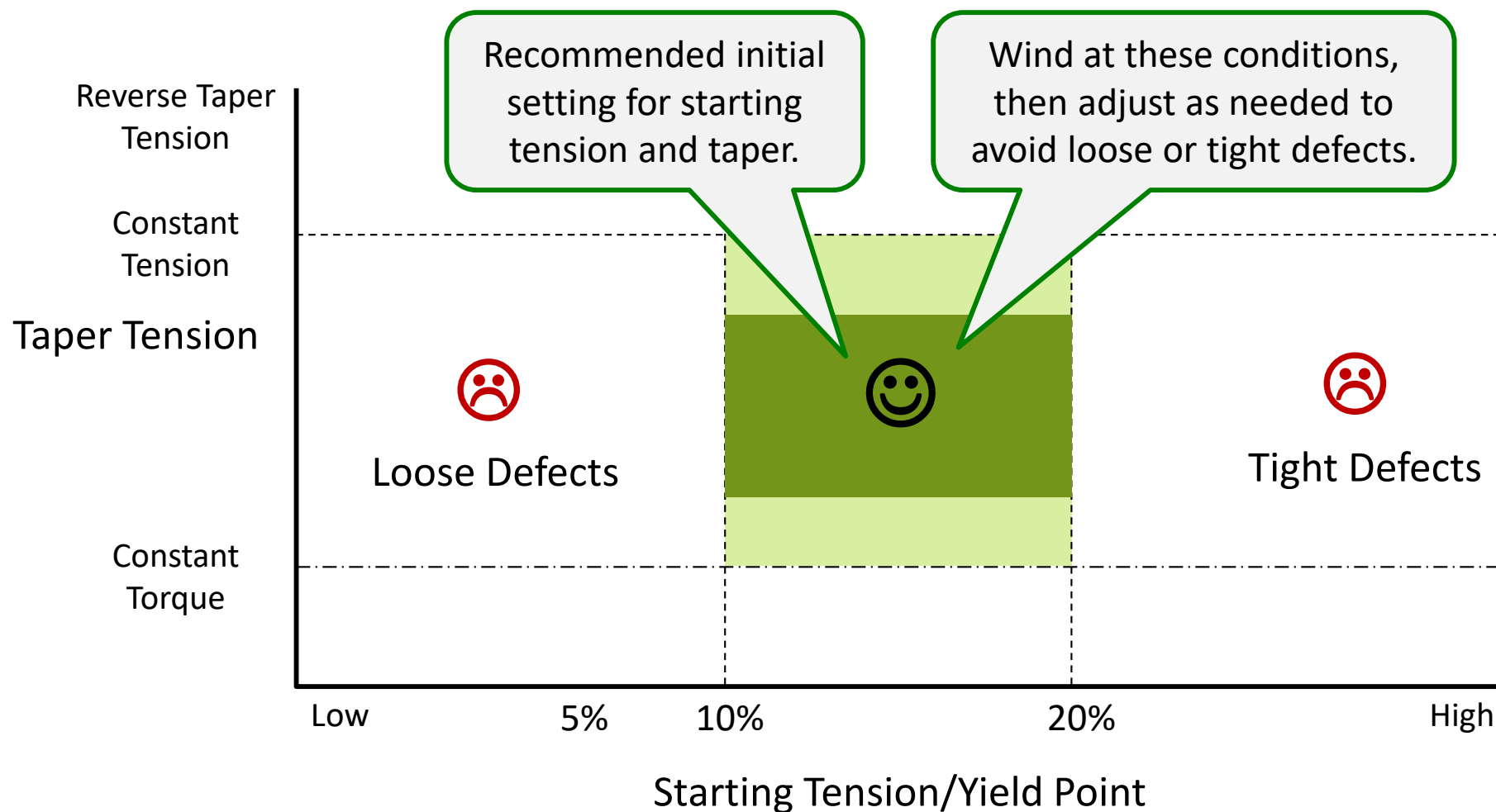
- Roll hardness emphasizes the end of roll winding conditions.
- Roll hardness will fail to detect tension or roll formation upsets near the core or first half of winding.

Roll Quality vs. Roll Hardness

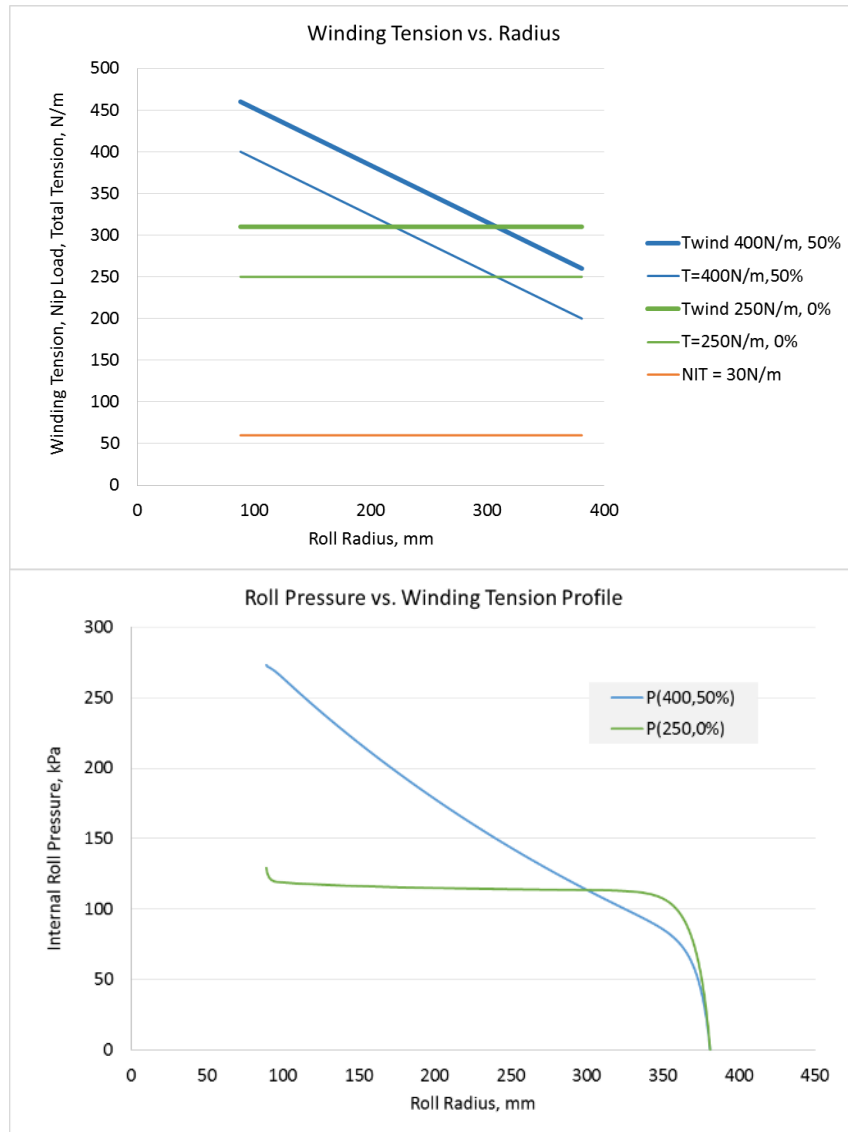
Roll hardness usually has a target range



Roll Quality vs. Roll Hardness



Comparison of Winding Profiles vs Hardness



Case 1:

Starting at 400 N/m and taper 50%,

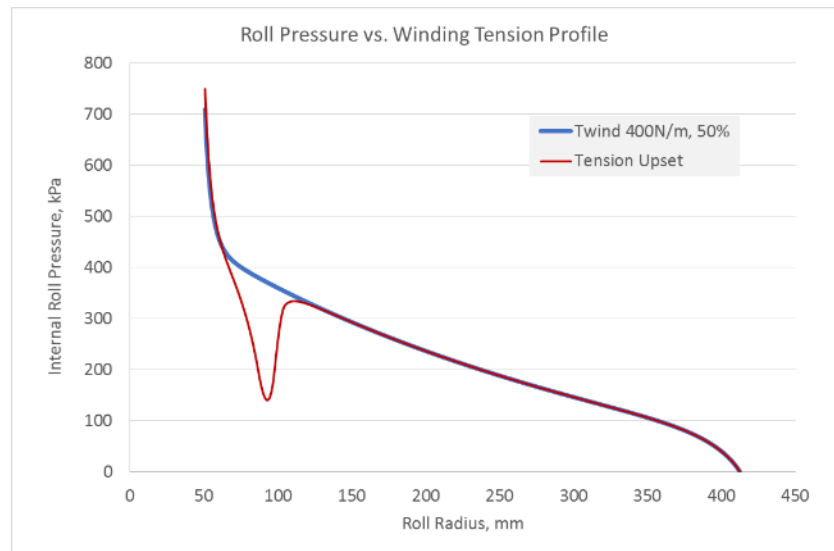
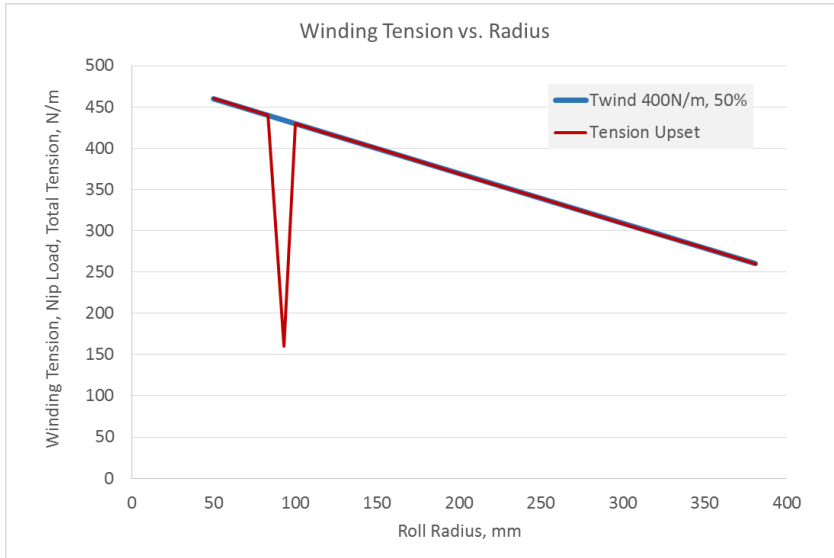
Case 2:

Starting at 250 N/m with constant tension
(Both rolls have constant nip load of 200 N/m,
creating 60 N/m of nip-induced tension.)

Both rolls end with similar interlayer pressure and similar roll hardness.

However, Case 1 with the higher starting tension would have significantly higher near-core pressures, higher torque capacity 😊 or higher risk of blocking 😞.

Comparison of Winding Profiles vs Hardness



Case 1:

Starting at 400 N/m and taper 50%,

Case 2:

Same as Case 1, but with temporary tension upset.

Both rolls end with similar interlayer pressure and similar roll hardness.

However, Case 2 with the tension upset is likely to have slippage in roll handling, cinching-enabled telescoping, or starring in layer wound outside of 'soft' layers.

6.0 Winding

6.1. Winding Success

6.2. Material Properties (Web and Core)

6.3. Winding a Roll

- ▶ Starting a Roll: Roll Transfers
- ▶ Managing Initial Contact
- ▶ Tension and Torque
- ▶ Winder Tensioning
- ▶ Co-Axial Winding

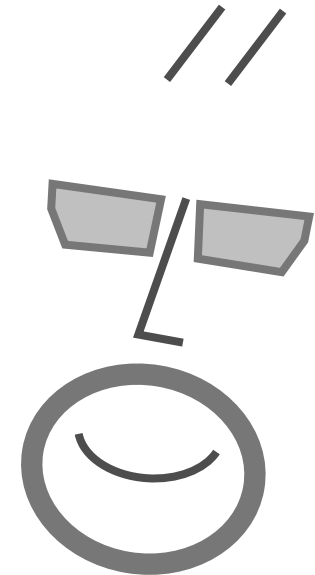
6.4. Roll Structure and Web Quality

- ▶ Wound-In Stresses
- ▶ Lateral Errors
- ▶ Cross Machine Variations (Gauge Bands and Bagginess)
- ▶ In-Roll Buckling

6.6. Wind-Ability

Winding Solutions Qs

What makes a product difficult to wind?



What Makes Winding Difficult?

The BIG 3 Challenges

1. Modulus Ratio
2. Roll Buildup Ratio
3. Traction

Easy

$$E_t/E_r \gg 1$$

$$R_f/R_c < 3$$

$$\mu = 0.2 - 0.5$$

Difficult

$$E_t/E_r \sim 1$$

$$R_f/R_c > 3$$

$$\mu < 0.1, \mu > 0.8, \\ \mu = f(P)$$

Having only one of these three difficult conditions is OK.
If you have two or all three, winding is very difficult.

What Makes Winding Difficult?

4 More Challenges

Difficult

- **Visco-Elasticity** Viscoelastic webs or coatings will cause rolls to loosen or tighten over time.
- **Thickness Variations** Problem begin when % diameter variations similar to web strain.
- **Entrained Air** Too much air causes lubrication-related telescoping and rolls to loosen over time.
- **Roll Transfer Rate (V/L)** When roll changes happen too quickly, cost of handling and waste go up.

Why is Winding Optimization Difficult?

Winding Experiment Costs:

- ▶ Time to wind a single roll may be long.
- ▶ Each data point is costly in terms of time, material, and customer happiness.
- ▶ Many winding defects have at low occurrence rates (sometimes less than 5%).
- ▶ Winding and measuring quality of a large number of rolls is required to find statistically significant difference.
- ▶ Some winding defects are immediately known (lateral shifting), other defects are only revealed at unwinding, often after a lengthy storage and transport time, and likely at a customer location.

Why is Winding Optimization Difficult?

Winding Defect Prediction:

- ▶ Modeling process and roll structure is complex with many variables to consider.
- ▶ Models require iterations of roll pressures and stack modulus.
- ▶ Models rely on variables that are often unknown.
(stack modulus, coefficient of friction, surface roughness).
- ▶ Models may predicts stresses of ideal conditions, but are poor at predicting many defects.
- ▶ Defects are a strong function of thickness profile, but thickness profile is either poorly measured or excluded in simple models.

Why is Winding Optimization Difficult? (continued)

Winding Equipment Challenges:

- ▶ Winding equipment has many different designs.
- ▶ Winding equipment suppliers are often poor at providing winding process variables in pertinent engineering units and may have poor documentation on how the controls work.
- ▶ There is no universal standard for taper tension. In linear taper tension system, the set percent taper is never achieved if winding stops before maximum roll diameter.

Why is Winding Optimization Difficult? (continued)

Equipment Specific Variations:

- ▶ At-speed roll transfer systems are prone to starting rolls with wrinkled or folded over layers near the core.
- ▶ Zero speed roll transfer systems may have control variations during acceleration and deceleration and have changing air lubrication with speed changes.
- ▶ Winding may be dependent on variations in core properties (especially some slip shaft winding systems) and how cores are supported (chucks vs. shafts).
- ▶ Winding may be dependent on winding position (spindle A vs B on turret winders, top vs. bottom shaft on slitters, inboard vs outboard position on slitters).

Winding Control Recommendations – Torque + Nip

- ▶ Winding tension is the combined tension from center torque and nipping.
 - ▶ For example, if COF is 0.25, each 4 N/m of nip load adds 1 N/m of tensioning.
 - ▶ A target winding tension can come from many combinations of high torque + low nip load to low torque + high nip load.
- ▶ Adjust tension balance of torque and nipping based on material and process limits.
 - ▶ Nip more for high speed, large diameter winding.
 - ▶ Nip less for deflection sensitive winding (small diameter core or nipping roll + wide widths). Avoid increasing torque in cinching-sensitive winding.
- ▶ Do not change winding tension or nip load in step functions, preferring gradual linear or curvilinear changes vs. diameter.

Winding Control Recommendations – Nipping

- ▶ Add nip rollers when web porosity and roughness are unable to prevent lubrication.
- ▶ Choose a nip load that prevents shifted layers at the end of winding (final radius, final speed before decelerations, final tension) and run the entire roll at that constant nip load.
 - ▶ Nip load may be set high during the first 10-25mm of winding to create firm near-core layers.
 - ▶ Nip load may be set in to increase with diameter, especially if high load on small diameter, wide cores creates undesired deflection. Higher nip loads may be applied after the core is reinforced with 25-50mm radial buildup.
- ▶ Avoid speed-based nip load tapering.
 - ▶ It makes sense to increase nip load with higher speed when considering air control, but often speed-functions creates more damage than benefit.

Winding Control Recommendations – Torque

- ▶ Many winders are controlled to tension targets, but always consider the torque required to create the tensions (i.e. $\text{torque} = \text{force of tension} \times \text{radius}$).
- ▶ Choose a starting torque to create a tension 10-20% of the product damaging stress (i.e. force per cross-sectional area).
- ▶ Set the final torque to decrease tension 25-50% from the starting core tension (ending at 5-10% of damaging stress).
 - ▶ Final tension can be anywhere between constant tension or constant torque for small roll buildup ratios ($R_{\text{final}}/R_{\text{core}} < 2$).
 - ▶ Taper tension may be close to constant torque for cinching sensitive winding (low coefficient of friction, low strains, low stack modulus).
- ▶ Avoid speed-based tension tapering.
 - ▶ At extreme speeds (> 1000 mpm), tension may need to increase with speed, but first use nip load to avoid speed tapering.

Winding Control Recommendations – Speed

- ▶ Avoid any non-continuity in torque, nip, or speed in winding.
- ▶ Optimize winding for a specific target speed or limited speed range.
 - ▶ Optimized tensions (from torque vs radius) and nip load will be based on a planned speed. Don't expect the same winding conditions are optimal for 2X speed increase or decrease.
- ▶ Wind the entire roll at the same speed, avoiding speed changes until reaching the target length or diameter.
 - ▶ At-speed transfers are preferred to zero-speed transfers to avoid acceleration inertial torques (except when at-speed transfers create wrinkling and near core softness in the roll structure).
 - ▶ For zero-speed roll starts, accelerate at a pre-set rate (not operator controlled) with an 'S' ramp function, typically below 30-60 mpm/s [100-200 fpm/s].

Winding Control Recommendations – Preparation

- ▶ **Verify your winder's capabilities**
 - ▶ What is minimum and maximum nip load and torque?
 - ▶ Verify that control settings equal actual performance.
 - ▶ Use nip load geometry, verified roller weight and supplied air pressure to confirm nip load and uniformity are as expected.
 - ▶ Use load cell rollers or static measurements of tension or torque to confirm created tensions and torque match machine set points.
- ▶ **Confirm winder range allows running at 'default' winding conditions. Default conditions equal:**
 - ▶ Constant nip load (that prevent air-related shifting at target speed and maximum diameter).
 - ▶ Starting tension at 10-20% of damage stress (i.e. force / area).
 - ▶ Final tension = 50-75% of starting tension (5-10% damage stress).

Winding Control Recommendations – Optimize

- ▶ **Step 1: Winding a number of rolls at default values**
 - ▶ Constant nip load (without air-related shifting).
 - ▶ Starting tension at 10-20% of damage stress (i.e. force / area).
 - ▶ Final tension of 25-50% of starting tension.
- ▶ **Step 2: Evaluate roll and sheet quality vs radial position**
 - ▶ Does the roll show tight or loose defects at the outside of the roll?
 - ▶ Does the roll show tight or loose defects at the near core layers?
 - ▶ Re-evaluate roll and web quality at unwinding after expected storage and transport conditions.
- ▶ **Step 3: If needed, select new winding conditions.**
 - ▶ Where tight roll defects form, decrease torque or nip load;
 - ▶ Where loose roll defects form, increase torque or nip load
- ▶ **Repeat Step 1-3 to reach minimum roll and web defects.**

Winding Control Recommendations – Optimizing

- ▶ Thinner, low modulus, and tacky webs need lower than average tensions. Thicker, high modulus, and slippery webs need higher than average tensions.
- ▶ Repeat Steps 1-3 for winding with good and poor thickness profile ranges. (Many defects will appear with poor thickness profile. Some will appear with good thickness profile.)
- ▶ Consider whether a design thickness profile is best (smile, frown, W, or M shaped thickness profile vs. width may improve winding).
- ▶ Consider that the best winding conditions may be a strong function of good vs poor thickness profile, winding at lower tension when thickness profile is poor.

Winding Control Recommendations – Nip Limited?

If roll quality indicates winding is improved outside of existing winder nip capabilities:

- ▶ If more nip load is needed:
 - ▶ Upgrade to larger air cylinders
 - ▶ Install pressure boosting (to run above 6 bar)
 - ▶ If nipping from above, change to a heavier nip roller.
 - ▶ Higher loads may lead to deflection problems, so increase roller and core stiffness may be needed.
- ▶ If less nip load is needed:
 - ▶ Upgrade to smaller air cylinders
 - ▶ If nipping from below, change to a lighter nip roller.

Winding Control Recommendations – Torque Limited?

If roll quality indicates winding is improved outside of existing winder torque capabilities:

- ▶ If more torque load is needed:
 - ▶ Upgrade to larger motor or clutch.
 - ▶ For friction clutch winding, change to higher coefficient of friction surfaces, increase the number of friction surfaces, or install pressure boosting (to run above 6 bar).
 - ▶ Ensure all components have the needed increase torque transmission capacity (without excessive torsion, slip, or failure), including core chuck or shaft coupling.
 - ▶ Changing to lower gear ratio can create higher torque with the sacrifice of lower maximum speed.
- ▶ If less torque is needed:
 - ▶ Change to smaller motors, clutches, system drag, friction components, or high gear ratios.

Winding Control Recommendations – Tension Limited?

If the closed-loop tension feedback components limits winding:

- ▶ **Load cell rollers:**
 - ▶ Upgrade to larger or small load cell.
 - ▶ Increase or decrease wrap angle or change orientation of the load cell to wrap angle to change tension to measure load relationship.
 - ▶ Load cell rollers usually have greater range of measurement (50:1 or 100:1) than torque ranges (20:1 or 40:1), so load cell should be able to successfully provide feedback over the full applied torque range.
- ▶ **Dancer rollers:**
 - ▶ Change to larger or smaller air cylinders.
 - ▶ Change the mechanical leverage of air cylinder to roller (pivot length change).
 - ▶ Ensure low system friction and inertia for low tension ranges.